

COMBINATORIAL AND TOPOLOGICAL GAMES: BAIRE SPACES AND APPLICATIONS IN CLASSICAL MATHEMATICS

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INTRODUCTION

In 1935 Stanisław Mazur wrote a problem in the Scottish Book, the notebook kept at the Scottish Café (see Figure 2), in the beautiful city of Lwów, where the Polish mathematicians of the time did a good part of their thinking. Two players take turns choosing nested intervals of the real line, each inside the previous one, and the question is when the first player can force the point where the intervals close down to land in a given set, and when the second player can keep it out. Banach found the answer but never published a proof; the first published proof is Oxtoby's, of 1957. The game has been called the Banach–Mazur game ever since.

The game turns out to be a precise way of measuring *how big* a set is: a set is “large” if one of the players can always steer the game into it, and “small” if the other player can always steer the game away from it. Over the last sixty years people have changed the rules in all sorts of ways (prescribing the sizes of the moves, limiting how much the players may compute, letting them play vectors, points or covers instead of intervals), and each new version has cracked open a problem in some unexpected corner of mathematics. These notes first develop the theory of the game and then tell twelve of those stories.

Part I develops the theory: the Banach–Mazur game, its strengthening the strong Choquet game, and what the two games say about Baire spaces and complete metrizable spaces. It is self-contained, and it can be read as a short course on its own.

Part II is about applications of the Banach–Mazur game itself and its close relatives, the games of Choquet type, in which the two players take turns shrinking a set:

Number theory: Schmidt's games and badly approximable numbers (Schmidt, 1966).

Dynamical systems and Lie theory: Bounded orbits of flows on spaces of lattices: Margulis's conjecture, proved by Kleinbock and Margulis (1996).

Theoretical computer science: Baire category inside complexity classes (Mehlhorn, 1973; Lutz, 1990).

Topological groups: When separate continuity implies joint continuity (Ellis, 1957; Namioka, 1974; Kenderov, Korteov and Moors, 2001).

Infinite-dimensional Ramsey theory: The Ellentuck topology (Ellentuck, 1974) and the Kastanas game (Kastanas, 1983).

Functional analysis: Asplund spaces (Namioka–Phelps and Stegall, 1975; the game of Kenderov and Moors, 1999) and Gowers's dichotomy for Banach spaces (Gowers, 1996).

Set theory: Nonprincipal ultrafilters on ω , the Baire property and the axiom of choice (Sierpiński, 1938; Talagrand, 1980), and Galvin's games for selective ultrafilters and P-points.

Part III is about other famous topological games, in which the players do not simply shrink a set:

43) Definicja nowej gry Mazur
 Dany jest zbiór E liczb rzeczywistych. Gra między graczami A i B polega na tym,
 że: A wybiera dowolny odciinek δ_1 , B następnie wybiera dowolny odciinek δ_2 za-
 wartą w δ_1 , A następnie wybiera dowolny odciinek δ_3 zawarty w δ_2 i t. d.; A wy-
 grywa jeżeli przekrój $\delta_1, \delta_2, \dots, \delta_n, \dots$ zawiera punkt z zbioru E , w przeciwnym przypadku
 przegrywa.
 Jeżeli E jest dopełnieniem zbioru 1-jej kategorii, to istnieje metoda prz. której A
 wygrywa; jeżeli E jest zbiorem 1-jej kategorii, to istnieje metoda prz. której B
 wygrywa.
 Problemat (Nagroda: 1 flaszka wina J. Mazur)
 Czy prawda jest, że dla A istnieje metoda wygrania tylko dla takich zbiorów
 E , których dopełnienie jest w jakimś odciunku 1-jej kategorii; podobnie czy meto-
 da wygrania istnieje dla B tylko wówczas gdy E jest zbiorem 1-jej kategorii.
 ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35 Banach

FIGURE 1. Mazur's Problem 43 in the Scottish Book, in his own hand, with the prize he offered: "Nagroda: 1 flaszka wina" (a bottle of wine). Below, from the opposite page: "ad 43) Przypuszczenie Mazura jest prawdziwe. 4/VIII 35" ("Mazur's conjecture is true," 4 August 1935), signed by Banach. Courtesy of the Banach family, via the Stefan Banach Portal, Adam Mickiewicz University, Poznań (kielich.amu.edu.pl/Stefan_Banach).

Set-theoretic topology: The Menger game and the refutation of Menger's conjecture (Fremlin and Miller, 1988).

Descriptive set theory: Borel determinacy (Martin, 1975) and the Wadge hierarchy (Wadge, 1972).

Part IV collects three beautiful applications of games that are not topological at all:

Probabilistic combinatorics: The Erdős–Selfridge theorem and derandomization (Erdős and Selfridge, 1973).

Fixed point theory: Hex and the Brouwer fixed point theorem (Gale, 1979).

Model theory: Ehrenfeucht–Fraïssé games (Fraïssé, 1954; Ehrenfeucht, 1961).

Part V, an epilogue, tells the story of the Scottish Book: how it survived the war, how it travelled to Los Alamos, Denton and Toronto, and how the games it started came back to settle problems of Banach himself.

Most sections of Parts II–IV can be read on their own, after Part I; the opening of Part II says which of its sections lean on which. In the last section of Part II the Banach–Mazur game confronts the axiom of choice directly: a nonprincipal ultrafilter on ω defines a game

that nobody wins, and the same game shows that determinacy and nonprincipal ultrafilters cannot live together.

Since every game in Part II is the original one with a few rules changed, we record the original here, in the form in which we will keep modifying it. Mazur's question was about a set E that the first player aims at; we write $S = \mathbb{R} \setminus E$ for the set that the second player aims at, and give the payoff to her (see the conventions below). Part I proves a version of the theorem in the box that is valid in every topological space (see Theorem 3.5).

The Banach–Mazur game. Fix a set $S \subseteq \mathbb{R}$. Player I chooses a closed interval U_0 of positive length; Player II chooses a closed interval $V_0 \subseteq U_0$ of positive length; Player I chooses $U_1 \subseteq V_0$; and so on, forever. Player II wins if every point of $\bigcap_k V_k$ belongs to S . Theorem (Banach; Oxtoby, 1957): Player II has a winning strategy if and only if $\mathbb{R} \setminus S$ is meager, that is, if and only if S is “large” in the sense of Baire category.

Behind all the variations of the rules, two features of games recur. A winning strategy is built one move at a time, so a big global question (“is this set large?”) turns into a small local one (“what should I play next?”). And strategies can be *combined*: a player can pursue several goals at once by taking turns between them, one round for this goal, the next round for that one. The second feature is why sets defined by games survive countable intersections. Comeager sets and sets of full measure survive them too, but largeness in the sense of dimension does not: two sets of full Hausdorff dimension, or of positive measure, can have empty intersection. The winning sets of Schmidt's game have full dimension and are closed under countable intersections (see Section 7).

Conventions. We follow the conventions of Part I throughout. Player I moves first; in games of Banach–Mazur type, Player II answers inside Player I's move; and the payoff set belongs to Player II: she is the one trying to produce an object with a given property, and he is the one trying to stop her. Where an author used other names for the players (Spoiler and Duplicator, Maker and Breaker, ONE and TWO, β and α , Alice and Bob), we say which of them is our Player I. Each section of Parts II–IV has roughly the same shape: the problem, its history, the game, the main result, and how the game solves the problem.

Part I. Topological games and Baire spaces

This part studies *topological games*: infinite games in which two players alternately choose open sets, and in which topological properties of a space are characterized by which player has a winning strategy. The two main results are that a space is a Baire space exactly when the first player has no winning strategy in the Banach–Mazur game (see Theorem 3.3 below), and that a metrizable space is completely metrizable exactly when the second player has a winning strategy in the strong Choquet game (see Theorem 4.4 below). The easy half of the second result, together with the first, already gives a game-theoretic proof of the Baire category theorem (see Corollary 4.5 below). The hard half, that a winning strategy for the second player produces a complete metric, is proved in Section 5; it uses three classical tools that are worth knowing for their own sake: Alexandrov's theorem on G_δ sets, Stone's theorem on locally finite refinements, and König's lemma.

What you need to know. We assume a first course in metric spaces and the basic vocabulary of point-set topology: open and closed sets, closure and interior, dense sets, continuity, and Cauchy sequences and completeness. We also use the following notions.



FIGURE 2. The Scottish Café. (a) Plac Akademicki in Lwów, 1925; the café occupied the ground floor of the turreted corner building on the right, at ul. Akademicka 27 (today Prospekt Shevchenka 27). Photograph by Marek Münz; National Library of Poland (Polona), public domain. (b) The interior of the café in the 1930s; photographer unknown, public domain.

- (1) A topological space X is *metrizable* if there is a metric on X whose open sets are exactly the open sets of X ; such a metric is called *compatible*. The space is *completely metrizable* if it has a compatible metric that is complete. Being completely metrizable is a property of the topology: the irrational numbers are not complete in the usual metric of $\mathbb{R}$, but they have a complete compatible metric (see Section 5).
- (2) A G_δ set is a countable intersection of open sets.
- (3) Every metric space (X, d) is a dense subspace of a complete metric space, its *completion*, which is unique up to an isometry that fixes X ; for instance, $\mathbb{R}$ is the completion of $\mathbb{Q}$.
- (4) *Zorn's lemma*: if every chain in a non-empty partially ordered set has an upper bound, then the set has a maximal element. We use it to find maximal families of pairwise disjoint open sets.

1. BAIRE CATEGORY

In this section we recall the notions of category and show that a space is a Baire space exactly when no non-empty open subset of the space is of the first category (see Theorem 1.3 below).

Definition 1.1. Let X be a topological space, and let $E \subseteq X$ be given. The closure of E will be denoted by $\overline{E}$, and the interior of E by E° .

- (1) The set E is said to be *nowhere dense* in X if $(\overline{E})^\circ = \emptyset$. Equivalently, the set E is nowhere dense if every non-empty open set $U \subseteq X$ contains a non-empty open set V such that $V \cap E = \emptyset$.
- (2) The set E is said to be of the *first category* in X (or *meager*) if E is the union of a countable family of nowhere dense sets. A set which is not of the first category is said to be of the *second category* (or *non-meager*).

- (3) A set $C \subseteq X$ is said to be *comeager* (or *residual*) if the complement $X \setminus C$ is of the first category.
- (4) The space X is called a *Baire space* if the intersection of every countable family of dense open subsets of X is dense in X .
- (5) The set E has the *Baire property* if there is an open set O such that the symmetric difference $E \triangle O$ is of the first category; that is, E is an open set up to a negligible error. Every open set and every closed set has the Baire property, and so does every Borel set.



Remark 1.2. The notions in Definition 1.1 classify subsets according to topological “size,” providing a qualitative analogue of measure theory.

- (1) A nowhere dense set is thin everywhere: every non-empty open set contains a non-empty open set that misses the nowhere dense set entirely. Every finite subset of $\mathbb{R}$ is nowhere dense, and so is the Cantor ternary set.
- (2) A set of the first category is topologically negligible, and a countable union of such sets is again of the first category, just as a countable union of Lebesgue-null sets is null. The rationals $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$ form a set of the first category in $\mathbb{R}$. The analogy has limits: the line $\mathbb{R}$ is the union of a set of the first category and a set of Lebesgue measure zero [Oxt80, Chapter 1], so “small” in one sense can be “large” in the other.
- (3) A comeager set contains “most” points of the space; the elements of a comeager set are called the *generic* points of X . In $\mathbb{R}$, the set of irrational numbers is comeager.
- (4) In Theorem 1.3 below we show that X is a Baire space exactly when no non-empty open subset of X is negligible. This *local* condition is stronger than asking that X itself be of the second category in X : the disjoint union $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\}) \subseteq \mathbb{R}^2$ is of the second category in itself, because the open subset $\mathbb{R} \times \{1\}$ of X is, but X is not a Baire space, because the open subset $\mathbb{Q} \times \{0\}$ of X is of the first category (see Figure 3).



FIGURE 3. The space $X = (\mathbb{Q} \times \{0\}) \cup (\mathbb{R} \times \{1\})$ of Remark 1.2(4). Each line is an open subset of X . The upper line makes X of the second category in itself; the lower line, a countable union of points, is a non-empty open set of the first category, so X is not a Baire space.

Theorem 1.3. *A topological space X is a Baire space if and only if no non-empty open subset of X is of the first category.*

Proof. Suppose that X is a Baire space, and let $G \subseteq X$ be open and non-empty. If $G = \bigcup_{n=1}^{\infty} E_n$ with each E_n nowhere dense, then $V_n = X \setminus \overline{E_n}$ is open and dense in X . Hence $\bigcap_{n=1}^{\infty} V_n$ is dense in X , so $\bigcap_{n=1}^{\infty} V_n$ must intersect G . However, we have

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction. Thus G is of the second category.

Conversely, suppose no non-empty open subset of X is of the first category. Let $\{V_n\}$ be a sequence of dense open subsets of X , and let $O \subseteq X$ be open and non-empty. Write $E_n = X \setminus V_n$. Each E_n is closed, so $\overline{E_n} = E_n$, and E_n has empty interior because V_n is dense; thus each E_n is nowhere dense. If $O \cap \bigcap_{n=1}^{\infty} V_n = \emptyset$, then $O \subseteq \bigcup_{n=1}^{\infty} E_n$. This makes O a non-empty open set of the first category, contrary to our hypothesis. Hence $O \cap \bigcap_{n=1}^{\infty} V_n \neq \emptyset$, and the intersection is dense. ☺

Exercises.

- (1) Let X be a topological space, and let $E \subseteq X$ be given. Show that the two descriptions of a nowhere dense set in Definition 1.1(1) are equivalent, and that the closure of a nowhere dense set is nowhere dense.
- (2) Show that the union of two nowhere dense subsets of a topological space is nowhere dense. Show by an example in $\mathbb{R}$ that the union of countably many nowhere dense sets need not be nowhere dense.
- (3) Let X be a topological space. Show that X is a Baire space if and only if every comeager subset of X is dense in X . Deduce that in a non-empty Baire space no comeager set is of the first category.
- (4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Show that U , with the subspace topology, is a Baire space. (Hint: show that a subset of U that is nowhere dense in U is nowhere dense in X , and use Theorem 1.3.)
- (5) (*) Let X be a Baire space, and let $D \subseteq X$ be a dense G_δ set. Show that D , with the subspace topology, is a Baire space. (Hint: a dense open subset of D has the form $W \cap D$ with W open and dense in X .)

2. INFINITE GAMES

In this section we fix the formal framework for the games of these notes: trees of positions, plays, payoff sets, and strategies.

Definition 2.1. Let A be a non-empty set. We denote by $A^{<\omega}$ the collection of all finite sequences $s = (a_0, \dots, a_{n-1})$ of elements of A . The length n of s is denoted by $|s|$; the empty sequence $\langle \rangle$ has length 0. If $t \in A^{<\omega}$ and $|t| \leq |s|$ with $t_i = s_i$ for $0 \leq i < |t|$, we write $t \subseteq s$; if sequences are regarded as functions defined on initial segments of ω , this is literally inclusion. For $s \in A^{<\omega}$ and $a \in A$, we write $s \frown a = (s_0, \dots, s_{|s|-1}, a)$ for the extension of s by a .

A non-empty subset $T \subseteq A^{<\omega}$ is called a *tree* if $s \in T$ and $t \subseteq s$ implies $t \in T$. For each $s \in T$, we write

$$\text{Succ}(s) = \{a \in A : s \frown a \in T\}.$$

We assume throughout that $\text{Succ}(s) \neq \emptyset$ for every $s \in T$. An infinite sequence $x = (a_n)_{n=0}^{\infty} \in A^\omega$ is a *play* in T if $(a_0, \dots, a_{n-1}) \in T$ for all $n \geq 1$. The set of all such plays is denoted by $[T]$.

Definition 2.2. An *infinite game* is a pair (T, W) , where $T \subseteq A^{<\omega}$ is a tree and $W \subseteq [T]$. Two players, I and II, alternately choose elements $a_n \in A$ such that $(a_0, \dots, a_n) \in T$. Player I chooses a_{2n} ; Player II chooses a_{2n+1} ($n = 0, 1, 2, \dots$). Player II *wins* the resulting play $x = (a_n) \in [T]$ if $x \in W$; otherwise, Player I wins. The elements of T are called *positions*.

A *strategy* for Player I is a function σ defined on the positions of even length such that $\sigma(s) \in \text{Succ}(s)$. A play or position is *consistent* with σ if $a_{2n} = \sigma(a_0, \dots, a_{2n-1})$ whenever

a_{2n} is defined. We say that σ is a *winning strategy* for Player I if every play consistent with σ lies in $[T] \setminus W$.

A strategy τ for Player II and the notion of a winning strategy for Player II are defined analogously.



Remarks 2.3. (1) In these notes W is the set of plays won by Player II. Many texts, including [Kec95], use the opposite convention, in which the payoff set is Player I's. Ours suits the games below, where Player II is the player who wants a non-empty intersection.

- (2) The two players need not choose from the same kind of object: if Player I chooses from a set A_I and Player II from a set A_{II} , take $A = A_I \cup A_{II}$ and let the tree T record which moves are legal.
- (3) At most one player has a winning strategy: if σ and τ were winning strategies for Players I and II, the unique play consistent with both would lie both outside and inside W .
- (4) The game is called *determined* if one of the two players has a winning strategy. Not every game is determined: there are Baire spaces X in which neither player has a winning strategy in the Banach–Mazur game $\text{BM}(X)$ of Section 3 (see the remark after Theorem 3.3), and the axiom of choice produces undetermined games in which the players choose finite binary strings (see Section 13). On the other hand, every game in which the players choose natural numbers and whose payoff set is a Borel subset of ω^ω is determined; this is Martin's theorem (see Section 15).
- (5) A strategy may use the whole position, not only the last move: the move prescribed by σ at a position $(a_0, \dots, a_{2n-1})$ may depend on everything played so far.

Exercises.

- (1) Let $A = \{0, 1\}$ and $T = A^{<\omega}$. For each of the following payoff sets $W \subseteq [T]$, decide which player has a winning strategy in (T, W) and describe one: (a) the plays containing at least one 1; (b) the plays with $a_n = 0$ for every n ; (c) the plays with $a_1 = a_0$; (d) the plays in which Player I plays 1 infinitely often.
- (2) Let (T, W) be an infinite game, and let σ and τ be strategies for Player I and Player II. Show that there is exactly one play consistent with both σ and τ , as used in the Remarks following Definition 2.2.
- (3) Let T be a tree, and let $W \subseteq W' \subseteq [T]$ be given. Show that a winning strategy for Player II in (T, W) is winning in (T, W') , and that a winning strategy for Player I in (T, W') is winning in (T, W) .
- (4) Let (T, W) be an infinite game, and suppose that there is an integer N such that whether a play lies in W depends only on its first N moves. Show that the game is determined. (Hint: decide who wins from each position of length N , and work backwards to $\langle \rangle$.)
- (5) (*) Let $A = \{0, 1\}$, $T = A^{<\omega}$, and let W be the set of plays with $a_{2n+1} = a_0$ for every n . Show that Player II has a winning strategy, but that no strategy for her whose move depends only on the last move of Player I is winning.

3. THE BANACH–MAZUR GAME

In this section we define the Banach–Mazur game and show that a space is a Baire space exactly when the first player has no winning strategy (see Theorem 3.3 below).

Definition 3.1. Let X be a topological space, and let $\mathcal{O}^+(X)$ be the collection of all non-empty open subsets of X . The *Banach–Mazur game* on X , denoted by $\text{BM}(X)$, is the game (T, W) where:

- (1) The set of moves is $A = \mathcal{O}^+(X)$.
- (2) The tree T consists of all finite sequences $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \dots$ of sets in $\mathcal{O}^+(X)$.
- (3) The payoff set is $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^{\infty} U_n \neq \emptyset\}$.

Thus Player I plays the sets U_n , Player II plays the sets V_n , and Player II wins if the sets played have a point in common.

Example 3.2. (1) In $\text{BM}(\mathbb{Q})$, Player I has a winning strategy. Fix an enumeration $q_0, q_1, \dots$ of $\mathbb{Q}$, and let Player I play $U_0 = \mathbb{Q} \setminus \{q_0\}$ and $U_n = V_{n-1} \setminus \{q_n\}$ for $n \geq 1$. These sets are open and non-empty, and every q_n is missing from U_n , so $\bigcap_n U_n = \emptyset$ (see Figure 4).
 (2) In $\text{BM}(\mathbb{R})$, Player II has a winning strategy: in answer to U_n , she plays a bounded open interval V_n with $\overline{V_n} \subseteq U_n$. The compact sets $\overline{V_n}$ are nested, so these sets have a common point, and that point lies in every U_n .

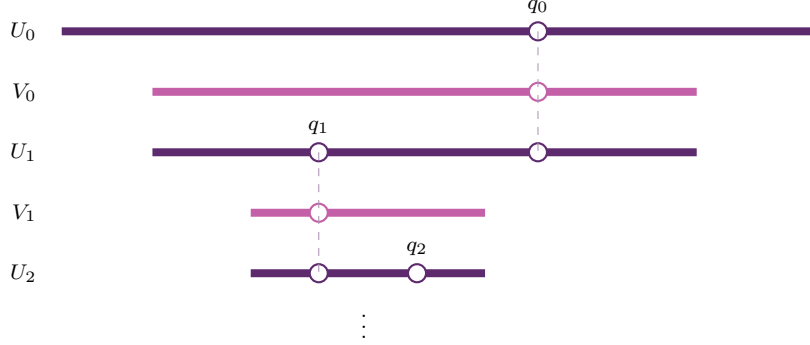


FIGURE 4. Player I wins $\text{BM}(\mathbb{Q})$ (see Example 3.2(1)). The moves of Player I are drawn in plum and those of Player II in pink, as intervals of rationals. At step n Player I removes the rational q_n (a hole), and a hole, once made, persists in every later set. Every rational is removed at some step, so the sets have no point in common.

Theorem 3.3. Let X be a topological space.

- (1) Player I has a winning strategy in $\text{BM}(X)$ if and only if X contains a non-empty open set of the first category.
- (2) The space X is a Baire space if and only if Player I has no winning strategy in $\text{BM}(X)$.

Proof. By Theorem 1.3, part (2) is an immediate consequence of part (1).

Suppose first that $G \subseteq X$ is open, non-empty, and of the first category. Then $G = \bigcup_{n=0}^{\infty} E_n$, where each E_n is nowhere dense. Player I plays $U_0 = G$. If Player II responds with $V_0 \subseteq U_0$, the set $V_0 \setminus \overline{E_0}$ is open and non-empty. Player I may therefore choose a non-empty open set $U_1 \subseteq V_0 \setminus \overline{E_0}$. In general, given V_{n-1} , Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \quad U_n \neq \emptyset.$$

This defines a strategy σ for Player I. For any play consistent with σ , we have $U_n \cap E_{n-1} = \emptyset$ for all $n \geq 1$, so that

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus σ is a winning strategy for Player I.

Conversely, suppose Player I has a winning strategy σ , and put $U_0 = \sigma(\langle \rangle)$. We show that U_0 is of the first category. For a position p of odd length $2n+1$ consistent with σ , we shall write U^p for the last entry of p , the set U_n played by Player I.

We build, for each $n \geq 0$, a family $\mathcal{P}_n$ of positions of length $2n+1$ consistent with σ , such that:

- (a) the sets U^p , $p \in \mathcal{P}_n$, are pairwise disjoint;
- (b) their union $D_n = \bigcup_{p \in \mathcal{P}_n} U^p$ is dense in U_0 ;
- (c) every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$.

Let $\mathcal{P}_0 = \{(U_0)\}$. Given $\mathcal{P}_n$, fix $q \in \mathcal{P}_n$. Each non-empty open $V \subseteq U^q$ is a legal move for Player II at q , and leads to the position $q \frown V \frown \sigma(q \frown V)$ of length $2n+3$. By Zorn's lemma, choose a family $\mathcal{P}(q)$ of such positions, maximal with respect to the property that their last entries are pairwise disjoint. The union of these last entries is dense in U^q : otherwise some non-empty open $O \subseteq U^q$ misses all of them, and then the position $q \frown O \frown \sigma(q \frown O)$, whose last entry lies in O , could be added to $\mathcal{P}(q)$, contradicting maximality. Put $\mathcal{P}_{n+1} = \bigcup_{q \in \mathcal{P}_n} \mathcal{P}(q)$. Property (c) holds by construction. Property (a) holds because last entries of positions in the same $\mathcal{P}(q)$ are disjoint, and those in $\mathcal{P}(q)$ and $\mathcal{P}(q')$ with $q \neq q'$ lie in the disjoint sets U^q and $U^{q'}$. For (b), argue by induction on n , the case $n=0$ being trivial. A non-empty open $O \subseteq U_0$ meets D_n , hence $O \cap U^q \neq \emptyset$ for some $q \in \mathcal{P}_n$. The union of the last entries of the positions in $\mathcal{P}(q)$ is dense in U^q , so that union meets $O \cap U^q$; since that union is contained in D_{n+1} , the set O meets D_{n+1} . Figure 5 shows the first two levels.

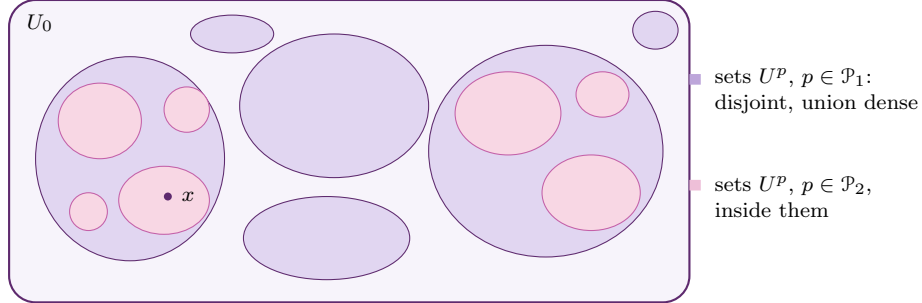


FIGURE 5. The proof of Theorem 3.3: the first two levels of the construction. Inside U_0 , the answers of σ at the first level form a maximal family of pairwise disjoint open sets, whose union D_1 is dense in U_0 ; inside each of them, the second level does the same. A point x lying in every D_n picks out exactly one set at each level, and these sets are the moves of Player I in a single play consistent with σ .

Each D_n is open and dense in U_0 , so $U_0 \setminus D_n$ is nowhere dense. Suppose that U_0 were of the second category. Then U_0 is not contained in the first-category set $\bigcup_n (U_0 \setminus D_n)$, so there is a point $x \in \bigcap_n D_n$. By (a), for each n there is exactly one $p_n \in \mathcal{P}_n$ with $x \in U^{p_n}$. By (c), the position p_{n+1} extends some $q \in \mathcal{P}_n$, and $x \in U^{p_{n+1}} \subseteq U^q$, so $q = p_n$ by (a). Thus the

positions $p_0 \subseteq p_1 \subseteq \dots$ fit together into a single play consistent with σ , and $x \in \bigcap_n U^{p_n}$ lies in every set played by Player I. This contradicts the assumption that σ is winning. Hence U_0 is a non-empty open set of the first category, and the proof is complete. ☺



Remark 3.4. Part (2) does not say that Player II has a winning strategy in $\text{BM}(X)$ when X is a Baire space; in general neither player need have one (see [Tel87]).

In Mazur's original game, and in most applications in Part II, the players aim at a given set. Let X be a topological space, and let $S \subseteq X$ be given. The game $\text{BM}(X, S)$ has the same moves as $\text{BM}(X)$, but Player II wins the play $(U_0, V_0, U_1, \dots)$ if

$$\bigcap_{n=0}^{\infty} U_n \subseteq S,$$

that is, if no point outside S survives. The sets $U_0 \supseteq V_0 \supseteq U_1 \supseteq \dots$ are nested, so $\bigcap_n U_n = \bigcap_n V_n$. Unlike in $\text{BM}(X)$, Player II also wins when the intersection is empty; Player I wins only if he produces a point outside S .

Theorem 3.5 (Banach; Oxtoby). *Let X be a topological space, and let $S \subseteq X$ be given.*

- (1) *Player II has a winning strategy in $\text{BM}(X, S)$ if and only if $X \setminus S$ is of the first category in X .*
- (2) *Suppose that X is a complete metric space. Then Player I has a winning strategy in $\text{BM}(X, S)$ if and only if there is a non-empty open set U such that $U \cap S$ is of the first category in X .*

Proof. (1) Suppose first that $X \setminus S = \bigcup_n E_n$ with each E_n nowhere dense. In answer to U_n , Player II plays a non-empty open set $V_n \subseteq U_n \setminus \overline{E_n}$. A point of $\bigcap_n V_n$ lies outside every E_n , hence in S , so this strategy is winning.

Conversely, let τ be a winning strategy for Player II. We repeat the construction in the proof of Theorem 3.3, now with the moves of Player II. For a position p of even length consistent with τ , write V^p for its last entry. As in that proof, Zorn's lemma gives families $\mathcal{P}_n$ of positions of length $2n+2$ consistent with τ , such that the sets V^p , $p \in \mathcal{P}_n$, are pairwise disjoint, their union D_n is dense in X , and every $p \in \mathcal{P}_{n+1}$ extends some $q \in \mathcal{P}_n$. At a position $q \in \mathcal{P}_n$, every non-empty open $U \subseteq V^q$ is a legal move for Player I, and the answer of τ lies in U , so the maximality argument goes through unchanged. (For $\mathcal{P}_0$, take a maximal family of positions $(U, \tau(U))$ whose last entries are pairwise disjoint; for every non-empty open O , the set $\tau(O)$ lies in O , so the union of their last entries is dense.) Let $x \in \bigcap_n D_n$ be given. Exactly as before, the positions $p_n \in \mathcal{P}_n$ with $x \in V^{p_n}$ fit together into a single play consistent with τ , and x lies in the intersection of this play. Since τ is winning, $x \in S$. Thus $\bigcap_n D_n \subseteq S$, so $X \setminus S \subseteq \bigcup_n (X \setminus D_n)$, and each $X \setminus D_n$ is closed with empty interior, hence nowhere dense. Therefore $X \setminus S$ is of the first category.

(2) Let d be a complete metric on X . Suppose first that $U \cap S = \bigcup_n E_n$ with each E_n nowhere dense. Player I plays $U_0 = U$ and, in answer to V_{n-1} , a non-empty open set U_n with $\overline{U_n} \subseteq V_{n-1} \setminus \overline{E_{n-1}}$ and $\text{diam } U_n < 2^{-n}$. The closed sets $\overline{U_n}$ are nested and their diameters tend to 0, so by completeness they have a point x in common. Then $x \in U_n$ for every n , so $x \in U$, and x lies in no E_n ; hence $x \notin S$, and Player I has won.

Conversely, let σ be a winning strategy for Player I, and put $U_0 = \sigma(\langle \rangle)$. We show that $U_0 \cap S$ is of the first category. Repeat the construction in the proof of Theorem 3.3, with one change: at level n , the families $\mathcal{P}(q)$ use only moves V of Player II with $\text{diam } V < 2^{-n}$.

Every non-empty open set contains such a V , so the unions D_n are still dense in U_0 . For $x \in \bigcap_n D_n$ we obtain, as before, a play consistent with σ whose intersection contains x . The diameters of the moves tend to 0, so this intersection is exactly $\{x\}$. Since σ is winning, the intersection contains a point outside S ; hence $x \notin S$. Thus $U_0 \cap \bigcap_n D_n \subseteq U_0 \setminus S$, and $U_0 \cap S$ is contained in the first-category set $\bigcup_n (U_0 \setminus D_n)$. $\odot$



Remark 3.6. Taking $X = \mathbb{R}$ and S the set of irrational numbers, part (1) shows that Player II wins $\text{BM}(\mathbb{R}, S)$: she removes the rationals one at a time, just as Player I did in Example 3.2(1). Part (1) holds in every topological space. In part (2), only the “if” direction needs completeness, since Player I must produce a point: in $X = \mathbb{Q}$ with $S = \emptyset$, the set $U \cap S$ is empty for every U , yet Player II wins by part (1), because $\mathbb{Q}$ is of the first category in itself.

Exercises.

- (1) Let C be the Cantor ternary set. For each of the sets S below, decide which player has a winning strategy in $\text{BM}(\mathbb{R}, S)$, and describe one explicitly: (a) the irrational numbers; (b) the rational numbers; (c) the set C ; (d) the set $\mathbb{R} \setminus C$. Check your answers against Theorem 3.5.
- (2) Let X be a non-empty compact Hausdorff space. Show that Player II has a winning strategy in $\text{BM}(X)$. (Hint: such a space is regular, so every point of a non-empty open set U has an open neighborhood whose closure lies in U .)
- (3) Let X be a topological space, and let $S_0, S_1, \dots \subseteq X$ be given. Suppose that Player II has a winning strategy τ_k in $\text{BM}(X, S_k)$ for every k . Show directly, by letting her use τ_k in the rounds of an infinite set R_k of rounds, with the sets R_k pairwise disjoint, that she has a winning strategy in $\text{BM}(X, \bigcap_k S_k)$. Then give a second proof using Theorem 3.5.
- (4) Let X be a Baire space, and let $U \subseteq X$ be a non-empty open set. Use Theorem 3.3 to show that U is a Baire space, by turning a winning strategy for Player I in $\text{BM}(U)$ into one in $\text{BM}(X)$.
- (5) (*) Let X be a complete metric space, and let $S \subseteq X$ have the Baire property. Show that $\text{BM}(X, S)$ is determined. (Hint: if S is of the first category in no non-empty open set, show that the open set O with $S \triangle O$ of the first category is dense.)

4. THE STRONG CHOQUET GAME

In this section we add points to the moves of Player I, and show that the resulting game characterizes complete metrizable spaces among metrizable spaces (see Theorem 4.4 below). Here we prove the easy half of the theorem and derive from it the Baire category theorem (see Corollary 4.5 below); the hard half occupies Section 5.

Definition 4.1. Let X be a topological space. The *strong Choquet game* on X , denoted by $\text{Ch}(X)$, is played as follows. Player I chooses a pair (x_0, U_0) with $U_0 \in \mathcal{O}^+(X)$ and $x_0 \in U_0$. Player II responds with $V_0 \in \mathcal{O}^+(X)$ such that $x_0 \in V_0 \subseteq U_0$. At step $n \geq 1$, Player I chooses (x_n, U_n) such that $x_n \in U_n \subseteq V_{n-1}$, and Player II responds with $V_n \in \mathcal{O}^+(X)$ such that $x_n \in V_n \subseteq U_n$. Player II wins the play if $\bigcap_{n=0}^{\infty} U_n \neq \emptyset$.

In the format of Definition 2.2, Player I chooses from $X \times \mathcal{O}^+(X)$ and Player II from $\mathcal{O}^+(X)$. If the points x_n are omitted, what remains is exactly the Banach–Mazur game $\text{BM}(X)$, which in this context is also called the *Choquet game*. Following Choquet, who called the two players β (the first) and α (the second), the space X is called α -favorable if Player II has a winning

strategy in $\text{BM}(X)$, and *strongly α -favorable* if Player II has a winning strategy in $\text{Ch}(X)$; Kechris [Kec95] calls such spaces *Choquet* and *strong Choquet* spaces.

Lemma 4.2. *If X is strongly α -favorable, then X is α -favorable.*

Proof. Let τ be a winning strategy for Player II in $\text{Ch}(X)$. In $\text{BM}(X)$, when Player I plays U_n , Player II picks any point $x_n \in U_n$ and answers with the set that τ gives in reply to (x_n, U_n) . The moves (x_n, U_n) are legal in $\text{Ch}(X)$, because $x_n \in U_n \subseteq V_{n-1}$, so the sets U_n played are those of a play of $\text{Ch}(X)$ consistent with τ , and their intersection is non-empty. ☺

Example 4.3. Player I wins $\text{Ch}(\mathbb{Q})$: he follows the strategy of Example 3.2(1), adding any point of each set he plays.

Theorem 4.4 (G. Choquet). *Let X be a metrizable space. Then X is completely metrizable if and only if X is strongly α -favorable.*

Proof. Suppose first that X is completely metrizable, and fix a compatible complete metric d on X . We define a strategy τ for Player II. When Player I plays (x_n, U_n) , Player II chooses r_n with $0 < r_n < 2^{-n}$ such that the closed ball $B[x_n, r_n] = \{y \in X : d(x_n, y) \leq r_n\}$ is contained in U_n , and plays $V_n = B(x_n, r_n)$.

Consider a play consistent with τ . For $m > n$ we have $x_m \in U_m \subseteq V_n = B(x_n, r_n)$, so $d(x_m, x_n) < 2^{-n}$. Thus (x_n) is a Cauchy sequence, converging to some $x \in X$. For each n , the points x_m with $m > n$ lie in $B[x_{n+1}, r_{n+1}]$ (since $x_m \in V_{n+1}$ for $m > n + 1$), so $x \in B[x_{n+1}, r_{n+1}] \subseteq U_{n+1} \subseteq U_n$. Hence $x \in \bigcap_n U_n$, and τ is a winning strategy for Player II (see Figure 6).

The converse, that a strongly α -favorable metrizable space is completely metrizable, is proved in Section 5. ☺

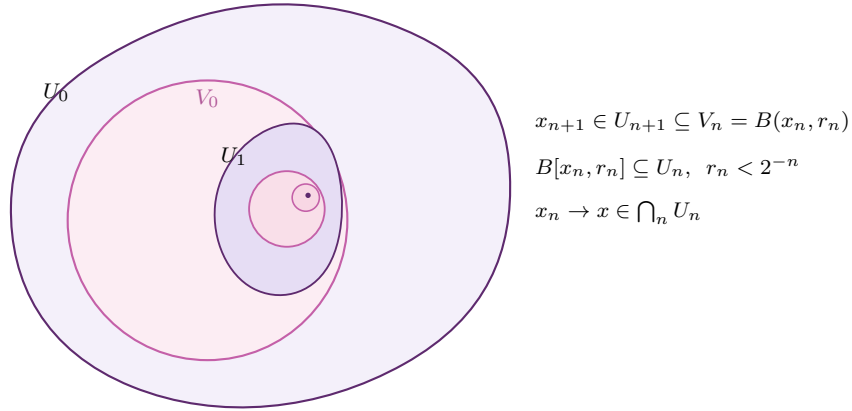


FIGURE 6. The easy half of Theorem 4.4. Player II answers each point x_n with a ball about it whose closure lies inside the set just played and whose radius is less than 2^{-n} . The points x_n form a Cauchy sequence, and its limit lies in every closed ball, hence in every U_n .

Corollary 4.5 (Baire category theorem). *Every completely metrizable space is a Baire space.*

Proof. Let X be completely metrizable. By Theorem 4.4 and Lemma 4.2, Player II has a winning strategy in $\text{BM}(X)$. Then Player I has none, and X is a Baire space by Theorem 3.3. ☺



Remark 4.6. The points x_n cannot be dropped: an α -favorable metrizable space need not be completely metrizable. Let $X = \{(a, b) \in \mathbb{R}^2 : b > 0\} \cup (\mathbb{Q} \times \{0\})$. The open upper half plane H is dense in X , and Player II wins $\text{BM}(X)$ by always answering with an open disk whose closure lies in $U_n \cap H$ and whose radius is less than 2^{-n} . But X is not completely metrizable: a closed subspace of a completely metrizable space is completely metrizable (a Cauchy sequence in the subspace converges in the space, and its limit lies in the subspace), and hence a Baire space by Corollary 4.5, whereas the closed subspace $\mathbb{Q} \times \{0\}$ of X is not a Baire space. Hence, by the hard half of Theorem 4.4 (proved in Section 5), Player II has no winning strategy in $\text{Ch}(X)$. This can also be seen directly, without Section 5: in $\text{Ch}(X)$ Player I can choose his points on the axis, and Player II must answer with sets containing them; Figure 7 shows how Player I then wins.

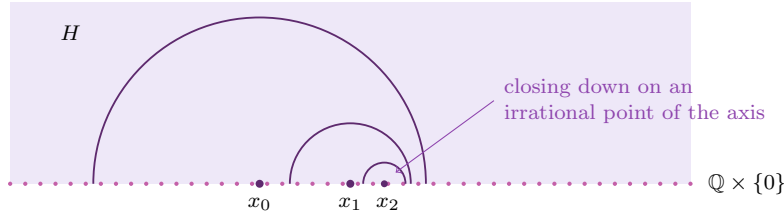


FIGURE 7. Player I wins $\text{Ch}(X)$ for the space X of Remark 4.6. He plays rational points of the axis, with small balls of X about them, inside the previous answer of Player II, and avoids one more rational each time. The balls close down on a point of the axis that is not rational, hence not in X .

Exercises.

- (1) Let X be a non-empty compact Hausdorff space. Show that Player II has a winning strategy in $\text{Ch}(X)$, so that X is strongly α -favorable.
- (2) Let X be a non-empty completely metrizable space with no isolated points. Show that X is uncountable. (Hint: each singleton is closed with empty interior; use Corollary 4.5.)
- (3) The interval $(0, 1)$ is not complete in the usual metric. Show directly that Player II has a winning strategy in $\text{Ch}((0, 1))$. (Hint: imitate the proof of Theorem 4.4, choosing $r_n < 2^{-n}$ so that the closed interval $[x_n - r_n, x_n + r_n]$ lies in U_n .)
- (4) Let X be the space of Remark 4.6. Write out in full the winning strategy for Player I in $\text{Ch}(X)$ indicated in Figure 7, and check that it is winning. (Hint: enumerate $\mathbb{Q} \times \{0\}$ as $p_0, p_1, \dots$, and at step n make sure that p_n lies outside a closed disk of radius less than 2^{-n} that contains U_n and lies in the previous such disk.)
- (5) (*) Let $P = \mathbb{R} \setminus \mathbb{Q}$ be the space of irrational numbers, with the usual metric, and enumerate $\mathbb{Q}$ as $q_0, q_1, \dots$. Show directly that Player II has a winning strategy in $\text{Ch}(P)$. (Hint: answer (x_n, U_n) with $V_n = (x_n - r_n, x_n + r_n) \cap P$, where the closed interval $[x_n - r_n, x_n + r_n]$ misses q_n and its points in P lie in U_n .)

5. THE CONVERSE OF CHOQUET'S THEOREM

In this section we prove the harder half of Theorem 4.4: if X is metrizable and Player II has a winning strategy in $\text{Ch}(X)$, then X is completely metrizable. The proof is long, so we first explain where a complete metric can come from and what the strategy has to do with it.

Where to look. Fix a compatible metric d on X . Every metric space is a dense subspace of a complete metric space, its *completion* $\tilde{X}$. If X is complete, then $\tilde{X} = X$; in general the completion contains new points, and the question is how X sits inside it. Recall that a countable intersection of open sets is called a G_δ set. Two subspaces of $\mathbb{R}$ show the difference. The irrationals form the set $\bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$, a countable intersection of open sets, and they are completely metrizable (for instance, by continued fractions they are homeomorphic to the space of sequences of positive integers). The rationals are not a countable intersection of open subsets of $\mathbb{R}$: such an intersection would be a dense G_δ , and so would the irrationals, and the two disjoint dense G_δ sets would contradict the Baire category theorem. In Lemma 5.2 below we show that every G_δ subset of a complete metric space is completely metrizable. Our goal is therefore to find open sets $G_1, G_2, \dots$ of $\tilde{X}$ with $X = \bigcap_n G_n$.

Where the open sets come from. The strategy of Player II is a machine that answers points with small open sets. At the first stage, Player I may play any point x of X (with the open set X), and the strategy answers with an open set $V \ni x$. These sets cover X , and slightly enlarged into $\tilde{X}$, their union is an open set $G_1 \supseteq X$. At the second stage, Player I continues each of these positions with each point of the set just played, and so on. The open set G_n is, roughly, the union of the (enlarged) sets played by Player II at stage n , over all positions, so $X \subseteq G_n$ for every n . Suppose now that a point y of the completion lies in every G_n . If the sets containing y at the various stages all came from *one* play, then, since Player II wins, that play would have a common point, and, since we shall arrange the sets to shrink, y would be that point; in particular, y would belong to X .

The difficulty. At each stage, y may lie in sets coming from many different positions, and choosing one position at each stage does not, in general, produce a single play: the position chosen at stage $n+1$ must extend the one chosen at stage n . If at every stage only *finitely* many positions are involved, a compactness argument, König's lemma (see Lemma 5.4 below), makes the choices fit together. The finiteness is supplied by a theorem of A. H. Stone on metric spaces (see Lemma 5.3 below): every open cover can be refined to one in which each point has a neighborhood meeting only finitely many sets. Figure 8 shows how the two ingredients combine.

5.1. A preliminary lemma and three classical tools. Throughout this subsection, (X, d) is a metric space, and $\tilde{X}$ is a metric space, with metric $\tilde{d}$ extending d , in which X is dense. Closures and diameters of subsets of $\tilde{X}$ are taken in $\tilde{X}$. The preliminary lemma enlarges open sets of X to open sets of $\tilde{X}$ without making them bigger in diameter (see Figure 9).

Lemma 5.1. *For an open set $V \subseteq X$, put $\tilde{V} = \tilde{X} \setminus \tilde{X} \setminus \tilde{V}$. Then the following hold:*

- (1) *the set $\tilde{V}$ is open in $\tilde{X}$, and $\tilde{V} \cap X = V$;*
- (2) *if $V \subseteq V'$, then $\tilde{V} \subseteq \tilde{V}'$;*
- (3) *the diameters satisfy $\text{diam } \tilde{V} = \text{diam } V$.*

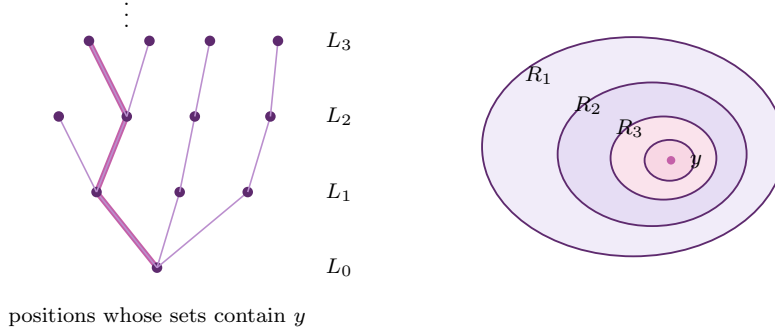


FIGURE 8. The end of the proof of Theorem 4.4. Left: for a point y of the completion lying in every G_n , the set L_n of positions of stage n whose sets contain y is finite and non-empty, and each member of L_{n+1} extends a member of L_n . König's lemma finds a branch (pink), that is, a single play. Right: along that play the sets R_n containing y shrink to a point; since Player II wins, the play has a common point in X , which must be y .

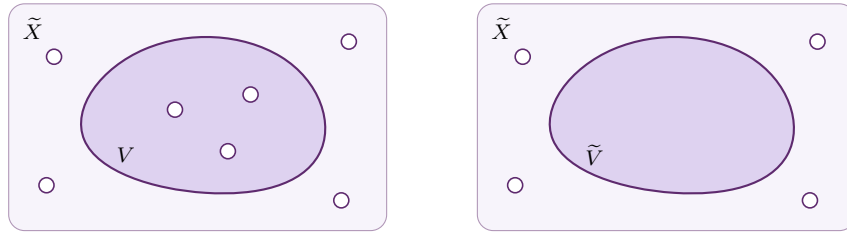


FIGURE 9. Lemma 5.1. The white dots are points of $\tilde{X}$ that are not in X . Left: an open set V of X , which misses the white dots inside it. Right: the open set $\tilde{V}$ of $\tilde{X}$, which also contains them. The two sets have the same points of X and the same diameter.

Proof. (1) The set $\tilde{V}$ is the complement of a closed set. If $x \in V$, then, since V is open in X , some ball $B(x, \varepsilon)$ of X lies in V , so the ball of radius ε about x in $\tilde{X}$ misses $X \setminus V$, and $x \in \tilde{V}$. If $x \in X \setminus V$, then $x \in \overline{X \setminus V}$, and $x \notin \tilde{V}$.

(2) If $V \subseteq V'$, then $X \setminus V' \subseteq X \setminus V$, and the complements of the closures reverse the inclusion.

(3) Let $y \in \tilde{V}$ and $\varepsilon > 0$ be given. The set of points of $\tilde{V}$ at distance less than ε from y is open and non-empty, so it meets the dense set X , at a point of $\tilde{V} \cap X = V$. Thus $\tilde{V} \subseteq \overline{V}$, and $\text{diam } \tilde{V} \leq \text{diam } \overline{V} = \text{diam } V$. The reverse inequality holds because $V \subseteq \tilde{V}$. ☺

The first tool is the theorem of Alexandrov announced above; it reduces our problem to finding a G_δ set.

Lemma 5.2 (P. S. Alexandrov). *Every G_δ subset of a complete metric space is completely metrizable.*

Proof. Let (Y, ρ_0) be a complete metric space, and let $G = \bigcap_{n \geq 1} G_n$ with each G_n open in Y . We may assume that each closed set $F_n = Y \setminus G_n$ is non-empty, discarding the indices with $G_n = Y$ (if no index remains, then $G = Y$ and there is nothing to prove). The idea is to

add to ρ_0 terms that blow up near the sets F_n , so that a Cauchy sequence cannot approach F_n . For $x \in G$ put $f_n(x) = 1/\rho_0(x, F_n)$, a continuous function on G , and for $x, y \in G$ put

$$\rho(x, y) = \rho_0(x, y) + \sum_{n \geq 1} 2^{-n} \min\{1, |f_n(x) - f_n(y)|\}.$$

Each term satisfies the triangle inequality, so ρ is a metric on G .

The metric ρ is compatible with the topology of G . Indeed, $\rho \geq \rho_0$, so ρ -convergence implies ρ_0 -convergence. Conversely, if $x_k \rightarrow x$ in G with respect to ρ_0 , then each term of the series tends to 0 by continuity of f_n , the series is dominated by $\sum_n 2^{-n}$, and hence $\rho(x_k, x) \rightarrow 0$.

The metric ρ is complete. Let (x_k) be ρ -Cauchy. Then (x_k) is ρ_0 -Cauchy, so it converges to some $y \in Y$. For each n , the sequence $(f_n(x_k))_k$ is Cauchy, hence bounded, say by c_n , so $\rho_0(x_k, F_n) \geq 1/c_n$ for every k , and therefore $\rho_0(y, F_n) \geq 1/c_n > 0$. Thus $y \in G_n$ for every n , that is, $y \in G$, and $\rho(x_k, y) \rightarrow 0$ by the compatibility just proved. ☺

The second tool supplies the finiteness. A family $\mathcal{V}$ of subsets of a space Y , indexed by a set I , is *locally finite* if every point of Y has a neighborhood that meets V_i for only finitely many indices $i \in I$. A family $\mathcal{V}$ *refines* a family $\mathcal{U}$ if every member of $\mathcal{V}$ is contained in some member of $\mathcal{U}$.

Lemma 5.3 (A. H. Stone). *Every open cover of a metric space has a locally finite open refinement that covers the space.*

The proof below is due to M. E. Rudin. The idea is to cover the space by small balls, assigning each ball to the first member of the cover that contains it generously, and to stop using small balls around points that are already covered by larger ones.

Proof. Let (Y, ρ) be a metric space, write $B(c, r)$ for open balls, and let $\{U_s : s \in \Lambda\}$ be an open cover of Y . By the well-ordering theorem we may assume that Λ is well ordered. For $s \in \Lambda$ and $n \geq 1$, let $V_{s,n}$ be the union of the balls $B(c, 2^{-n})$ over all centers $c \in Y$ such that:

- (i) the index s is the least one with $c \in U_s$;
- (ii) the center c lies in no set $V_{t,j}$ with $t \in \Lambda$ and $j < n$;
- (iii) the ball $B(c, 3 \cdot 2^{-n})$ is contained in U_s .

The definition is a recursion on n , because (ii) involves only the sets with smaller second index. Each $V_{s,n}$ is open, and $V_{s,n} \subseteq U_s$ by (iii).

The sets $V_{s,n}$ cover Y . Let $c \in Y$ be given, let s be the least index with $c \in U_s$, and choose n with $B(c, 3 \cdot 2^{-n}) \subseteq U_s$. Either $c \in V_{t,j}$ for some t and some $j < n$, or c satisfies (i)–(iii) and $c \in B(c, 2^{-n}) \subseteq V_{s,n}$.

The family is locally finite. Let $c \in Y$ be given, and choose s and n with $c \in V_{s,n}$. Choose $j \geq 1$ with $B(c, 2^{-j}) \subseteq V_{s,n}$. We show that the ball $B(c, 2^{-n-j})$ meets no set $V_{t,i}$ with $i \geq n+j$, and at most one set $V_{t,i}$ for each $i < n+j$; then it meets only finitely many sets of the family.

Let $i \geq n+j$, and let c' be a center used for $V_{t,i}$. Since $n < i$, condition (ii) gives $c' \notin V_{s,n}$, so $\rho(c, c') \geq 2^{-j}$. Every point of $B(c', 2^{-i})$ is therefore at distance at least $2^{-j} - 2^{-i} \geq 2^{-j} - 2^{-n-j} \geq 2^{-n-j}$ from c , and $B(c, 2^{-n-j})$ misses $V_{t,i}$.

Let $i < n+j$, and suppose that $p \in V_{t,i}$ and $q \in V_{t',i}$ with $t < t'$. Choose centers c_1 for $V_{t,i}$ and c_2 for $V_{t',i}$ with $\rho(p, c_1) < 2^{-i}$ and $\rho(q, c_2) < 2^{-i}$. By (iii), $B(c_1, 3 \cdot 2^{-i}) \subseteq U_t$, and by (i), $c_2 \notin U_t$ because $t < t'$. Hence $\rho(c_1, c_2) \geq 3 \cdot 2^{-i}$ and

$$\rho(p, q) \geq \rho(c_1, c_2) - \rho(p, c_1) - \rho(q, c_2) > 2^{-i} \geq 2 \cdot 2^{-n-j}.$$

Two points of $B(c, 2^{-n-j})$ are at distance less than $2 \cdot 2^{-n-j}$, so the ball cannot meet both $V_{t,i}$ and $V_{t',i}$. ☺

The third tool turns finitely many choices at each stage into one coherent choice.

Lemma 5.4 (D. König). *Let $L_0, L_1, \dots$ be non-empty finite sets, and for each n let $\pi_n: L_{n+1} \rightarrow L_n$ be a function. Then there are $\ell_n \in L_n$, for $n \geq 0$, with $\pi_n(\ell_{n+1}) = \ell_n$ for every n .*

Proof. For $m \geq n$ write $\pi_{n,m}: L_m \rightarrow L_n$ for the composite $\pi_n \circ \dots \circ \pi_{m-1}$ (the identity when $m = n$). Call $\ell \in L_n$ *good* if for every $m \geq n$ some element of L_m is mapped to ℓ by $\pi_{n,m}$. If $\ell \in L_n$ is the image of an element of L_m , then ℓ is also the image of an element of L_k for every k with $n \leq k \leq m$. Hence ℓ is good as soon as it is the image of elements of L_m for infinitely many m .

Some element of L_0 is good: every L_m is non-empty, so each m supplies an element of L_0 in the image of $\pi_{0,m}$, and since L_0 is finite, one element is supplied by infinitely many m . If $\ell \in L_n$ is good, then some element of the finite set $\pi_n^{-1}(\ell)$ is good: for each $m > n$, an element of L_m mapped to ℓ passes through some element of $\pi_n^{-1}(\ell)$, and one element of $\pi_n^{-1}(\ell)$ serves infinitely many m . Choosing recursively a good ℓ_0 and, for each n , a good $\ell_{n+1} \in \pi_n^{-1}(\ell_n)$ proves the lemma. ☺

5.2. The proof.

Proof of the converse of Theorem 4.4. Let X be metrizable, and let τ be a winning strategy for Player II in $\text{Ch}(X)$. Fix a compatible metric d on X and the completion $(\tilde{X}, \tilde{d})$ of (X, d) , in which X is dense, and use the notation $\tilde{V}$ of Lemma 5.1. We write $B(x, r)$ for the open ball of radius r about x in X . A *position of stage n* is a position of length $2n$ of $\text{Ch}(X)$ consistent with τ ; it ends with the move V_{n-1} of Player II when $n \geq 1$, and it is the empty position when $n = 0$.

Step 1: the families $\mathcal{S}_n$. By recursion on $n \geq 0$ we construct families $\mathcal{S}_n$ of pairs (R, p) , where R is a non-empty open subset of $\tilde{X}$ and p is a position of stage n . Write G_n for the union of the sets R with $(R, p) \in \mathcal{S}_n$. We require the following:

- (P1) if $n \geq 1$ and $(R, p) \in \mathcal{S}_n$, then $R \subseteq \tilde{V}$, where V is the last move of p , and $\text{diam } R \leq 2^{-n}$;
- (P2) the set G_n contains X ;
- (P3) the family of sets R , $(R, p) \in \mathcal{S}_n$, is locally finite in G_n ;
- (P4) each $(R, p) \in \mathcal{S}_{n+1}$ has a designated *parent* $(R', p') \in \mathcal{S}_n$ such that $R \subseteq R'$ and p extends p' by two moves: Player I plays $(x, R' \cap X)$ for some point x , and Player II answers as τ prescribes.

Put $\mathcal{S}_0 = \{(\tilde{X}, \langle \rangle)\}$; conditions (P2) and (P3) hold trivially.

Suppose that $\mathcal{S}_n$ has been constructed. Let $(R', p') \in \mathcal{S}_n$ and $x \in R' \cap X$ be given. The move $(x, R' \cap X)$ is legal for Player I at p' : for $n = 0$ it is a first move, and for $n \geq 1$ condition (P1) and Lemma 5.1(1) give $R' \cap X \subseteq \tilde{V} \cap X = V$, where V is the last move of p' . Let V_x be the answer of τ , so that $x \in V_x \subseteq R' \cap X$, let q_x be the resulting position of stage $n+1$ (the sets V_x , the positions q_x and the sets Q_x and O_x below depend also on the pair (R', p') , which we suppress from the notation), and put

$$Q_x = V_x \cap B(x, 2^{-n-2}) \quad \text{and} \quad O_x = R' \cap \tilde{Q}_x.$$

The set O_x is open in $\tilde{X}$, contains x , is contained in R' and in $\tilde{V}_x$ (see Lemma 5.1), and has diameter at most 2^{-n-1} by Lemma 5.1(3). By (P2), every point of X lies in some R' , so the sets O_x , over all pairs $(R', p') \in \mathcal{S}_n$ and all $x \in R' \cap X$, cover X . Their union H is open in

$\tilde{X}$, and the sets O_x form an open cover of the metric space H . By Lemma 5.3, this cover has a locally finite open refinement $\mathcal{R}$ that covers H ; we discard its empty members. For each $R \in \mathcal{R}$ choose a set O_x containing it, coming from the pair (R', p') and the point x , and put (R, q_x) into $\mathcal{S}_{n+1}$, with designated parent (R', p') .

Conditions (P1)–(P4) hold for $n+1$. Indeed, $R \subseteq O_x \subseteq \tilde{V}_x$, where V_x is the last move of q_x , and $\text{diam } R \leq \text{diam } O_x \leq 2^{-n-1}$; the union G_{n+1} equals H , which contains X ; the family $\mathcal{R}$ is locally finite in $H = G_{n+1}$; and $R \subseteq O_x \subseteq R'$, with q_x extending p' by the moves $(x, R' \cap X)$ and V_x .

Step 2: a G_δ set containing X . The set $G = \bigcap_{n \geq 1} G_n$ is a G_δ subset of $\tilde{X}$, and $X \subseteq G$ by (P2). It remains to be shown that $G \subseteq X$.

Step 3: every point of G is the common point of a play. Let $y \in G$ be given. For $n \geq 0$, let L_n be the set of pairs $(R, p) \in \mathcal{S}_n$ with $y \in R$. Each L_n is non-empty, because $y \in G_n$ (and L_0 is the single pair $(\tilde{X}, \langle \rangle)$). Each L_n is finite, by (P3): some neighborhood of y meets only finitely many of the sets R , and every R containing y is among them. By (P4), the parent of a member of L_{n+1} contains y , so it lies in L_n ; let $\pi_n: L_{n+1} \rightarrow L_n$ be the parent map. By Lemma 5.4, there are $(R_n, p_n) \in L_n$ such that (R_n, p_n) is the parent of (R_{n+1}, p_{n+1}) for every n .

By (P4), each position p_{n+1} extends p_n by a move $(x_n, R_n \cap X)$ of Player I and the answer of τ . Hence the positions p_n fit together into a single play of $\text{Ch}(X)$, consistent with τ , in which Player I plays the sets $R_n \cap X$. Since τ is winning, there is a point $z \in \bigcap_n (R_n \cap X)$. For every $n \geq 1$, the points y and z both lie in R_n , which has diameter at most 2^{-n} by (P1). Hence $\tilde{d}(y, z) = 0$, and $y = z \in X$ (see Figure 8).

Step 4: conclusion. By Steps 2 and 3, we have $X = G$, a G_δ subset of the complete metric space $\tilde{X}$; since $\tilde{d}$ extends d , the topology that X inherits from $\tilde{X}$ is its own. By Lemma 5.2, the space X is completely metrizable. ☺



Remark 5.5. The points x_n of Player I are what make the proof work: at every stage, Player I plays every point of X , and this is why each G_n contains all of X . Without the points the theorem fails (see Remark 4.6). For separable spaces, a different proof is given in [Kec95, Section 8].

5.3. Summary of Part I. The main results of this part fit into one chain of implications. For a metrizable space X ,

$$\begin{array}{c}
 X \text{ is completely metrizable} \\
 \Updownarrow \text{Theorem 4.4} \\
 \text{Player II has a winning strategy in } \text{Ch}(X) \\
 \Downarrow \text{Lemma 4.2} \\
 \text{Player II has a winning strategy in } \text{BM}(X) \\
 \Downarrow \text{Remarks 2.3(3)} \\
 \text{Player I has no winning strategy in } \text{BM}(X) \\
 \Updownarrow \text{Theorem 3.3} \\
 X \text{ is a Baire space} \\
 \Updownarrow \text{Theorem 1.3} \\
 \text{no non-empty open subset of } X \text{ is of the first category.}
 \end{array}$$

Reading the chain from top to bottom gives the Baire category theorem (see Corollary 4.5). The second implication cannot be reversed: the space of Remark 4.6 is metrizable and

Player II wins $\text{BM}(X)$, but X is not completely metrizable. The games with a target set measure the size of sets inside X : Player II wins $\text{BM}(X, S)$ exactly when S is comeager, and, in a complete metric space, Player I wins exactly when S is of the first category in some non-empty open set (see Theorem 3.5).

Exercises.

- (1) Let $X = \mathbb{R} \setminus \{0\}$ with the usual metric, so that $\tilde{X} = \mathbb{R}$. Compute $\tilde{V}$ for $V = (-1, 0) \cup (0, 1)$ and for $V = (0, 1)$, and check the three statements of Lemma 5.1 in both cases.
- (2) Show that the open cover $\{(-n, n) : n \geq 1\}$ of $\mathbb{R}$ is not locally finite, and find a locally finite open refinement of it that covers $\mathbb{R}$, as promised by Lemma 5.3.
- (3) Apply the proof of Lemma 5.2 to $G = (0, 1)$, a single open subset of $Y = \mathbb{R}$, and write down the metric ρ explicitly. Show that the sequence $(1/k)_{k \geq 2}$ is Cauchy for the usual metric but not for ρ .
- (4) Show that the conclusion of Lemma 5.4 can fail if the sets L_n are allowed to be infinite. (Hint: let L_0 have one element, let L_n consist of the pairs (i, n) with $i \geq n$ for $n \geq 1$, let π_0 be constant, and let $\pi_n(i, n+1) = (i, n)$ for $n \geq 1$.)
- (5) (*) Let (X, d) be a metric space whose topology is completely metrizable, possibly by a metric other than d . Use the proof of Theorem 4.4 to show that X is a G_δ subset of the completion of (X, d) . Deduce that $\mathbb{Q}$ is not completely metrizable.

6. NOTES AND HISTORICAL REMARKS

✂ 6.1. **Baire category.** Theorem 1.3 is an elementary reformulation of the definition and carries no attribution. This theorem should not be confused with the *Banach category theorem* [Ban30], which is deeper: in any topological space, the union of an arbitrary family of open sets of the first category is again of the first category. The title of Oxtoby's paper [Oxt57] refers to this theorem.

✂ 6.2. **The Banach–Mazur and strong Choquet games.** As told in the Introduction, the Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81] (see also Section 20). In Mazur's formulation two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets a given set E . Mazur observed that the first player wins if E is comeager in some interval and that the second player wins if E is of the first category, and conjectured the converses. Banach recorded on 4 August 1935 that the conjecture is true, but his proof was never published. The first published proof is Oxtoby's [Oxt57]. He showed that in an arbitrary topological space the second player has a winning strategy exactly when E is of the first category; the other converse, that the first player wins exactly when E is comeager in some non-empty open set, requires further hypotheses on the space, and holds for instance in complete metric spaces. This is Theorem 3.5, under the following dictionary:

- (1) Mazur's first player is our Player I, and Mazur's second player is our Player II.
- (2) Mazur's set E , at which the first player aims, is $X \setminus S$, where S is the set at which Player II aims in $\text{BM}(X, S)$.
- (3) "The intersection meets E " becomes "the intersection is not contained in S ", so Mazur's first player wins exactly when our Player I wins.

Whether the moves are open or closed intervals makes no difference to the proofs. The game $\text{BM}(X)$, in which Player II only needs a non-empty intersection, is a different game from

$\text{BM}(X, X)$, which Player II wins trivially; Theorem 3.3 is Oxtoby’s argument adapted to it. See [Tel87] for a survey.

The strong Choquet game goes back to Choquet’s work on Baire spaces [Cho58], and is expounded, together with Theorem 4.4, in his *Lectures on Analysis* [Cho69], where the players are called β and α . Kechris states Theorem 4.4 for separable metrizable spaces, where the theorem characterizes the Polish spaces [Kec95, Section 8]. The proof of Section 5, for arbitrary metrizable spaces, rests on three classical results. Lemma 5.2 is due to Alexandrov [Ale24]. Lemma 5.3 is A. H. Stone’s theorem that metric spaces are paracompact [Sto48]; the short proof given here is M. E. Rudin’s [Rud69]. Lemma 5.4 is Kőnig’s infinity lemma [Kőn27], first stated for graphs.

Part II. Applications of Banach–Mazur and Choquet games

Each section of this part takes the Banach–Mazur game of Section 3, or a game of Choquet type in which the players shrink sets (see Section 4), changes a rule or two, and uses the new game to solve a problem from outside topology. The idea from Part I that recurs most often is the one recorded in the box of the Introduction: a set is large in the sense of Baire category exactly when Player II can steer the play into it (see Theorem 3.5).

The sections can be read in any order, with three exceptions: Section 8 builds on the proof in Section 7; Section 12 builds on the Ramsey theory of Section 11; and Section 13 uses the Ramsey theory of Section 11. They also differ in what they ask of the reader. The sections on Diophantine approximation, computational complexity and ultrafilters need little beyond Part I and give complete proofs of their main points. The sections on homogeneous dynamics and Banach spaces assume some graduate background (matrix groups, functional analysis). The sections on topological groups and Ramsey theory sketch the main arguments rather than proving every step. Subsections marked *optional* may be skipped on a first reading. Table 1 lists the games of this part side by side.

7. APPLICATIONS TO DIOPHANTINE APPROXIMATION

The problem. Every real number can be approximated by fractions, but some numbers are much harder to approximate than others. For $t \in \mathbb{R}$ let $\|t\|$ be the distance from t to the nearest integer. A vector $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ is *badly approximable* if there is a constant $c = c(\mathbf{x}) > 0$ such that

$$\max_{1 \leq i \leq n} \|qx_i\| \geq cq^{-1/n} \quad \text{for every integer } q \geq 1.$$

For $n = 1$ this says $|x - p/q| \geq c/q^2$ for every fraction p/q . Let $\mathbf{Bad}(n)$ be the set of badly approximable vectors. Dirichlet’s theorem (1842) says that $\max_i \|qx_i\| < q^{-1/n}$ for infinitely many q , so badly approximable vectors are the ones for which Dirichlet’s theorem cannot be improved by more than a constant: they are the most stubborn vectors there are. How big is $\mathbf{Bad}(n)$? In particular, is the intersection of countably many translates of $\mathbf{Bad}(n)$ still big?

✂ **7.1. History of the problem and of the solution using topological games.** For $n = 1$ there is a classical description: a real number is badly approximable exactly when the partial quotients in its continued fraction expansion are bounded. This makes the golden ratio $(1+\sqrt{5})/2 = [1; 1, 1, 1, \dots]$, whose partial quotients are as small as they can be, the most badly approximable number of all. Badly approximable numbers are at once rare and common.

Section	Moves	Player II wins if	Main result
7. Diophantine approximation	closed balls whose radii shrink by the fixed factors α and β	the limit point lies in S	Theorem 7.1
8. Homogeneous dynamics	Schmidt's game on an expanding leaf of the flow	the orbit of the limit point is bounded	Theorem 8.1
9. Complexity	finite binary strings, each extending the last; Player II must compute her moves quickly	the language produced is not in $\mathcal{C}$	Theorem 9.1
10. Topological groups	open sets, as in $\text{BM}(X)$	the intersection is non-empty and sequences from D cluster	Theorem 10.1
11. Ramsey theory	Ellentuck basic sets; Player II adds one element at each move	the set produced lies in $\mathcal{A}$	Theorem 11.1
12. Banach spaces	arbitrary sets against relatively weak*-open sets; block subspaces against vectors	the sets shrink to a point in norm; the vectors lie in Σ	Theorems 12.1 and 12.2
13. Ultrafilters	finite binary strings; sets in $\mathcal{U}$ against points or finite sets	the sequence lies in Z ; the set produced lies in $\mathcal{U}$	Theorem 13.1

TABLE 1. The games of Part II. In every row Player I moves first and the payoff set belongs to Player II.

Khinchine proved in 1924 that $\mathbf{Bad}(n)$ has Lebesgue measure zero, so a number chosen at random is almost surely *not* badly approximable; Jarník, on the other hand, proved in 1928 that $\mathbf{Bad}(1)$ has Hausdorff dimension 1, the largest possible. The set $\mathbf{Bad}(1)$ is thus tiny for measure and huge for dimension.

Neither notion, however, says anything useful about intersections, since two sets of full dimension can be disjoint, and so a question as innocent as “is there a number x such that $x + \sqrt{2}, x + \sqrt{3}, x + \sqrt{5}, \dots$ are all badly approximable?” was out of reach.

The problem was solved by Wolfgang Schmidt in 1966 [Sch66]. He invented a new version of the Banach–Mazur game, now called Schmidt’s game, proved that the winning sets of this game have full Hausdorff dimension and survive countable intersections, and showed that $\mathbf{Bad}(1)$ is a winning set. In 1969 he did the same for $\mathbf{Bad}(n)$ for every n [Sch69]. The game had more to give: in 1983 Schmidt conjectured that for weights $i, j > 0$ with $i + j = 1$, the sets $\mathbf{Bad}(i, j)$ and $\mathbf{Bad}(j, i)$ of vectors in $\mathbb{R}^2$ that are badly approximable with weights (i, j) and (j, i) have a point in common, and he observed that a counterexample would prove the famous Littlewood conjecture. Badziahin, Pollington and Velani proved Schmidt’s conjecture in 2011 [BPV11], and in 2016 An showed that each $\mathbf{Bad}(i, j)$ is winning for Schmidt’s game [An16], which gives the conjecture, in a stronger form, from the intersection property.

Recognition. Schmidt received the Cole Prize in Number Theory in 1972, and he is one of the few number theorists invited to address the International Congress of Mathematicians three times. The solution of Schmidt’s conjecture by Badziahin, Pollington and Velani appeared in the *Annals of Mathematics*.

7.2. The game. Schmidt's game is the Banach–Mazur game of Section 3 with one new rule: the sizes of the moves are prescribed in advance. Let (X, d) be a complete metric space, let $S \subseteq X$ be given, and fix two numbers $\alpha, \beta \in (0, 1)$. The game is played with closed balls $B[x, r] = \{y \in X : d(x, y) \leq r\}$ of prescribed radii. Schmidt called the players Black and White: Black, who moves first, is our Player I, and White is our Player II. In Choquet's terminology our Player I is β and our Player II is α , so the letter α belongs to Player II in both; but the numbers α and β below are Schmidt's shrinking factors, and α -winning sets have nothing to do with the α -favorable spaces of Section 4.

- (1) Player I chooses a closed ball $U_0 = B[x_0, r_0]$ of any radius he likes.
- (2) Player II chooses a closed ball $V_0 \subseteq U_0$ of radius αr_0 .
- (3) Player I chooses a closed ball $U_1 \subseteq V_0$ of radius $\beta \alpha r_0$.
- (4) And so on: the play produces nested balls $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \dots$, where U_k has radius $r_k = (\alpha\beta)^k r_0$ and V_k has radius αr_k .

The radii go to 0 and X is complete, so the balls close down on a single point x_∞ (as in the proof of Theorem 4.4). Player II wins if $x_\infty \in S$. We call this game the (α, β) -game for S . The set S is α -winning if Player II has a winning strategy for every $\beta \in (0, 1)$. Notice the order of the quantifiers: Player II declares her handicap α first, promising to shrink by the factor α every time, and only then does Player I pick β , as aggressive as he likes. Her strategy is allowed to depend on β , but α is fixed once and for all. As the proof of Theorem 7.1(2) below shows, this order of quantifiers is what lets α -winning sets survive countable intersections. On the real line the balls are just closed intervals, and we speak of lengths instead of radii (see Figure 10).

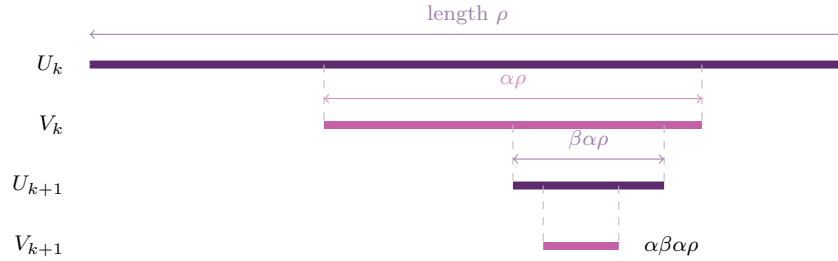


FIGURE 10. Two rounds of Schmidt's (α, β) -game on the line, drawn with $\alpha = \frac{1}{2}$ and $\beta = \frac{2}{5}$. Player I (plum) and Player II (pink) take turns choosing closed intervals, each inside the previous one, and the lengths are prescribed: Player II shrinks by the factor α , Player I by the factor β . Only the positions of the intervals are up to the players.

Because the sizes of the moves are prescribed, Player I can no longer shrink everything to a point in one move, and, as the proof of Theorem 7.1(1) below shows, this small change lets the game detect Hausdorff dimension.

7.3. The Main Result of this section. Schmidt's theorems say three things about α -winning sets: they are large, they survive countable intersections, and the badly approximable vectors form one of them.

Theorem 7.1 (Schmidt, 1966 and 1969). *Let $0 < \alpha < 1$ be given.*

- (1) *If $S \subseteq \mathbb{R}^n$ is α -winning, then $S \cap U$ has Hausdorff dimension n for every non-empty open set $U \subseteq \mathbb{R}^n$.*
- (2) *The intersection of countably many α -winning subsets of $\mathbb{R}^n$ is α -winning.*
- (3) *The set $\mathbf{Bad}(1)$ is $\frac{1}{2}$ -winning, and $\mathbf{Bad}(n)$ is α -winning for suitable $\alpha > 0$.*

Together, the three parts answer both questions of the problem. Since a translate of an α -winning set is α -winning (translate the strategy), part (2) shows that the set $\bigcap_k (\mathbf{Bad}(n) + \mathbf{t}_k)$ is α -winning for every sequence of vectors $\mathbf{t}_k$, and part (1) shows that this intersection has full dimension. In particular, there are (lots of) numbers x such that $x + \sqrt{p}$ is badly approximable for every prime p .

7.4. How the game solves the problem. We go through the three parts of Theorem 7.1, starting with the one that is pure number theory. For simplicity we prove that $\mathbf{Bad}(1)$ is α -winning for every $\alpha < \frac{1}{2}$; Schmidt's sharper bookkeeping also handles $\alpha = \frac{1}{2}$.

Badly approximable numbers: the plan. Fix $\alpha < \frac{1}{2}$ and $\beta \in (0, 1)$, and let Player I open with an interval U_0 of length ρ_0 . Put $R = 1/(\alpha\beta) > 1$, so that every round shrinks the intervals by the factor R : the interval U_k has length $\rho_k = \rho_0 R^{-k}$. We will find a constant $c > 0$ and a strategy for Player II such that the final point x satisfies

$$\left| x - \frac{p}{q} \right| \geq \frac{c}{q^2} \quad \text{for every fraction } p/q \text{ with } q \geq 1.$$

Call the interval $(\frac{p}{q} - \frac{c}{q^2}, \frac{p}{q} + \frac{c}{q^2})$ the *no-go zone* of p/q . Player II's job is to keep the final point out of every no-go zone. There are infinitely many zones, but she has infinitely many moves, and the whole proof is about giving each move a manageable amount of work.

Idea 1: fractions with small denominators keep their distance. If $p/q \neq p'/q'$, then

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{qq'},$$

because $pq' - p'q$ is a non-zero integer. Two fractions whose denominators are both small therefore cannot be close together, and this little inequality drives the whole argument.

Idea 2: sort the fractions into generations. Let $A = 1/(3\rho_0)$. Say that p/q belongs to *generation 0* if $q^2 < A$, and to *generation $k \geq 1$* if

$$AR^{k-1} \leq q^2 < AR^k.$$

Every fraction belongs to exactly one generation, and generation k is the one that Player II will deal with in round k . Grouping the denominators by size in this way keeps the fractions of a single generation far apart: by Idea 1, two different fractions of generation k are at distance more than $1/(AR^k) = 3\rho_0 R^{-k} = 3\rho_k$. If we fatten Player I's interval U_k by ρ_k on each side, we obtain an interval of length $3\rho_k$, which therefore contains *at most one* fraction of generation k . If there is one, call that fraction the *dangerous fraction* of round k .

Idea 3: the no-go zones are small. A fraction p/q of generation $k \geq 1$ has $q^2 \geq AR^{k-1}$, so the no-go zone of p/q has length

$$\frac{2c}{q^2} \leq \frac{2c}{AR^{k-1}} = 6cR\rho_k.$$

Choose c so small that $6cR \leq 1 - 2\alpha$ and $2c \leq (1 - 2\alpha)\rho_0$ (the second condition takes care of generation 0, whose zones have length at most $2c$). Then every zone of generation k has length at most $(1 - 2\alpha)\rho_k$, which is less than ρ_k . In particular, a zone of generation k that

reaches into U_k must belong to a fraction in the fattened interval, that is, to the dangerous fraction.

The strategy. In round k , Player II looks at Player I's interval U_k and at the no-go zone of the dangerous fraction of round k , if there is one. Removing the zone from U_k leaves at most two intervals, of total length at least $\rho_k - (1 - 2\alpha)\rho_k = 2\alpha\rho_k$, so one of them has length at least $\alpha\rho_k$. Player II plays her interval V_k inside that one, and that is the whole strategy.

Why the strategy wins. Take any fraction p/q , and let k be the generation of p/q . The final point x lies in $V_k \subseteq U_k$. If the zone of p/q reaches into U_k , then p/q is the dangerous fraction of round k , and V_k avoids the zone of p/q , so x does too. If the zone does not reach into U_k , then $x \in U_k$ is outside the zone anyway. Either way $|x - p/q| \geq c/q^2$, so $x \in \mathbf{Bad}(1)$. The constant c depends on α , β and ρ_0 , which is fine, since the definition of $\mathbf{Bad}(1)$ allows each number a constant of its own.

The game has done for us exactly what was promised in the introduction. The condition “ $|x - p/q| \geq c/q^2$ for all fractions” involves infinitely many fractions at once; the game splits this condition into infinitely many small jobs, each of which (dodge one interval) is easy.

Higher dimensions. For $\mathbf{Bad}(n)$ the same plan works, with Idea 1 replaced by the *simplex lemma*: if $n + 1$ rational points in $\mathbb{R}^n$ with denominators $q_0, \dots, q_n$ do not lie on a common hyperplane, then the simplex they span has volume at least $1/(n! q_0 \cdots q_n)$. (The volume is $1/n!$ times a determinant whose entries are fractions with denominators q_i , and a non-zero determinant of that kind is at least $1/(q_0 \cdots q_n)$ in absolute value, exactly as in Idea 1.) Consequently, all the rational points of one generation that lie in a small ball sit on a single hyperplane, and in each round Player II has to dodge a thin slab around one hyperplane instead of one interval.

Full dimension. Suppose Player II has a winning strategy in the (α, β) -game for S . Inside Player II's ball V_k , of radius αr_k , Player I has room for roughly β^{-n} disjoint balls of radius $\beta \alpha r_k$, and he may play any of them. Let him try them all, while Player II keeps following her strategy. The result is a tree of plays: every ball has about β^{-n} children, and the radii shrink by the factor $\alpha\beta$ from one generation to the next. The endpoints of all these plays form a Cantor set, and every point of this Cantor set lies in S , because Player II won every one of those plays. A Cantor set built this way has Hausdorff dimension at least about

$$\frac{\log(\beta^{-n})}{\log(1/(\alpha\beta))} = \frac{n \log(1/\beta)}{\log(1/\alpha) + \log(1/\beta)},$$

and this tends to n as $\beta \rightarrow 0$. Since an α -winning set is winning for *every* β , letting $\beta \rightarrow 0$ shows that such a set has dimension n , and starting the construction inside an arbitrary open set gives the local version.

Countable intersections. Let $S_1, S_2, \dots$ be α -winning sets. Player II now shares her time among them: she splits the rounds into infinitely many infinite arithmetic progressions, and in the rounds of the k -th progression she plays her strategy for S_k , ignoring everything else. From the point of view of S_k , everything that happens between two consecutive rounds of the k -th progression (her moves for the other sets and Player I's moves) just looks like one very big shrinking move by Player I, so the game for S_k is an (α, β_k) -game with a much smaller β_k . Since S_k is α -winning, Player II wins that game too. The final point therefore lies in every S_k . Her time-sharing strategy is a fugue: each voice enters at a time of its own and follows a line of its own, and in the end all of them sound together. Measure and dimension have no such device, and this is the real reason games are useful here.

Exercises.

- (1) Let $0 < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α -winning. Show that S meets every non-empty open subset of $\mathbb{R}^n$; in particular, every α -winning set is dense.
- (2) Let $S \subseteq \mathbb{R}^n$ be α -winning, and let $\lambda \neq 0$ and $\mathbf{t} \in \mathbb{R}^n$ be given. Show that the set $\lambda S + \mathbf{t}$ is α -winning. (Hint: the map $\mathbf{x} \mapsto \lambda \mathbf{x} + \mathbf{t}$ carries $B[\mathbf{x}, r]$ onto $B[\lambda \mathbf{x} + \mathbf{t}, |\lambda|r]$; transport the strategy.)
- (3) Let $\alpha \in (0, 1)$ and $\beta \in (0, \frac{1}{2})$ be given, and let $S \subseteq \mathbb{R}$ be countable. Show, without using Theorem 7.1, that Player I has a winning strategy in the (α, β) -game for S . Conclude that $\mathbb{Q}$ is not α -winning for any α . (Hint: since $2\beta < 1$, a closed interval contains two disjoint closed subintervals of β times its length.)
- (4) Let $0 < \alpha < \frac{1}{2}$ be given. Show directly that the set of irrational numbers is α -winning, and more generally that the complement of every countable subset of $\mathbb{R}$ is α -winning.
- (5) (*) Let $0 < \alpha' < \alpha < 1$ be given, and let $S \subseteq \mathbb{R}^n$ be α -winning. Show that S is α' -winning. (Hint: given β' , put $\beta = \alpha' \beta' / \alpha$, so that $\alpha \beta = \alpha' \beta'$; in the (α', β') -game, Player II plays a ball of radius $\alpha' r_k$ inside the ball of radius αr_k prescribed by her strategy for the (α, β) -game.)

8. APPLICATIONS TO HOMOGENEOUS DYNAMICS

The problem. A *lattice* in $\mathbb{R}^m$ is a set of the form $g\mathbb{Z}^m$ with g an invertible matrix, and the lattice $g\mathbb{Z}^m$ is *unimodular* if $\det g = \pm 1$. The unimodular lattices form a space, which can be identified with $\mathrm{SL}(m, \mathbb{R})/\mathrm{SL}(m, \mathbb{Z})$. Flows on this space move lattices around by stretching some directions and squeezing others, for example

$$g_t = \mathrm{diag}(e^t, \dots, e^t, e^{-nt}), \quad t \geq 0,$$

acting on $\mathbb{R}^{n+1}$. For a flow of this kind almost every orbit is dense, so wanders all over the space, and the lattices whose orbits stay in a bounded region form a set of measure zero. How big is this set of measure zero?

The words “Lie group”, “lattice” and “horospherical subgroup” will appear, but the heart of the matter is a computation with 2×2 matrices acting on the plane, which we do by hand below; the Lie group language is just the coordinate system that lets the same computation run in higher dimensions. A reader of Section 7 already knows the main idea, since this section shows that the game played there is a game about lattices being stretched and squeezed.

A small glossary. The reader may keep the example $G = \mathrm{SL}(m, \mathbb{R})$, $\Gamma = \mathrm{SL}(m, \mathbb{Z})$ in mind throughout.

- A *Lie group* is a group that is also a smooth manifold, such as $\mathrm{SL}(m, \mathbb{R})$, the group of real $m \times m$ matrices of determinant 1.
- A *lattice* Γ in a Lie group G is a discrete subgroup such that the quotient G/Γ has finite volume. The word has two meanings here: a lattice in $\mathbb{R}^m$ is a grid of points, and a lattice in G is a subgroup. They are linked by the identification of the coset $g\mathrm{SL}(m, \mathbb{Z})$ with the grid $g\mathbb{Z}^m$, so that $\mathrm{SL}(m, \mathbb{R})/\mathrm{SL}(m, \mathbb{Z})$ is the space of unimodular grids.
- “Almost every” and “measure zero” refer to the natural measure on G/Γ inherited from G (its Haar measure), which the flow preserves.
- A set of points of G/Γ is *bounded* if its closure is compact; for grids this is made concrete by Mahler’s criterion in the history below.
- A flow g_t is *quasi-unipotent* if all the eigenvalues of the matrices g_t have absolute value 1, so that no direction is stretched exponentially.

Other terms (nilpotent, semisimple, mixing) appear only in the general statement of the main theorem, and the reader may skip them.

✂ **8.1. History of the problem and of the solution using topological games.** The space of lattices is not compact: one can squash a lattice to create a very short non-zero vector, and then squash further. “Going to infinity” in this space is usually called “going into the cusp”, and a lattice deep in the cusp is a very thin needle-like grid with a vector of length nearly zero. Mahler’s compactness criterion (1946) says that this is the only way to escape to infinity: a set of unimodular lattices is bounded exactly when the lengths of their shortest non-zero vectors are bounded away from zero.

The link with the previous section was found by Dani, who showed in 1985 [Dan85] that badly approximable vectors correspond exactly to bounded orbits of the flow above, and in 1986 used Schmidt’s games to show that bounded orbits are abundant for several families of flows [Dan86]. In his plenary address at the International Congress of Mathematicians in Kyoto in 1990 [Mar91], Margulis conjectured that this abundance is universal: for every one-parameter flow that is not quasi-unipotent (roughly, every flow that really stretches something), the set of points with bounded orbits is *thick*, that is, this set meets every non-empty open set in a set of full Hausdorff dimension.

The conjecture was proved by Kleinbock and Margulis in 1996 [KM96], by playing Schmidt’s game on the expanding leaves of the flow. Later, Kleinbock and Weiss [KW13] introduced *modified Schmidt games*, adapted to flows that stretch different directions at different rates, and showed in 2013 that sets of bounded orbits are winning for them, which adds the countable intersection property to thickness. Winning has also been established for badly approximable points on planar curves, by An, Beresnevich and Velani in 2018 [ABV18].

Recognition. Margulis received the Fields Medal in 1978, the Wolf Prize in 2005 and the Abel Prize in 2020. Dani received the Shanti Swarup Bhatnagar Prize in 1990 and was an invited speaker at the International Congress of Mathematicians in Zürich in 1994. The dynamical approach to Diophantine approximation that grew out of this work led Kleinbock and Margulis to their proof of the Baker–Sprindžuk conjectures, by non-divergence estimates rather than games, in the *Annals of Mathematics* in 1998 [KM98].

8.2. The game. The game is Schmidt’s game of Section 7, played not on the whole space but on an *expanding leaf* of the flow. The expanding horospherical subgroup H consists of the elements h with $g_{-t}hg_t \rightarrow e$ as $t \rightarrow \infty$; these are the directions that the flow stretches. Since distances in these directions grow exponentially, a tiny ball soon becomes a ball of unit size, and this is what lets Player II deal with the problem one scale at a time. Each orbit Hx is a copy of a nilpotent Lie group with a natural metric, and the players play Schmidt’s game with closed balls in Hx ; Player II wins if the final point hx has a bounded orbit. For the diagonal flow above, the subgroup H consists of the matrices $\begin{pmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{pmatrix}$, so the leaf through $\Lambda_0 = \mathbb{Z}^{n+1}$ is parametrized by $\mathbf{y} \in \mathbb{R}^n$, and the game on this leaf is literally Schmidt’s game on $\mathbb{R}^n$.

8.3. The Main Result of this section. The main result has two parts: a dictionary between numbers and orbits, due to Dani, and the theorem of Kleinbock and Margulis.

Theorem 8.1 (Dani, 1985; Kleinbock–Margulis, 1996).

- (1) Dani’s correspondence. *For $\mathbf{x} \in \mathbb{R}^n$ let $\Lambda_{\mathbf{x}} = \begin{pmatrix} I_n & \mathbf{x} \\ 0 & 1 \end{pmatrix} \mathbb{Z}^{n+1}$. Then $\mathbf{x} \in \mathbf{Bad}(n)$ if and only if the forward orbit $\{g_t \Lambda_{\mathbf{x}} : t \geq 0\}$ is bounded.*
- (2) Thickness of bounded orbits. *For every non-quasi-unipotent one-parameter subgroup of a connected semisimple Lie group G without compact factors, and every lattice Γ in G , the set of points of G/Γ with bounded orbits is thick.*

Part (1) says that Schmidt’s theorem of Section 7 is a theorem about one particular flow; part (2) says the same phenomenon happens for all flows that stretch.

8.4. How the game solves the problem. Let us see the whole mechanism in the simplest case $n = 1$, where everything can be computed by hand, and then explain what changes in general.

Lattices in the plane. For $n = 1$ we have $g_t = \text{diag}(e^t, e^{-t})$ and $u_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, so

$$\Lambda_x = u_x \mathbb{Z}^2 = \{(p + qx, q) : p, q \in \mathbb{Z}\}, \quad g_t \Lambda_x = \{(e^t(p + qx), e^{-t}q) : p, q \in \mathbb{Z}\}.$$

The flow stretches the first coordinate and squeezes the second, and the *product* of the two coordinates of a vector, namely $q(p + qx)$, does not change along the orbit. The whole argument rests on this invariant product (see Figure 11).

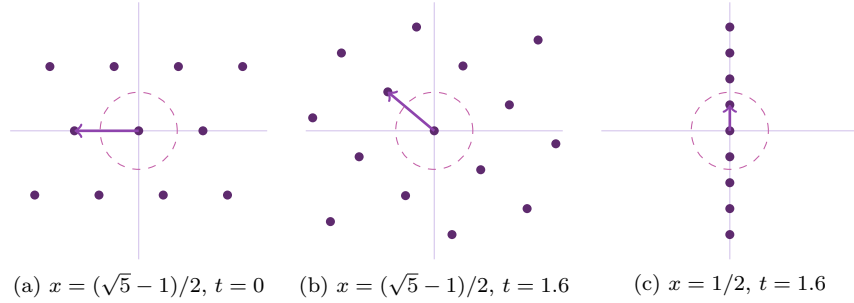


FIGURE 11. The grid $g_t \Lambda_x$ near the origin, with a shortest non-zero vector drawn as an arrow and a disk of radius 0.6 dashed. The flow stretches the grid horizontally by e^t and squeezes it vertically by e^{-t} . For the badly approximable number $x = (\sqrt{5} - 1)/2$, in (a) and (b), the grid is sheared but no non-zero point ever comes close to the origin, so the orbit stays bounded. For the rational number $x = 1/2$, in (c), the points $(0, 2qe^{-t})$ with $q \in \mathbb{Z}$ crowd towards the origin, and the orbit goes into the cusp.

Dani’s correspondence, by hand. Suppose first that $x \in \mathbf{Bad}(1)$, so $|p + qx| \geq c/|q|$ whenever $q \neq 0$. Take a non-zero vector $(e^t(p + qx), e^{-t}q)$ of $g_t \Lambda_x$. If $q \neq 0$, the product of the coordinates of this vector has absolute value $|q| |p + qx| \geq c$, so the two coordinates cannot both be smaller than $\sqrt{c}$, and the vector has length at least $\sqrt{c}$. If $q = 0$, the vector is $(e^t p, 0)$ with $p \neq 0$, of length at least $e^t \geq 1$. So every lattice on the orbit has no non-zero vector shorter than $\min(1, \sqrt{c})$, and by Mahler’s criterion the orbit is bounded.

Conversely, suppose that $x \notin \mathbf{Bad}(1)$. Then for every $\varepsilon > 0$ there are p and $q \geq 1$ with $|p + qx| < \varepsilon/q$. Choose the time t with $e^{-t}q = \sqrt{\varepsilon}$. At that time the vector coming from (p, q) has second coordinate $\sqrt{\varepsilon}$ and first coordinate $e^t |p + qx| < (q/\sqrt{\varepsilon})(\varepsilon/q) = \sqrt{\varepsilon}$, so this vector is shorter than $\sqrt{2\varepsilon}$. Since ε is arbitrary, the orbit dives deeper and deeper into the cusp and is unbounded.

So a good rational approximation p/q of x is *the same thing* as an excursion of the orbit into the cusp at time $t \approx \log q$.

The dictionary between scales and times. A one-line computation gives

$$g_t u_x g_{-t} = u_{e^{2t}x},$$

so the flow stretches the expanding leaf $\{u_x \Lambda : x \in \mathbb{R}\}$ by the factor e^{2t} . An interval of length ρ in the leaf is stretched to length about 1 at the time t with $e^{2t} \approx 1/\rho$. Now look again at the proof that **Bad**(1) is α -winning in Section 7. In round k , Player I's interval has length $\rho_k = \rho_0 R^{-k}$, and the fractions Player II worries about are those of generation k , with $q^2 \approx R^k$; by the computation above, these are exactly the fractions that send the orbit into the cusp at the time t with $e^{2t} \approx R^k \approx 1/\rho_k$. In other words, *round k of the game corresponds to the time at which the current interval has been stretched to unit size, and in that round Player II dodges the points whose orbits visit the cusp at that time.* Read in this dictionary, Schmidt's proof is a proof about orbits, and the formula $g_t u_x g_{-t} = u_{e^{2t}x}$ translates each step of one proof into the corresponding step of the other.

The general case. Kleinbock and Margulis play the same game for an arbitrary flow that stretches, with Player II working on an expanding leaf Hx . In each round she looks at Player I's ball, waits (in her head) until the flow has stretched that ball to unit size, and asks which points of the ball are deep in the cusp at that time. For the diagonal flow on the plane this bad set was a single short interval, the no-go zone of one fraction. In general there is no single dangerous fraction, and this is where the real work is: Kleinbock and Margulis show, using the mixing properties of the flow, that the bad set occupies only a small proportion of the ball, small enough that Player II can always find a ball of the prescribed size that avoids this region. The final point then never goes far into the cusp, so its orbit is bounded by Mahler's criterion, and the set of points with bounded orbits is winning on every expanding leaf, hence of full dimension there.

From leaves to the whole space. Roughly, near any point the space looks like a product of a piece of an expanding leaf with the directions that the flow does not stretch, and whether an orbit stays bounded depends only mildly on those other directions. So full dimension on every expanding leaf gives full dimension in the whole space, that is, thickness.

Exercises.

- (1) Let $x \in \mathbb{R}$ and $t \geq 0$ be given. Check by multiplying matrices that $g_t u_x g_{-t} = u_{e^{2t}x}$. Show that Λ_x is unimodular, and that the product of the two coordinates of a vector of $g_t \Lambda_x$ coming from (p, q) does not depend on t .
- (2) Let $x \in \mathbb{R}$ be given. Show that $\Lambda_{x+1} = \Lambda_x$. Explain why this is consistent with Theorem 8.1(1), by checking directly that $x \in \mathbf{Bad}(1)$ if and only if $x+1 \in \mathbf{Bad}(1)$.
- (3) Let $x = a/b$ be rational, with $b \geq 1$. Find a non-zero vector of $g_t \Lambda_x$ of length $b e^{-t}$, and deduce that the orbit $\{g_t \Lambda_x : t \geq 0\}$ is unbounded. Check that this agrees with Theorem 8.1(1), by showing that no rational number is badly approximable.
- (4) (*) Let $\varphi = (1+\sqrt{5})/2$ and $\varphi' = (1-\sqrt{5})/2$. Show that $(p+q\varphi)(p+q\varphi') = p^2 + pq - q^2$ for all integers p, q , and deduce that $|q| |p+q\varphi| \geq 1/(1+\sqrt{5})$ whenever $q \neq 0$. Conclude that, for every $t \geq 0$, every non-zero vector of $g_t \Lambda_\varphi$ has length at least $(1+\sqrt{5})^{-1/2}$.

9. APPLICATIONS TO COMPUTATIONAL COMPLEXITY

The problem. A *language* is a set L of finite binary strings, and a complexity class is a set of languages, such as P (decidable in polynomial time), the class $E = \text{DTIME}(2^{O(n)})$ (decidable in time $2^{O(n)}$ on inputs of length n) or $\text{ESPACE} = \text{DSpace}(2^{O(n)})$. We would like to say that “most” languages in E are hard, or that “most” languages in ESPACE need big circuits. Ordinary Baire category and Lebesgue measure on the space of all languages are useless for this: every complexity class is countable, hence meager and of measure zero, so that P and E look equally small. The problem is to find a notion of “small” that works *inside* a complexity class.

✂ 9.1. **History of the problem and of the solution using topological games.** The problem was solved by Kurt Mehlhorn in 1973 [Meh73], who introduced resource-bounded Baire category, in which the descriptions of meager sets must themselves be computable with limited resources. In 1990 Jack Lutz reformulated the theory in terms of the Banach–Mazur game played with resource-bounded strategies [Lut90]. He proved, among other results, that languages with high circuit complexity are typical in ESPACE in this sense, and in 1992 he introduced the companion notion of resource-bounded measure [Lut92]. These notions became standard tools for quantitative questions about complexity classes.

Recognition. Mehlhorn received the Leibniz Prize of the Deutsche Forschungsgemeinschaft in 1986.

9.2. **The game.** List all binary strings in the standard order, namely $\lambda, 0, 1, 00, 01, 10, 11, 000, \dots$ (shorter strings first, and strings of the same length in dictionary order). A language L is then the same thing as an infinite 0–1 sequence, the *characteristic sequence* of L , whose N -th bit says whether the N -th string belongs to L . A finite 0–1 string w is a partial description of a language, since w decides membership for the first $|w|$ strings. Note that the strings of length n occupy positions of size about 2^n in this list, so to decide all strings of length $< n$ we need a description of length about 2^n .

Let $\mathcal{C}$ be a class of languages.

- (1) The players alternately extend finite binary strings, $w_0 \sqsubset w_1 \sqsubset w_2 \sqsubset \dots$, with Player I playing w_{2k} and Player II playing w_{2k+1} .
- (2) The play produces the language $L = \bigcup_k w_k$.
- (3) Player II wins if $L \notin \mathcal{C}$.

This is the Banach–Mazur game on the Cantor space, and by the theorem of Banach and Oxtoby (see Theorem 3.5; its proof works just as well when the moves are restricted to basic open sets N_w) Player II has a winning strategy exactly when $\mathcal{C}$ is meager. The resource-bounded version adds one rule: *Player II is not allowed to think for too long*. Her strategy must compute each move in time polynomial in the length of the current string (or in space polynomial in that length, when we work in ESPACE). A class is *meager in E* if Player II has such a strategy with the time bound, and *meager in ESPACE* if she has one with the space bound. Player I has no restrictions at all.

9.3. **The Main Result of this section.** A notion of smallness is useful only if it does not declare everything small. The theory rests on the fact that resource-bounded category satisfies a version of the Baire category theorem, and this is what allows it to tell subclasses of a complexity class apart.

Theorem 9.1 (Mehlhorn, 1973; Lutz, 1990).

- (1) Baire category theorem. *The class E is not meager in itself in the resource-bounded sense.*
- (2) *The class P is meager in E in the resource-bounded sense.*
- (3) *The class SIZE($2^n/n$) of languages with circuits of size $2^n/n$ is meager in ESPACE in the resource-bounded sense.*

Part (3) says that big circuit complexity is typical in ESPACE. Lutz also proved the measure version: in resource-bounded measure, almost every language in ESPACE has circuit-size complexity above $2^n/n$ [Lut92]. Here category and measure agree, although, as on the real line, they can disagree in general.

9.4. How the game solves the problem. The game formulation makes the definition of category effective in the most natural way imaginable: one simply limits the resources of the winning player. The proofs then become constructions of efficient strategies, and they are pleasantly concrete.

The Baire category theorem: the class E is not meager in itself. Suppose that Player II has a polynomial-time strategy. We show that Player I can make the play produce a language in E, so that the strategy does not win the game for the class E. His strategy is the simplest possible: at each of his moves he appends a single 0. The whole play is then computable: to decide whether the N -th string is in L , run the game until the current string has length at least N and read off bit N . Each round adds at least one bit, so at most N rounds are needed, and each of Player II's moves takes time polynomial in the current length, which is at most polynomial in N (she cannot write more bits than she has time for), and hence bit N is computed in time polynomial in N . Since the strings of length n sit at positions $N \approx 2^n$, membership of a string of length n is decided in time polynomial in 2^n , that is, in time $2^{O(n)}$. So $L \in E$.

P is meager in E: diagonalization, one move at a time. Let $M_1, M_2, \dots$ be a list of Turing machines in which M_i halts within $n^i + i$ steps on inputs of length n , so that every language in P is decided by some M_i . Player II wants to make sure that L disagrees with every M_i on at least one string, and each of her moves takes care of a few more machines. Let w be the current string, and let s be the least length such that w decides no string of length s ; then w decides only strings of length less than s , so $|w| < 2^s$, and all 2^s strings of length s are still free. For each $i \leq \log s$, Player II picks the i -th string x_i of length s , runs M_i on x_i , and extends w so that

$$x_i \in L \iff M_i \text{ rejects } x_i,$$

filling all the other new positions with 0. Each simulation takes at most $s^{\log s} + \log s$ steps, which is less than 2^s once s is large, so the whole move takes time polynomial in 2^s . (Here $|w| \geq 2^{s-1} - 1$, because w decides all strings of length less than $s - 1$; so a bound polynomial in 2^s is polynomial in $|w|$, as the rules require.) Player I may then extend the string as much as he likes, but he cannot undo what she wrote. Since Player II moves infinitely often and $s \rightarrow \infty$, every M_i is eventually defeated on some string, so $L \notin P$.

The one clever point is the restriction $i \leq \log s$. Player II cannot afford to simulate M_i on short strings when i is large, since n^i steps would exceed her time budget: if she tried to defeat M_{1000} in her very first move, the n^{1000} steps of the simulation would exceed any fixed polynomial bound. She waits instead until $s \geq 2^{1000}$, when the strings are long enough that

simulating M_{1000} costs little relative to $|w|$. This “diagonalization with delay” is a standard trick of complexity theory, and the game makes the trick look natural.

Circuit size: a counting argument, one length at a time. Here the resource bound is space instead of time, and the key fact is Shannon’s counting argument (1949): there are 2^{2^n} Boolean functions of n variables, one for each truth table. A circuit with s gates can be described by listing the operation and the two inputs of each gate, and a careful count (taking into account that renaming the gates does not change the circuit) shows that for large n there are fewer circuits of size $2^n/n$ than there are truth tables. So some Boolean function of n variables has no circuit of size $2^n/n$.

Player II turns this count into a strategy. At each of her moves she takes the smallest length n such that no string of length n has been decided yet. She finds the first truth table on n variables that has no circuit of size $2^n/n$, by brute force: go through the truth tables one by one, and for each one go through all circuits of size $2^n/n$ and check whether any of these circuits computes that truth table. Then she defines L on the 2^n strings of length n to be that truth table. The search takes an astronomical amount of *time*, but very little *space*: at any moment she only needs to remember one truth table and one circuit, which takes space polynomial in 2^n , hence polynomial in the length of the current string. Space can be reused and time cannot, and Player II is charged only for space. Since she does this for infinitely many n , the final language has no circuits of size $2^n/n$ at infinitely many lengths, so $L \notin \text{SIZE}(2^n/n)$.

Exercises.

- (1) Number the binary strings in the standard order starting from 0, so that λ is string number 0. Show that the strings of length n are those numbered $2^n - 1$ through $2^{n+1} - 2$, and that a string w of length n has number $m - 1$, where m is the number whose binary expansion is $1 \frown w$. Deduce that a finite 0–1 string decides all strings of length less than n exactly when its length is at least $2^n - 1$.
- (2) Consider the game of this section without resource bounds. Show that Player II has a winning strategy for every countable class $\mathcal{C}$, and that Player I has a winning strategy for the class of languages containing the empty string λ .
- (3) Fill in the proof that E is not meager in itself. Let p be an increasing polynomial such that Player II computes her answer to a string w in time at most $p(|w|)$, and let Player I always append a single 0. Show that bit N of the resulting language can be computed in time polynomial in N . (Hint: every move that has to be simulated is made from a string of length at most N .)
- (4) Show that $s^{\log_2 s} + \log_2 s < 2^s$ for every integer $s \geq 17$, but not for $s = 16$. Explain why the proof that P is meager in E needs such an inequality only for large s .
- (5) (*) A circuit with s gates on n variables is a list of s gates, each labelled AND, OR or NOT and given two inputs among the variables and the earlier gates (a NOT gate ignores its second input), the last gate being the output. Show that there are at most $s(3(n+s)^2)^s$ circuits with between 1 and s gates, and deduce that, for all sufficiently large n , some Boolean function of n variables has no circuit with at most $2^n/(10n)$ gates.

10. APPLICATIONS TO TOPOLOGICAL GROUPS

The problem. A *semitopological group* is a group with a topology in which multiplication is *separately* continuous: the translations $x \mapsto xa$ and $x \mapsto ax$ are continuous. A semitopological group is a *topological group* if multiplication is jointly continuous and inversion is continuous. Separate continuity is much weaker than joint continuity (think of $f(x, y) = xy/(x^2 + y^2)$ on the plane, which is continuous in each variable separately but not at the origin). When does separate continuity of multiplication force a semitopological group to be a topological group? Behind this lies a more general question: when is a separately continuous function $f: X \times Y \rightarrow \mathbb{R}$ jointly continuous at many points?

✂ 10.1. **History of the problem and of the solution using topological games.** Baire proved in his thesis of 1899 that a separately continuous function on $\mathbb{R}^2$ is jointly continuous at the points of a dense G_δ set. In 1957 Ellis proved that every locally compact Hausdorff semitopological group is a topological group [Ell57], and in 1974 Namioka proved a general joint continuity theorem for functions on $X \times K$, with X complete in a suitable sense and K compact, from which he deduced Ellis's theorem [Nam74].

The game-theoretic solution came in three steps. In 1981 Christensen replaced the completeness hypothesis on X in Namioka's theorem by the existence of a winning strategy in a topological game [Chr81]. In 1983 Saint-Raymond showed that the weaker hypothesis that the opponent has no winning strategy suffices, and deduced that every metrizable Baire space has the joint continuity property [SR83]. For groups, Bouziad proved in 1996 that every Čech-analytic Baire semitopological group is a topological group [Bou96], and in 2001 Kenderov, Kortezov and Moors introduced a game that describes a class of spaces, the strongly Baire spaces, on which every semitopological group is a topological group [KKM01].

10.2. **The game.** Let X be a space and $D \subseteq X$ a dense set. In the game $\text{SB}(D)$ of Kenderov, Kortezov and Moors, their β is our Player I and their α is our Player II.

- (1) Player I chooses a non-empty open set $U_1 \subseteq X$.
- (2) Player II chooses a non-empty open set $V_1 \subseteq U_1$.
- (3) In general, Player I chooses a non-empty open $U_{k+1} \subseteq V_k$ and Player II a non-empty open $V_{k+1} \subseteq U_{k+1}$.
- (4) Player II wins if $\bigcap_k V_k \neq \emptyset$ and every sequence (a_k) with $a_k \in V_k \cap D$ has a cluster point in X .

A regular space is *strongly Baire* if, for some dense D , Player I has no winning strategy. This is the Banach–Mazur game of Section 3 with a more demanding winning condition for Player II; the games of Christensen and Saint-Raymond are variants in which Player II also picks points as she goes.

The motivating examples are the Čech-complete spaces. Let X be Čech-complete, say with $X = \bigcap_k G_k$ for sets G_k open in a compactification bX . Player II wins with $D = X$ by choosing V_k whose closure in bX lies inside $G_k \cap U_k$. The closures form a decreasing sequence of non-empty compact sets, their intersection lies in X and equals $\bigcap_k V_k$, and every sequence with $a_k \in V_k$ clusters in $\bigcap_k V_k$.

10.3. **The Main Result of this section.** The main result goes from functions to groups: Namioka's joint continuity theorem, Ellis's theorem on locally compact groups, and the theorem of Kenderov, Kortezov and Moors, which contains Ellis's.

Theorem 10.1 (Ellis, 1957; Namioka, 1974; Kenderov–Kortezov–Moors, 2001).

- (1) Namioka’s theorem. *Let X be a Čech-complete space, let K be a compact space, and let $f: X \times K \rightarrow \mathbb{R}$ be separately continuous. Then there is a dense G_δ set $A \subseteq X$ such that f is jointly continuous at every point of $A \times K$.*
- (2) Ellis’s theorem. *Every locally compact Hausdorff semitopological group is a topological group.*
- (3) *Every semitopological group whose underlying space is strongly Baire is a topological group.*

By the argument of the preceding subsection, every regular Čech-complete space, in particular every locally compact Hausdorff space and every complete metric space, is strongly Baire, so part (3) contains part (2). There is a useful way to read part (1): the function f is jointly continuous at every point of $\{x\} \times K$ exactly when the map $x \mapsto f(x, \cdot)$ from X to $C(K)$, which is continuous for the topology of pointwise convergence, is continuous at x for the norm topology. So Namioka’s theorem is about upgrading pointwise convergence to uniform convergence, at least at many points.

10.4. How the game solves the problem: a sketch. The arguments of Christensen, Saint-Raymond and Kenderov–Kortezov–Moors share a pattern worth learning, and we only sketch it. Suppose, for instance, that multiplication is not jointly continuous at (e, e) . Then there is a neighbourhood W of e such that every pair of neighbourhoods of e contains elements u, v with $uv \notin W$. Player I uses these pairs to build a strategy: at each move he chooses an open set inside Player II’s last move that records a new pair of witnesses, chosen so that separate continuity at any cluster point of the witnesses would be violated. Since X is strongly Baire, this strategy cannot be a winning one, so some play against this strategy is won by Player II. That play hands us a cluster point of the witnesses, and separate continuity at that point gives a contradiction. (The witnesses must be taken from the dense set D , so that the winning condition of the game, which only speaks of sequences from D , applies to them; arranging this is the technical heart of the proofs, and we omit it.) In short, the failure of joint continuity is turned into a strategy for Player I, and the topological hypothesis says exactly that such strategies can be beaten. Continuity of inversion is handled separately, and for locally compact groups the argument gives Ellis’s theorem; Namioka’s original deduction, which uses translations to spread a single point of joint continuity over the whole group, follows the same idea.

Exercises.

- (1) Let $f(x, y) = xy/(x^2 + y^2)$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$. Show that f is continuous in each variable separately, that f is not jointly continuous at $(0, 0)$, and that f is jointly continuous at every other point of $\mathbb{R}^2$.
- (2) Show that every strongly Baire space is a Baire space. (Hint: a winning strategy for Player I in $\text{BM}(X)$ is also winning in $\text{SB}(D)$ for every dense set D .) Deduce that $\mathbb{Q}$ is not strongly Baire.
- (3) Let X be a complete metric space. Show directly that Player II has a winning strategy in $\text{SB}(X)$, by answering U_k with a non-empty open set V_k such that $\overline{V_k} \subseteq U_k$ and $\text{diam } V_k < 2^{-k}$.
- (4) Let $\mathbb{R}$ carry the Sorgenfrey topology, whose basic open sets are the intervals $[a, b)$. Show that addition is jointly continuous, that the translations are homeomorphisms,

and that $x \mapsto -x$ is not continuous. Assuming that the Sorgenfrey line is regular, deduce from Theorem 10.1(3) that it is not strongly Baire.

- (5) (*) Let X be a topological space, let K be a compact space, and let $f: X \times K \rightarrow \mathbb{R}$ be separately continuous. Show that f is jointly continuous at every point of $\{x\} \times K$ if and only if the map $x' \mapsto f(x', \cdot)$ from X to $C(K)$ is continuous at x for the supremum norm.

11. APPLICATIONS TO INFINITE-DIMENSIONAL RAMSEY THEORY

The problem. Ramsey's theorem (1930) says that if you colour the n -element subsets of $\omega = \{0, 1, 2, \dots\}$ with two colours, there is an infinite $H \subseteq \omega$ all of whose n -element subsets get the same colour. We now colour the *infinite* subsets instead. Call $\mathcal{A} \subseteq [\omega]^\omega$, the set of infinite subsets of ω , *Ramsey* if there is an infinite $H \subseteq \omega$ such that $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$. Using the axiom of choice one can build sets that are not Ramsey (a nonprincipal ultrafilter is one, as we shall see in Section 13). Which sets are Ramsey?

There is a Rothko hiding in every Pollock: however wildly two colours are splashed on the pairs of natural numbers, some infinite set has all of its pairs painted in a single colour. This section asks how much of that survives when we colour infinite sets instead of pairs.

- ✂ 11.1. **History of the problem and of the solution using topological games.** Ramsey's theorem appeared in 1930 [Ram30], and in 1952 Erdős and Rado showed, using the axiom of choice, that the infinite-dimensional analogue of Ramsey's theorem fails [ER52]. Nash-Williams proved in 1965 that clopen sets, in the topology inherited from the Cantor space, are Ramsey [NW65]; Galvin and Prikry extended this to Borel sets (published in 1973) [GP73], and Silver to analytic sets (published in 1970) [Sil70], by a proof that used metamathematical methods. Mathias developed the theory further, with applications to forcing, in his work on happy families [Mat77].

The problem was solved by Erik Ellentuck in 1974 [Ell74], who introduced a new topology on $[\omega]^\omega$, finer than the usual one, and proved that the sets that are Ramsey in a hereditary sense are exactly those with the Baire property in this topology. This gave a short proof of Silver's theorem. In 1983 Kastanas [Kas83] characterized the same sets by the determinacy of a game, which is a Banach–Mazur game in Ellentuck's topology. Gowers's game in Section 12 is the analogue of Kastanas's game for block sequences in Banach spaces.

11.2. **The game.** The game is played in a new topology whose open sets look strange at first, so we begin with the idea behind them. In the usual (Cantor space) topology on $[\omega]^\omega$, two infinite sets are close if they begin in the same way, that is, if they share the same finite initial segment s . Ellentuck's open sets ask for more: they fix the beginning s and restrict the rest of the set to come from a given reservoir M . For a finite $s \subseteq \omega$ and an infinite $M \subseteq \omega$ with $\max s < \min M$, let

$$[s, M] = \{X \in [\omega]^\omega : s \text{ is an initial segment of } X \text{ and } X \subseteq s \cup M\}.$$

Think of s as a stem that has already been decided and of M as the pool of candidates for the rest. These sets form a basis for the *Ellentuck topology*. A set $\mathcal{A}$ is *completely Ramsey* if for every $[s, M]$ there is an infinite $N \subseteq M$ with $[s, N] \subseteq \mathcal{A}$ or $[s, N] \cap \mathcal{A} = \emptyset$.

Let $\mathcal{A} \subseteq [\omega]^\omega$ be given, let $s \subseteq \omega$ be finite, and let $N \subseteq \omega$ be infinite with $\max s < \min N$. The Kastanas game $K(\mathcal{A}, s, N)$ is played as follows.

- (1) Player I chooses an infinite $M_0 \subseteq N$.

- (2) Player II chooses $n_0 \in M_0$ and an infinite $N_0 \subseteq M_0$ with $n_0 < \min N_0$.
- (3) In general, Player I chooses an infinite $M_k \subseteq N_{k-1}$, and Player II chooses $n_k \in M_k$ and an infinite $N_k \subseteq M_k$ with $n_k < \min N_k$.
- (4) Player II wins if $s \cup \{n_0, n_1, \dots\} \in \mathcal{A}$.

In Kastanas's paper the payoff set is Player I's; since the theorem is about determinacy and $\mathcal{A}$ is completely Ramsey exactly when the complement of $\mathcal{A}$ is, the switch does not matter. The moves are basic open sets in disguise: Player I's move M_k stands for $[s \cup \{n_0, \dots, n_{k-1}\}, M_k]$, and Player II answers with the smaller basic open set $[s \cup \{n_0, \dots, n_k\}, N_k]$. So this is a Banach–Mazur game in the Ellentuck topology, in which Player II must commit to one more element at each move (see Figure 12).

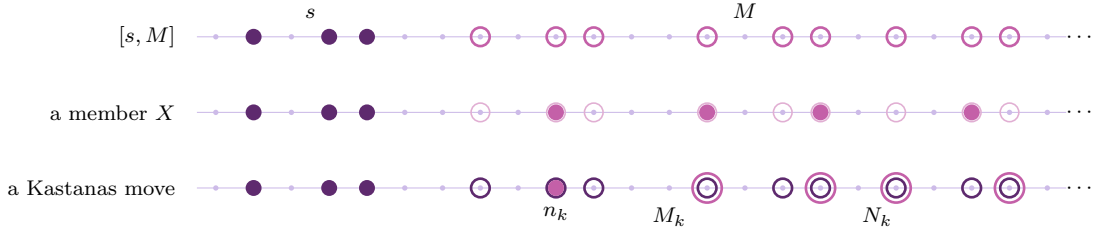


FIGURE 12. The Ellentuck topology on $[\omega]^\omega$. Top: a basic open set $[s, M]$ is given by a finite stem s (filled plum) and an infinite set M above it (circles). Middle: a member of $[s, M]$ is the stem together with an infinite subset of M (filled pink). Bottom: one round of the Kastanas game after the stem: Player I offers an infinite set M_k (plum circles), and Player II commits to one element $n_k \in M_k$ and keeps an infinite set $N_k \subseteq M_k$ above it (pink circles) for the rest of the play.

11.3. The Main Result of this section. The main result is Ellentuck's theorem, in the game form due to Kastanas.

Theorem 11.1 (Ellentuck, 1974; Kastanas, 1983). *For $\mathcal{A} \subseteq [\omega]^\omega$ the following are equivalent:*

- (1) *the set $\mathcal{A}$ is completely Ramsey;*
- (2) *the set $\mathcal{A}$ has the Baire property in the Ellentuck topology;*
- (3) *for every finite $s \subseteq \omega$ and every infinite $N \subseteq \omega$ with $\max s < \min N$, the Kastanas game $K(\mathcal{A}, s, N)$ is determined.*

In any topological space, the sets with the Baire property are closed under the Souslin operation, and the analytic sets are obtained from closed sets by that operation. Since the Ellentuck topology is finer than the usual one, every analytic set, and in particular every Borel set, is completely Ramsey by Theorem 11.1. This recovers the theorems of Galvin–Prikry and Silver in one stroke.

11.4. How the game solves the problem. The theorems of Galvin–Prikry, Silver and Ellentuck all rest on the same device, the *combinatorial forcing* of Galvin and Prikry. We say that M *accepts* s if $[s, M] \subseteq \mathcal{A}$, and that M *rejects* s if no infinite $N \subseteq M$ accepts s . In plain words, the set M accepts the stem s if, however you complete s with infinitely many elements of M , you can never escape from $\mathcal{A}$; and M rejects s if no amount of thinning out

M will make M accept s . By a diagonal argument one finds, inside any M , an infinite N that has decided every finite $s \subseteq N$, in the sense that N accepts or rejects each such s , and the decisions fit together. Ellentuck's insight was that these diagonal arguments are exactly the proof that the Ellentuck space is a Baire space, and that the sets with $[s, N] \cap \mathcal{A} = \emptyset$ inside every $[s, M]$ are exactly the nowhere dense sets. Kastanas's game turns the diagonalization into a play: a winning strategy for Player II, run against all possible moves of Player I and glued together by a diagonal fusion, produces an infinite $M \subseteq N$ with $[s, M] \subseteq \mathcal{A}$, and a winning strategy for Player I produces one with $[s, M] \cap \mathcal{A} = \emptyset$. So the Ramsey property is to Kastanas's game what the Baire property is to the Banach–Mazur game.

✂ **11.5. The notation $[\omega]^\omega$ and the partition calculus.** This section plays games of Banach–Mazur type on the Ellentuck space of infinite subsets of $\omega = \{0, 1, 2, \dots\}$, whose notation differs from that of Section 2; this subsection explains where the notation comes from. For a set S and a cardinal κ , one writes $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$ and $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$; thus $[\omega]^\omega$ and $[\omega]^{<\omega}$ are the families of infinite and of finite subsets of ω , as opposed to the sequence spaces A^ω and $A^{<\omega}$ of Section 2. This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which $\kappa \rightarrow (\lambda)_\mu^r$ means: for $|S| = \kappa$, every partition of $[S]^r$ into μ pieces has a set $H \subseteq S$ with $|H| = \lambda$ and $[H]^r$ inside a single piece. Ramsey's theorem becomes $\omega \rightarrow (\omega)_k^r$ for finite r and k . For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned $[\omega]^\omega$ into two pieces with no infinite homogeneous set [ER52]. Positive results recover $\omega \rightarrow (\omega)_2^\omega$ for definable partitions: Galvin and Prikry for Borel partitions [GP73], Silver for analytic ones [Sil70], and Ellentuck for partitions with the Baire property in the Ellentuck topology [Ell74]; in this section, games of Banach–Mazur type on the Ellentuck space are the bridge to these theorems. The partition calculus, and the school of infinite combinatorics that Erdős and his collaborators built around this calculus, are set out in the monograph of Erdős, Hajnal, Máté, and Rado [EHMR84].

Exercises.

- (1) Show that the sets $[s, M]$ form a basis of a topology on $[\omega]^\omega$. (Hint: if $[s, M] \cap [t, N]$ is non-empty, then one of s and t , say s , is an initial segment of the other, and the intersection is $[t, P]$, where P is the set of elements of $M \cap N$ greater than $\max t$.)
- (2) Show that the Ellentuck topology is finer than the usual one: for a finite $t \subseteq \{0, \dots, n-1\}$, the set of $X \in [\omega]^\omega$ with $X \cap \{0, \dots, n-1\} = t$ is a basic set $[s, M]$. Show also that, if $\omega \setminus M$ is infinite, the set $[\emptyset, M]$ has empty interior in the usual topology.
- (3) Let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ containing infinitely many even numbers, and let s and N be as in the definition of the Kastanas game. Show that Player II has a winning strategy in $K(\mathcal{A}, s, N)$ if N contains only finitely many odd numbers, and that Player I has one otherwise. Find in each case an infinite $N' \subseteq N$ with $[s, N'] \subseteq \mathcal{A}$ or $[s, N'] \cap \mathcal{A} = \emptyset$.
- (4) Show that a set $\mathcal{A} \subseteq [\omega]^\omega$ is completely Ramsey if and only if $[\omega]^\omega \setminus \mathcal{A}$ is, and that every completely Ramsey set is Ramsey.
- (5) (*) Color the two-element subsets of ω red or blue, and let $\mathcal{A}$ be the set of $X \in [\omega]^\omega$ whose two least elements form a red pair. Show that $\mathcal{A}$ is completely Ramsey. (Hint: treat separately the cases in which s has at least two elements, one element, or none; in the last case use Ramsey's theorem.)

12. APPLICATIONS TO BANACH SPACE THEORY

The problems. (1) *Differentiability of convex functions.* In calculus, a convex function of one variable is differentiable at all but countably many points. In infinite dimensions things are much less tidy. A Banach space X is an *Asplund space* if every continuous convex function on an open convex subset of X is Fréchet differentiable on a dense G_δ set. Which spaces are Asplund spaces?

(2) *The unconditional basic sequence problem.* Does every infinite-dimensional Banach space contain an unconditional basic sequence, that is, a sequence (e_n) that is a basis of the closed subspace spanned by (e_n) and for which every convergent expansion $\sum a_n e_n$ keeps converging when you flip the signs of the terms or shuffle their order?

A small glossary. Let X be a Banach space.

- The *dual* X^* is the Banach space of bounded linear functionals on X , and B_{X^*} is its closed unit ball. The *weak* topology* on X^* is the topology of pointwise convergence on the elements of X .
- A function f on an open subset of X is *Gâteaux differentiable* at x if there is $\varphi \in X^*$ with $f(x+th) = f(x) + t\varphi(h) + o(t)$ as $t \rightarrow 0$ for each fixed h , and *Fréchet differentiable* at x if $f(x+h) = f(x) + \varphi(h) + o(\|h\|)$, uniformly in the direction of h .
- A *basis* of X is a sequence (e_n) such that every $x \in X$ has a unique expansion $x = \sum_n a_n e_n$, and a *basic sequence* is a basis of its closed linear span. A basic sequence is *C -unconditional* if $\|\sum_n \varepsilon_n a_n e_n\| \leq C \|\sum_n a_n e_n\|$ for all signs $\varepsilon_n = \pm 1$, and *unconditional* if it is C -unconditional for some C .
- The space X is the *direct sum* of closed subspaces Y_1 and Y_2 if $Y_1 \cap Y_2 = \{0\}$ and $Y_1 + Y_2 = X$.

✂ 12.1. **History of the problems and their solutions using topological games.** Mazur proved in 1933 that continuous convex functions on separable Banach spaces are Gâteaux differentiable on dense G_δ sets, and Asplund introduced the spaces now named after him in 1968, in a paper in *Acta Mathematica* [Asp68]. The characterization problem was solved in 1975 by Namioka and Phelps [NP75] and by Stegall [Ste75]. The game-theoretic description of the property behind Asplundness, fragmentability, was found by Kenderov and Moors and published in 1999 [KM99].

The unconditional basic sequence problem was one of the central open problems of Banach space theory for decades. The problem was solved, negatively, by Gowers and Maurey in 1992 (published in 1993) [GM93]. They built a *hereditarily indecomposable* space: a Banach space so tangled that no infinite-dimensional subspace splits as a direct sum of two infinite-dimensional subspaces. Gowers then showed, in 1996 and 2002, using an infinite game played with subspaces and vectors, that such strange spaces are not an accident: every Banach space contains either a subspace with an unconditional basis or a hereditarily indecomposable subspace [Gow96, Gow02].

Recognition. Gowers received the Fields Medal in 1998, for research connecting functional analysis and combinatorics, which includes the solution of the unconditional basic sequence problem and the dichotomy theorem. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994 and received the Prize of the European Mathematical Society in 1996. Gowers's Ramsey theorem appeared in the *Annals of Mathematics* in 2002.

12.2. The games.

12.2.1. *The fragmentability game.* Let K be a set with two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$; for the dual ball B_{X^*} of a Banach space X , the topology $\mathcal{T}_1$ is the weak* topology and $\mathcal{T}_2$ is the norm topology.

- (1) Player I chooses a non-empty subset $U_0 \subseteq K$.
- (2) Player II chooses a non-empty relatively $\mathcal{T}_1$ -open subset $V_0 \subseteq U_0$.
- (3) In round $k \geq 1$, Player I chooses a non-empty subset $U_k \subseteq V_{k-1}$, and Player II a non-empty relatively $\mathcal{T}_1$ -open subset $V_k \subseteq U_k$.
- (4) Player II wins if $\bigcap_k U_k$ is empty, or if $\bigcap_k U_k$ is a single point x and every $\mathcal{T}_2$ -neighbourhood of x contains some U_k .

This is a Banach–Mazur game with an asymmetry: Player I may play any non-empty set at all, while Player II must play relatively open ones, and she wins if the sets shrink down to a point in the stronger topology.

12.2.2. *Gowers’s game.* Let X have a basis (e_n) . A *block vector* is a non-zero finite linear combination of the e_n , and a *block subspace* is the closed span of a sequence of block vectors $x_1, x_2, \dots$ with *successive supports*: every basis vector used in x_k comes before every basis vector used in x_{k+1} . Let Σ be a set of sequences of normalized block vectors. In round $k \geq 1$, Player I chooses an infinite-dimensional block subspace Y_k , and Player II chooses a normalized block vector $x_k \in Y_k$. Player II wins if $(x_1, x_2, \dots) \in \Sigma$. Player I’s subspaces play the role of open sets. In the Banach–Mazur game a small neighbourhood limits where the other player may look, and a block subspace does the same: it can be chosen to live as far out along the basis as Player I likes, so by choosing such a subspace he can force Player II’s next vector to be supported beyond any finite stage, and he can restrict the kind of vectors available to her.

12.3. **The Main Results of this section.** Differentiability is tied to the dual space for the following reason. The derivative of a function on X at a point is a linear functional, an element of X^* , so differentiability is a question about how the candidate derivatives (the subgradients) sit inside X^* . Roughly, Fréchet differentiability at a point means that subgradients at nearby points are close to the derivative *in norm*, and the norm is what fragmentability controls. The first result below collects several characterizations of Asplund spaces, one of which is exactly what the game of Kenderov and Moors detects; the second is Gowers’s dichotomy.

Theorem 12.1 (Asplund, 1968; Namioka–Phelps, 1975; Stegall, 1975). *For a Banach space X the following are equivalent:*

- (1) *the space X is an Asplund space;*
- (2) *every separable subspace of X has a separable dual;*
- (3) *the dual X^* has the Radon–Nikodým property, a property from the theory of vector-valued measures that we do not use below;*
- (4) *the dual ball B_{X^*} , with the weak* topology, is fragmented by the norm: every non-empty subset of B_{X^*} has non-empty relatively weak*-open subsets of arbitrarily small norm diameter.*

Kenderov and Moors [KM99] showed that a space $(K, \mathcal{T}_1)$ is fragmented by a metric whose topology contains $\mathcal{T}_2$ if and only if Player II has a winning strategy in the fragmentability game of the preceding subsection. In particular, if X is an Asplund space, Player II wins that game on B_{X^*} with the weak* and norm topologies.

Theorem 12.2 (Gowers’s dichotomy, 1996 and 2002). *Every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable.*

The two alternatives cannot happen at the same time, because a hereditarily indecomposable space contains no unconditional basic sequence. So every Banach space, however large, contains a subspace that is either perfectly civilized or thoroughly rebellious; there is no third option.

12.4. How the games solve the problems.

12.4.1. *Asplund spaces.* Problem (1) was solved in 1975 without games, by Namioka and Phelps and by Stegall. What the game of Kenderov and Moors adds is a new way to establish fragmentability, and with it the Asplund property: instead of producing a fragmenting metric, one describes how Player II should play. Here is the easy half of their theorem.

If K is fragmented by the norm, Player II has an easy strategy: answer each move U_k with a relatively weak*-open subset $V_k \subseteq U_k$ of norm diameter less than 2^{-k} . Then $\bigcap_k U_k$ has norm diameter 0 and is therefore empty or a single point x , and in the second case $U_{k+1} \subseteq V_k$ lies in the norm ball of radius 2^{-k} about x . Conversely, Kenderov and Moors build a fragmenting metric out of a winning strategy for Player II; roughly, two points are close when the strategy takes a long time to separate them. The advantage of the game is that a strategy is a local object, built one move at a time, while a fragmenting metric is a global one, and describing how to play is often much easier than writing down a metric.

12.4.2. *Gowers’s dichotomy.* Gowers’s infinite Ramsey theorem [Gow02] says that for every analytic set Σ and every sequence of positive tolerances, there is a block subspace Y such that either no block sequence of Y lies in Σ , or Player II has a winning strategy in the game played in Y for a slight enlargement of Σ , obtained by wiggling the members of Σ within the given tolerances. This is a game-theoretic version of the Galvin–Prikry and Ellentuck theorems on Ramsey sets (see Section 11), adapted to Banach spaces. An exact Ramsey statement is false for vectors, and the game is what can be rescued. Gowers’s proof uses a “combinatorial forcing” argument, in which subspaces are said to accept or reject finite block sequences.

Roughly, the dichotomy follows by applying the Ramsey theorem to the set of block sequences that fail to be C -unconditional. If some block subspace avoids these sequences for some constant C , every block basis in that subspace is C -unconditional, so the subspace has an unconditional basis. Otherwise, in every block subspace Player II can produce badly unconditional block sequences, and from her strategies Gowers extracts a subspace in which no two infinite-dimensional subspaces form a direct sum, that is, a hereditarily indecomposable subspace. The game is what locates this subspace; the existence of such spaces is the theorem of Gowers and Maurey.

Exercises.

- (1) Show that the standard unit vector bases of c_0 and of ℓ_p , for $1 \leq p < \infty$, are 1-unconditional.
- (2) Let $f(x, y) = x^3y/(x^4 + y^2)$ for $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$. Show that f is Gâteaux differentiable at the origin, with $\varphi = 0$, but not Fréchet differentiable there. (Hint: look at $f(x, x^2)$.)

- (3) Let (K, d) be a metric space, and let $\mathcal{T}_1 = \mathcal{T}_2$ be the topology of d . Show that Player II has a winning strategy in the fragmentability game, by answering U_k with the intersection of U_k and a ball of radius 2^{-k-1} about a point of U_k .
- (4) Let (e_n) be a 1-unconditional basis of X , and let $x_1, x_2, \dots$ be block vectors with successive supports. Show that $\|\sum_k \varepsilon_k b_k x_k\| \leq \|\sum_k b_k x_k\|$ for all finite sequences of scalars b_k and signs $\varepsilon_k = \pm 1$.
- (5) (*) Let X have a basis (e_n) , and consider Gowers's game in X . Show that Player II has a winning strategy when Σ is the set of all sequences of normalized block vectors with successive supports, and that Player I has a winning strategy when Σ is the set of sequences whose first term is $e_1/\|e_1\|$. (Hint: the coefficient functionals of a basis are continuous.)

13. APPLICATIONS TO ULTRAFILTERS

Definitions. A *filter* on ω is a non-empty family $\mathcal{F}$ of subsets of ω that does not contain $\emptyset$, is closed under finite intersections, and contains every superset of each of its members. An *ultrafilter* is a filter $\mathcal{U}$ such that for every $x \subseteq \omega$ either x or $\omega \setminus x$ belongs to $\mathcal{U}$. The ultrafilter of all sets containing a fixed point n is called *principal*; every other ultrafilter is *nonprincipal*, and contains every cofinite set. Let $\mathcal{U}$ be a nonprincipal ultrafilter, and consider a partition of ω into pieces that do not belong to $\mathcal{U}$. The ultrafilter is a *P-point* if for every such partition some element of $\mathcal{U}$ meets every piece in a finite set, and it is *selective* if for every such partition some element of $\mathcal{U}$ meets every piece in at most one point. The set of all ultrafilters on ω , with a natural compact topology, is the space $\beta\omega$, and $\omega^* = \beta\omega \setminus \omega$ is the set of nonprincipal ultrafilters; these spaces appear only in passing.

The problem. Identify every subset of ω with its characteristic function, so that the power set $\mathcal{P}(\omega)$ becomes the Cantor space 2^ω and a filter on ω becomes a set of reals. The axiom of choice provides nonprincipal ultrafilters on ω in abundance (there are $2^{2^{\aleph_0}}$ of them, the points of $\omega^* = \beta\omega \setminus \omega$), and yet nobody has ever exhibited one. The problem is to explain this: to determine how irregular a nonprincipal ultrafilter must be as a set of reals, how much choice its existence requires, and which ultrafilters are tame enough to carry the Ramsey theory of Section 11.

A principal ultrafilter is a point of ω in disguise: it collects the sets that contain the point, and its two-valued measure is the Dirac measure at the point. A nonprincipal ultrafilter is a point at infinity that passes judgement on every subset of ω , consistently and irrevocably, and always against the finite ones. Seen through Ulam's binary version of the Banach–Mazur game, such a judge is invisible to topology: the game it defines has no winner. Seen through the games of Galvin, the well-mannered judges, the selective ultrafilters and the P-points, are exactly those that cannot be outplayed.

- ✂ 13.1. **History of the problem and of the solution using topological games.** Ultrafilters came into mathematics through two doors. Tarski proved in 1930, using the axiom of choice, that every filter of subsets of a set extends to an ultrafilter [Tar30]; Stone made the ultrafilters of a Boolean algebra into the points of a compact space in 1936 [Sto36]; and Cartan introduced filters and ultrafilters in 1937 to describe convergence in general topology [Car37]. In 1938 Sierpiński observed that a nonprincipal ultrafilter on ω , regarded as a set of reals, is not Lebesgue measurable [Sie38]; the same symmetry shows that it does not have the Baire property. In 1962 Mycielski and Steinhaus proposed the Axiom of Determinacy,

which makes every set of reals regular and therefore rules out nonprincipal ultrafilters on ω altogether [MS62]. Feferman showed in 1965, by forcing, that the existence of a definable nonprincipal ultrafilter on ω cannot be proved in ZFC [Fef65], and Shelah showed in 1984 that ZF together with the axiom of dependent choices is consistent with the statement that every set of reals has the Baire property, relative to ZF alone [She84]. Some choice beyond dependent choice is therefore needed to produce a single nonprincipal ultrafilter on ω .

Meanwhile the ultrafilters themselves were being sorted. Walter Rudin proved in 1956 that under the continuum hypothesis there are P-points, and deduced that ω^* is not homogeneous [Rud56]; Frolík proved in 1967 that ω^* is not homogeneous in ZFC [Fro67]. Booth and Kunen isolated the selective ultrafilters [Boo70], and Kunen proved that an ultrafilter is selective exactly when every colouring of $[\omega]^2$ in two colours has a homogeneous set in the ultrafilter. Mathias showed in 1977 that selective ultrafilters carry the Ramsey theory of analytic sets [Mat77]. Kunen proved in 1976 that it is consistent with ZFC that there are no selective ultrafilters [Kun76], and Shelah proved shortly afterwards that it is consistent that there are no P-points at all [Wim82].

The topological side of the problem was settled by Jalali-Naini in his thesis of 1976 [JN76] and, independently, by Talagrand in 1980 [Tal80], who characterized the meager filters by a combinatorial condition on interval partitions; Talagrand's motivation came from pointwise compact sets of measurable functions, the subject of the theory of Rosenthal compacta created by Bourgain, Fremlin and Talagrand. The games enter twice. The irregularity of a nonprincipal ultrafilter is a strategy-stealing argument in Ulam's binary game, implicit in Sierpiński's proof; and Galvin and McKenzie characterized the selective ultrafilters and the P-points by the games described below (see the book of Bartoszyński and Judah [BJ95]).

Recognition. Talagrand received the Abel Prize in 2024, and Shelah the Wolf Prize in 2001.

13.2. The games.

Ulam's game. Let $Z \subseteq 2^\omega$ be given. In the game $\text{Ul}(Z)$, Player I chooses a nonempty finite binary string s_0 , Player II chooses a nonempty finite binary string t_0 , Player I chooses a nonempty s_1 , and so on, forever. Player II wins if the concatenation

$$x = s_0 \frown t_0 \frown s_1 \frown t_1 \frown \cdots$$

belongs to Z . A move is a basic clopen set $N_s = \{x \in 2^\omega : s \subseteq x\}$ inside the previous one, so this is the Banach–Mazur game on the Cantor space, in the binary form proposed by Ulam (Section 20). The theorem of Banach and Oxtoby (see Theorem 3.5, with moves restricted to basic open sets; the Cantor space is a compact metric space, so both parts apply) reads as follows here: Player II has a winning strategy in $\text{Ul}(Z)$ if and only if Z is comeager, and Player I has one if and only if $Z \cap N_s$ is meager for some finite binary string s .

Galvin's games. Let $\mathcal{U}$ be an ultrafilter on ω . In the *selective game* $S(\mathcal{U})$, Player I chooses $A_0 \in \mathcal{U}$, Player II chooses $n_0 \in A_0$, Player I chooses $A_1 \in \mathcal{U}$, Player II chooses $n_1 \in A_1$, and so on; Player II wins if $\{n_0, n_1, \dots\} \in \mathcal{U}$. In the *P-point game* $P(\mathcal{U})$, Player II chooses a finite set $F_k \subseteq A_k$ instead of a single element, and wins if $\bigcup_k F_k \in \mathcal{U}$. Player I plays the sceptic: he keeps narrowing the field to sets that $\mathcal{U}$ approves of, and challenges Player II to assemble, one element (or one finite batch) at a time, a set that $\mathcal{U}$ approves of as well.

Recall that P-points and selective ultrafilters were defined at the beginning of this section; every selective ultrafilter is a P-point.

13.3. The Main Result of this section. The main result records how nonprincipal ultrafilters behave in the two kinds of games: in Ulam's game they are invisible, and in Galvin's games the tame ones are exactly those that cannot be outplayed.

Theorem 13.1 (Sierpiński, 1938; Mycielski and Steinhaus, 1962; Galvin and McKenzie). *Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω , regarded as a subset of 2^ω . Then the following hold.*

- (1) *The game $\text{Ul}(\mathcal{U})$ is not determined.*
- (2) *The set $\mathcal{U}$ is neither meager nor comeager, it does not have the Baire property, and it is not measurable for the coin-tossing measure on 2^ω .*
- (3) *If the game $\text{Ul}(Z)$ is determined for every $Z \subseteq 2^\omega$, then every subset of 2^ω has the Baire property, and there is no nonprincipal ultrafilter on ω .*
- (4) *The ultrafilter $\mathcal{U}$ is selective if and only if Player I has no winning strategy in $S(\mathcal{U})$, and it is a P -point if and only if Player I has no winning strategy in $P(\mathcal{U})$.*

Part (2) explains why nobody has written down a nonprincipal ultrafilter on ω : analytic and coanalytic sets have the Baire property, so no nonprincipal ultrafilter is analytic or coanalytic. The proofs of (1) and (2), given below, use only two properties of $\mathcal{U}$:

- membership in $\mathcal{U}$ does not depend on finitely many digits, because $\mathcal{U}$ is nonprincipal;
- the complement $\bar{x} = \omega \setminus x$ belongs to $\mathcal{U}$ if and only if x does not, because $\mathcal{U}$ is an ultrafilter.

13.4. How the games solve the problem.

Stealing a strategy. The idea is to let Player I play two games against a strategy of Player II at once, the second always the mirror image of the first, so that the strategy is forced to put both a set and its complement into $\mathcal{U}$. For a finite binary string s , we write $\bar{s}$ for the string obtained by flipping every digit of s . Suppose that τ is a winning strategy for Player II in $\text{Ul}(\mathcal{U})$. In the first game he opens with the string 0, and τ answers t_0 . In the second game he opens with $\bar{0} \frown t_0$, and τ answers u_0 . Back in the first game, he plays $\bar{u}_0$, and τ answers t_1 ; in the second, he plays $\bar{t}_1$, and τ answers u_1 ; and so on (Figure 13). After each round, the position in the second game is the flip of the position in the first, so the two plays x and y satisfy $y = \bar{x}$. Both plays are won by τ , so $x \in \mathcal{U}$ and $\bar{x} \in \mathcal{U}$, which is impossible. Hence Player II has no winning strategy.

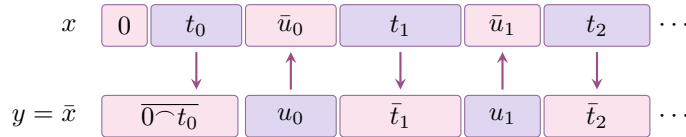


FIGURE 13. Stealing Player II's strategy in $\text{Ul}(\mathcal{U})$. Lavender blocks are answers of the strategy τ , blush blocks are moves of Player I, and each arrow copies an answer of τ in one game, flipped, as a move of Player I in the other.

Suppose now that Player I has a winning strategy σ , and let s be its first move. From the position s on, the strategy σ answers the moves of Player II and forces $x \notin \mathcal{U}$, that is, $\bar{x} \in \mathcal{U}$; flipping every move, we obtain a strategy that, from the position $\bar{s}$, answers its opponent and forces the play into $\mathcal{U}$. Player II can transplant this strategy into $\text{Ul}(\mathcal{U})$: she makes, if necessary, one dummy move, so that the position becomes longer than s ; at her next turn she pretends that the position is $\bar{s}$ followed by a single move of her opponent of the right length, and from then on she answers as the flipped strategy does. Her play agrees with a play that the flipped strategy wins, except on finitely many digits, so she wins $\text{Ul}(\mathcal{U})$, which we have just seen to be impossible. This proves (1).

Baire property and measure. If Z is comeager, then Player II wins $\text{Ul}(Z)$ by playing, at her k -th move, into the k -th of countably many dense open sets whose intersection lies in Z ; if Z is meager, then Player I wins in the same way. By (1), the set $\mathcal{U}$ is therefore neither meager nor comeager. A set $Z \subseteq 2^\omega$ that is invariant under finite changes and has the Baire property is meager or comeager. Indeed, if Z is not meager, then $N_s \setminus Z$ is meager for some s , and for every string t of the same length as s the homeomorphism of 2^ω that replaces the initial segment s by t carries N_s to N_t and preserves Z ; the finitely many sets N_t cover 2^ω . Hence $\mathcal{U}$ does not have the Baire property. Measure is no better. The flip $x \mapsto \bar{x}$ preserves the coin-tossing measure λ and carries $\mathcal{U}$ onto its complement, so if $\mathcal{U}$ were measurable we would have $\lambda(\mathcal{U}) = 1 - \lambda(\mathcal{U}) = \frac{1}{2}$, while Kolmogorov's zero-one law, applied to a measurable set that is invariant under finite changes, gives $\lambda(\mathcal{U}) \in \{0, 1\}$. This proves (2).

Determinacy kills ultrafilters. Assume that every game $\text{Ul}(Z)$ is determined, and let $Z \subseteq 2^\omega$ and a string s be given. Applying determinacy to the set $\{z : s \smallfrown z \in Z\}$ and the theorem of Banach and Oxtoby (see Theorem 3.5), we find a string $s' \supseteq s$ such that Z is comeager in $N_{s'}$ or meager in $N_{s'}$. Let O be the union of the sets $N_{s'}$ in which Z is comeager, and let O' be the union of those in which Z is meager. Then $O \cup O'$ is a dense open set, and $Z \triangle O$ is contained in the union of $O \setminus Z$, of $O' \cap Z$, and of the nowhere dense set $2^\omega \setminus (O \cup O')$, all of which are meager. Hence Z has the Baire property, and by (2) there is no nonprincipal ultrafilter on ω . The same argument, applied to the sets of a pointclass that is closed under continuous preimages, gives the further remarks on definable ultrafilters at the end of this section.

Selective ultrafilters. The proof of part (4) uses a theorem of Kunen, recorded in the history above, which we quote without proof.

Lemma 13.2 (Kunen). *Let $\mathcal{U}$ be a selective ultrafilter on ω . Then for every colouring of the two-element subsets of ω in two colours there is a set $H \in \mathcal{U}$ all of whose two-element subsets have the same colour.*

Suppose that $\mathcal{U}$ is selective, and let σ be a strategy for Player I in $S(\mathcal{U})$. For a finite set $s \subseteq \omega$, listed increasingly, let $A_s \in \mathcal{U}$ be the move of σ after Player II has played the elements of s , and let $B_n = \bigcap \{A_s : s \subseteq \{0, \dots, n-1\}\}$, a finite intersection, so that $B_n \in \mathcal{U}$. Colour a pair $m < n$ pink if $n \in B_{m+1}$ and purple otherwise. By Lemma 13.2 there is a homogeneous set $H \in \mathcal{U}$, and we may shrink it so that $H \subseteq A_\emptyset$. The colour of H is not purple, because for $m = \min H$ the sets $H \setminus \{m\}$ and B_{m+1} both belong to $\mathcal{U}$ and so they meet. So the colour is pink. List H increasingly as $h_0 < h_1 < \dots$. Then $h_0 \in A_\emptyset$, and $h_{k+1} \in B_{h_k+1} \subseteq A_{\{h_0, \dots, h_k\}}$ for every k . Hence Player II may answer σ with $h_0, h_1, h_2, \dots$ in turn, and the set she produces is $H \in \mathcal{U}$. So σ is not a winning strategy. Conversely, if $\mathcal{U}$ is not selective, fix a partition $\{P_j\}$ of ω into pieces outside $\mathcal{U}$ such that no element of $\mathcal{U}$ meets every piece in at most one point. Player I wins by always removing the pieces already visited: when Player II has played elements of $P_{j_0}, \dots, P_{j_{k-1}}$, he plays $\omega \setminus (P_{j_0} \cup \dots \cup P_{j_{k-1}})$, which belongs to $\mathcal{U}$; then the set produced by Player II meets each piece at most once, and so it is not in $\mathcal{U}$. The P-point game is handled by the same two arguments, with finite sets in place of single points and the P-point property in place of homogeneity; we refer to [BJ95].

Selectivity is exactly what the Ramsey theory of Section 11 needs. Mathias proved that if $\mathcal{U}$ is selective, then every analytic $\mathcal{A} \subseteq [\omega]^\omega$ has a homogeneous set in $\mathcal{U}$: there is $H \in \mathcal{U}$ with $[H]^\omega \subseteq \mathcal{A}$ or $[H]^\omega \cap \mathcal{A} = \emptyset$ [Mat77]. An ultrafilter cannot do this for itself: however we choose an infinite H , splitting H into two infinite halves yields two elements of $[H]^\omega$, exactly one of which belongs to $\mathcal{U}$. So every nonprincipal ultrafilter is a set that is not Ramsey, the simplest such set that the axiom of choice provides.

13.5. Further remarks (optional).

Definable ultrafilters. The proof of part (3) of Theorem 13.1 works one pointclass at a time, and under projective determinacy (which follows from the Woodin cardinals of Section 15) every projective set has the Baire property, so no nonprincipal ultrafilter is projective either. This bound is sharp in Gödel's constructible universe, where a transfinite recursion along the Σ_2^1 well-ordering of the reals produces a Δ_2^1 nonprincipal ultrafilter.

Meager filters. The game argument shows that a nonprincipal ultrafilter is not meager; the following theorem shows exactly how much of the ultrafilter property is responsible.

Theorem 13.3 (Jalali-Naini, 1976; Talagrand, 1980). *Let $\mathcal{F}$ be a filter on ω that contains every cofinite set. Then $\mathcal{F}$ is meager, as a subset of 2^ω , if and only if there is a partition of ω into consecutive finite intervals $I_0 < I_1 < \dots$ such that every element of $\mathcal{F}$ meets all but finitely many of the I_n .*

Proof. If such a partition exists, then $\mathcal{F}$ is contained in the union over m of the sets $\{x : x \cap I_n \neq \emptyset \text{ for all } n \geq m\}$. These sets are closed, and nowhere dense, because any finite string can be extended by zeros across a whole interval I_n . For the converse, let $C_0 \subseteq C_1 \subseteq \dots$ be closed nowhere dense sets whose union contains $\mathcal{F}$. We choose $0 = n_0 < n_1 < \dots$ and binary strings w_k of length $n_{k+1} - n_k$ such that $N_{u \frown w_k} \cap C_k = \emptyset$ for every string u of length n_k . This is possible because there are only finitely many such u , and we may extend a candidate for w_k once for each of them, each time avoiding C_k . Let $I_k = [n_k, n_{k+1})$ and let $e \subseteq \omega$ be the set whose characteristic function is $w_0 \frown w_1 \frown \dots$. If $z \in C_m$, then z does not agree with e on I_k for any $k \geq m$. Now let $x \in \mathcal{F}$ be given. Then $x \cup e \in \mathcal{F}$, so $x \cup e \in C_m$ for some m , and for every $k \geq m$ the sets $x \cup e$ and e differ somewhere in I_k , which means that x meets I_k . ☺

A nonprincipal ultrafilter $\mathcal{U}$ fails the condition of Theorem 13.3 for every partition: the union of the even intervals and the union of the odd intervals are complementary, so one of them belongs to $\mathcal{U}$, and it misses infinitely many intervals. This is a second proof that $\mathcal{U}$ is not meager. Theorem 13.3 also shows that nonmeagerness survives finite-to-one images: if $f: \omega \rightarrow \omega$ is finite-to-one and $f(\mathcal{F})$ is meager, witnessed by a partition $\{I_n\}$, then any partition of ω into consecutive intervals each of which contains one of the finite sets $f^{-1}(I_n)$ witnesses that $\mathcal{F}$ is meager.

Points that nobody can point at. The space ω^* is the Stone space of the Boolean algebra $\mathcal{P}(\omega)/\text{fin}$, and its points are the nonprincipal ultrafilters on ω . Pospíšil showed in 1937 that there are $2^{2^{\aleph_0}}$ of them, by means of an independent family of $2^{\aleph_0}$ subsets of ω in the sense of Fichtenholz, Kantorovich and Hausdorff: every Boolean combination of finitely many members of the family is infinite, so every assignment of signs to the members extends to an ultrafilter. A P-point is exactly a point $p \in \omega^*$ such that every G_δ subset of ω^* containing p is a neighbourhood of p , and no homeomorphism of ω^* can move a P-point to the limit of a countable discrete subset of ω^* ; this is Rudin's proof of non-homogeneity under the continuum hypothesis. Kunen found in ZFC, in 1978, *weak P-points*, points that are not limits of any countable subset of the rest of ω^* [Kun80]. All of these points are produced by choice, and the remarks on definable ultrafilters above say that none of them can be produced by a definition of low complexity. The large-cardinal picture is sharper still: Todorcevic proved that, in the presence of large cardinals, the selective ultrafilters are exactly the filters that are generic over $L(\mathbb{R})$ for the partial order $\mathcal{P}(\omega)/\text{fin}$ (see Farah [Far98]). A selective ultrafilter is thus a generic object for the very Boolean algebra whose Stone space it inhabits, and every

statement about it in $L(\mathbb{R})$ is decided by the forcing relation, the way a statement about a Cohen real is decided by finite conditions.

Exercises.

- (1) Let $\mathcal{U}$ be an ultrafilter on ω that contains a finite set. Show that $\mathcal{U}$ is principal. Deduce that a nonprincipal ultrafilter contains every cofinite set, and that if $x \in \mathcal{U}$ and $y \subseteq \omega$ differs from x in finitely many points, then $y \in \mathcal{U}$.
- (2) For each of the following sets $Z \subseteq 2^\omega$, decide which player has a winning strategy in $\text{Ul}(Z)$, and check the answer against Theorem 3.5: (a) the sequences with infinitely many terms equal to 1; (b) the sequences that are eventually 0; (c) the sequences x with $x(n) = 1$ for infinitely many even n ; (d) the sequences x with $x(0) = 0$.
- (3) Let $\mathcal{F}$ be the filter of cofinite subsets of ω . Show that $\mathcal{F}$ is meager in 2^ω , first directly and then by means of Theorem 13.3.
- (4) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω such that Player I has no winning strategy in $S(\mathcal{U})$. Show, without using Theorem 13.1, that Player I has no winning strategy in $P(\mathcal{U})$ either. (Hint: Player II may play one-element sets in $P(\mathcal{U})$.)
- (5) (*) Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω with the property in the conclusion of Lemma 13.2. Show that $\mathcal{U}$ is selective. (Hint: given a partition of ω into pieces outside $\mathcal{U}$, color a pair according to whether its two elements lie in the same piece.)

Part III. Applications of other topological games

In the games of this part the players no longer shrink a set. In the Menger game they choose open covers and finite subfamilies of them, and in the games of descriptive set theory they call out natural numbers, one at a time. What survives from Part I is the framework of Section 2 (see Definition 2.2): a game is given by a set of infinite plays, and the question is which player, if any, has a winning strategy.

14. APPLICATIONS TO SET-THEORETIC TOPOLOGY

The problem. Compact sets are the “small” sets of topology, and countable unions of compact sets (σ -compact sets) are almost as tame. In 1924 Karl Menger proposed a covering property that looked like a clean characterization of σ -compactness. A space X has the *Menger property* if for every sequence $\mathcal{U}_1, \mathcal{U}_2, \dots$ of open covers of X there are finite sets $\mathcal{F}_n \subseteq \mathcal{U}_n$ such that $\bigcup_n \mathcal{F}_n$ covers X . Every σ -compact space has the Menger property: if $X = \bigcup_n K_n$ with K_n compact, choose $\mathcal{F}_n$ to be a finite subset of $\mathcal{U}_n$ that covers K_n . Menger conjectured the converse for metric spaces. The problem is to decide Menger’s conjecture, in particular for sets of real numbers.

- ✂ 14.1. **History of the problem and of the solution using topological games.** Menger introduced the property in 1924 [Men24], in a different but equivalent form, while studying a basis property of metric spaces, and conjectured that this property characterizes σ -compactness. Hurewicz gave the property the form above in a paper of 1926 [Hur26], proved a theorem that is now stated as a characterization of the property by an infinite game (the game form is due to later authors, notably Scheepers), and introduced a stronger property for which he made the analogous conjecture. Under the continuum hypothesis the conjecture fails: a Luzin set, an uncountable set of reals meeting every meager set in a countable set, has the Menger property but is not σ -compact. In 1984 Telgársky characterized σ -compactness

of metrizable spaces by the same game [Tel84], so that Menger's conjecture became the statement that the Menger game on a set of reals is always determined.

The problem was solved by David Fremlin and Arnold Miller in 1988 [FM88], who constructed, in ZFC, a set of reals with the Menger property that is not σ -compact. The closely related property of Rothberger (1938), in which a single member of each cover is chosen, has a parallel game-theoretic story: Galvin introduced the point-open game in 1978 [Gal78], and in 1994 Pawlikowski proved that a set of reals is undetermined in this game if and only if the set is uncountable and has Rothberger's property [Paw94]. Borel conjectured in 1919 that every set of reals of strong measure zero is countable, and Laver proved in 1976 that this is consistent [Lav76]. Since Rothberger's property implies strong measure zero, in Laver's model every set of reals is determined in the point-open game, while under the continuum hypothesis a Luzin set gives an undetermined one [Rec94].

14.2. The games. In the *Menger game* on X , in round n Player I chooses an open cover $\mathcal{U}_n$ of the whole space X , and Player II chooses a finite set $\mathcal{F}_n \subseteq \mathcal{U}_n$. Player II wins if $\bigcup_n \mathcal{F}_n$ covers X . In the *Rothberger game* Player II must be even more frugal and chooses a single member of each cover. In the literature Player I is called ONE and Player II is called TWO. In Galvin's *point-open game*, Player I chooses a point $x_n \in X$ and Player II an open set $U_n \ni x_n$, and Player II wins if the sets U_n do not cover X ; the point-open game is dual to the Rothberger game, in the sense that a winning strategy for either player in one game corresponds to a winning strategy for the other player in the other game. Note the reversal: in the Menger and Rothberger games Player II wants her chosen sets to cover X , while in Galvin's game she chooses the open sets but wants them not to cover X .

14.3. The Main Result of this section. Hurewicz and Telgársky turned the Menger property and σ -compactness into statements about the two players of the Menger game, and the theorem of Fremlin and Miller shows that the gap between these two statements is real.

Theorem 14.1 (Hurewicz, 1926; Telgársky, 1984; Fremlin–Miller, 1988).

- (1) *A space X has the Menger property if and only if Player I has no winning strategy in the Menger game on X .*
- (2) *A separable metrizable space X is σ -compact if and only if Player II has a winning strategy in the Menger game on X .*
- (3) *There is a set of reals with the Menger property that is not σ -compact. Consequently, the Menger game on this set is not determined, and Menger's conjecture is false.*

14.4. How the games solve the problem.

Menger's property. If Player I has no winning strategy, then, given covers $\mathcal{U}_1, \mathcal{U}_2, \dots$, the strategy “play $\mathcal{U}_n$ in round n ” is not winning, so some play against this strategy is won by Player II, and her moves witness the Menger property. The converse is the substantial direction, and we only outline it. A space with the Menger property is Lindelöf (apply the property to a constant sequence of covers), so every open cover has a countable subcover. Player II may therefore restrict herself, at each position, to a fixed countable subcover of the cover that Player I's strategy plays; then only countably many finite positions can arise, and so the strategy plays only countably many covers. Hurewicz applies the Menger property to a suitable arrangement of these countably many covers to build a play that defeats the strategy; we omit the details.

σ -compactness. If $X = \bigcup_n K_n$ with K_n compact and increasing, Player II wins by choosing, in round n , a finite subfamily of $\mathcal{U}_n$ that covers K_n . Conversely, Telgársky unravels a winning strategy of Player II, position by position, into countably many compact sets that cover X .

Menger's conjecture. By parts (1) and (2), a separable metrizable space with the Menger property that is not σ -compact is exactly one on which neither player has a winning strategy. Fremlin and Miller construct such a set of reals from combinatorial properties of families of functions in ω^ω under eventual domination; the link is a theorem of Hurewicz, which says that a subset of ω^ω has the Menger property exactly when no continuous image of the subset in ω^ω is a dominating family. The point of view of games explains both the appeal and the failure of the conjecture. The appeal is that the conjecture asserts a form of determinacy, and determinacy holds for well-behaved sets (for analytic sets of reals the conjecture is in fact true). The failure comes from wild sets, on which games need not be determined, and Fremlin and Miller found a wild enough set without any extra axioms.

Exercises.

- (1) Let X be a countable space. Show that Player II has a winning strategy in the Rothberger game on X , and that Player I has a winning strategy in the point-open game on X .
- (2) Show that a space with the Menger property is Lindelöf. Deduce that an uncountable discrete space does not have the Menger property, and describe a winning strategy for Player I in the Menger game on such a space.
- (3) Show that the Menger property is inherited by closed subspaces and preserved by continuous surjections.
- (4) Show that Player II has a winning strategy in the Menger game on $[0, 1]$, but that Player I has a winning strategy in the Rothberger game on $[0, 1]$. (Hint: in round n , let Player I cover $[0, 1]$ by relatively open intervals of length at most 2^{-n-2} ; a finite family of intervals covering $[0, 1]$ has total length at least 1.)
- (5) (*) Show that ω^ω does not have the Menger property, and hence is not σ -compact. (Hint: let $\mathcal{U}_n$ consist of the sets $\{x \in \omega^\omega : x(n) = k\}$ for $k \in \omega$.) Deduce from Theorem 14.1(1) that Player I has a winning strategy in the Menger game on ω^ω , and find one directly.

15. APPLICATIONS TO DESCRIPTIVE SET THEORY

Definitions. The space ω^ω of infinite sequences of natural numbers carries the product topology: for a finite sequence s , the basic open set $N_s = \{x \in \omega^\omega : s \subseteq x\}$ consists of the sequences that begin with s , and a set is open if it is a union of such sets. (By tradition ω^ω is called “the Baire space”; this is a name, and should not be confused with the notion of Definition 1.1, although ω^ω , being completely metrizable, is also a Baire space in that sense.) The *Borel sets* form the smallest family of subsets that contains the open sets and is closed under complements and countable unions. They are arranged in levels: the open sets, the closed sets, then the F_σ sets (countable unions of closed sets) and the G_δ sets, and so on through the countable ordinals; the level at which a Borel set first appears is its *Borel rank*. A set is *analytic* if it is a continuous image of ω^ω . Measurable and Woodin cardinals, which appear in the history below, are large cardinals, and are mentioned only for orientation.

The problem. Here is the simplest infinite game imaginable: fix a set $E \subseteq \omega^\omega$ of infinite sequences of natural numbers. In the *Gale–Stewart game* $G(E)$, Player I and Player II take turns calling out natural numbers, Player I playing $x(0), x(2), \dots$ and Player II playing $x(1), x(3), \dots$, and Player II wins if the resulting sequence x belongs to E . In the language of Definition 2.2, $G(E)$ is the game $(\omega^{<\omega}, E)$ with moves in $A = \omega$. Recall that a game is determined if one of the players has a winning strategy (see Section 2). For finite games this is always the case (that is Zermelo’s theorem, which is why chess has an optimal strategy that nobody knows). For infinite games determinacy can fail: using the axiom of choice one can build sets E for which neither player has a winning strategy. A nonprincipal ultrafilter on ω gives one: Ulam’s game $\text{Ul}(\mathcal{U})$ of Section 13 is not determined (see Theorem 13.1), and it becomes a Gale–Stewart game once the countably many non-empty finite binary strings are numbered, so that calling out a number means playing a string. Which sets, and in particular which Borel sets, give determined games? And what does the answer tell us about the structure of the Borel sets?

✂ 15.1. **History of the problem and of the solution using infinite games.** The story of this problem is unusual: the game came first, and the applications came later. Gale and Stewart introduced these games in 1953 [GS53], proved that $G(E)$ is determined when E is open or closed, constructed an undetermined game from a well-ordering of the reals, and asked how far determinacy extends. Wolfe proved in 1955 that games with F_σ or G_δ payoff sets are determined [Wol55], and Davis extended this in 1964 to the next level of the Borel hierarchy [Dav64]. In 1970 Martin proved that analytic games are determined if there is a measurable cardinal [Mar70], and in 1971 Friedman showed that any proof of Borel determinacy must use uncountably many iterations of the power set operation [Fri71], so that the problem could not be solved by the methods of ordinary analysis. That a statement about sets of real numbers should need, for its proof, sets of sets of sets of $\dots$ of reals, iterated uncountably many times, was something new.

The problem was solved by Donald A. Martin in 1975 [Mar75], who proved that every Borel game is determined. Martin’s argument may well be the most expensive proof ever written of a statement about sets of real numbers. Even before Martin’s proof, Wadge had shown in his work of 1972, written up in his thesis of 1983 [Wad83], that determinacy arranges the Borel sets in a hierarchy that is almost linearly ordered and well-founded, by comparing sets through a game of his own. The theory was later extended far beyond the Borel sets: Martin and Steel proved in 1989 that projective games are determined if there are infinitely many Woodin cardinals [MS89].

Recognition. Martin’s proof appeared in the *Annals of Mathematics*. Martin, Steel and Woodin received the Karp Prize of the Association for Symbolic Logic in 1988 for their work on determinacy, and the proof of projective determinacy by Martin and Steel appeared in the *Journal of the American Mathematical Society*.

15.2. **The games.** Besides the Gale–Stewart game $G(E)$ of the problem, we need a game for comparing two sets. In the *Wadge game* $G_W(E, F)$, Player II tries to imitate Player I: whenever his sequence lands in E , hers should land in F , and vice versa. Player I plays $x(0), x(1), \dots$ and Player II plays $y(0), y(1), \dots$, alternately, Player I first, and Player II wins if

$$x \in E \iff y \in F.$$

The play is the single sequence $(x(0), y(0), x(1), y(1), \dots)$, so this is a Gale–Stewart game, and its payoff set is Borel when E and F are.

15.3. The Main Result of this section. The main result is Martin's theorem, together with Wadge's consequence for the comparison of Borel sets. For $E, F \subseteq \omega^\omega$ write $E \leq_W F$, and say that E is *Wadge reducible* to F , if there is a continuous $f: \omega^\omega \rightarrow \omega^\omega$ with $E = f^{-1}(F)$.

Theorem 15.1 (Martin, 1975; Wadge, 1972).

- (1) Borel determinacy. *For every Borel set $E \subseteq \omega^\omega$, the game $G(E)$ is determined.*
- (2) Wadge's lemma. *For all Borel sets $E, F \subseteq \omega^\omega$, either $E \leq_W F$ or $F \leq_W \omega^\omega \setminus E$.*

By part (2), any two Borel sets are comparable under $\leq_W$, up to complements. Martin and Monk showed that $\leq_W$ is also well-founded on the Borel sets. So the Borel sets, up to Wadge equivalence, form a well-founded hierarchy that refines the classical Borel hierarchy, and that is almost linear: two sets that are not comparable under $\leq_W$ are, up to Wadge equivalence, a set and its complement, so that antichains have at most two elements. Each level of the hierarchy is strictly more complex than the levels below it.

15.4. How the games solve the problem.

Open games. The idea is simply “never step into a position from which you are already lost”, and the openness of E guarantees that this cautious play is enough. Suppose that E is open and that Player II has no winning strategy in $G(E)$. We shall call a finite position *good for Player I* if Player II has no winning strategy from that position. If a position with Player I to move is good for him, some move keeps the position good, for otherwise Player II would have a winning strategy from every successor position and hence from the position itself; if a position with Player II to move is good for Player I, so is every position reached in one move. Player I can therefore play forever through good positions. If the resulting x were in E , then, since E is open, some initial segment of x would already force the play into E , and Player II would win trivially from that position, which would not be good for Player I. Hence $x \notin E$, and Player I has a winning strategy.

Borel games. Martin's proof reduces every Borel game to an open game on a larger tree. Roughly, for each Borel set E he builds a game G' whose moves are more complicated objects than natural numbers, together with a map from plays of G' to plays of $G(E)$ that sends strategies to strategies, such that the payoff set of G' is clopen and corresponds to E . Since G' is determined by the Gale–Stewart theorem, so is $G(E)$. The construction proceeds by induction on the Borel rank of E , and at each step the moves of the new game are sets of positions of the old one; this is why Martin's proof climbs through the iterated power sets of $\mathbb{R}$, in agreement with Friedman's theorem.

Wadge's lemma. By Borel determinacy, one of the players has a winning strategy in $G_W(E, F)$. If Player II has one, her strategy is literally a blueprint for a continuous function: Player I feeds in an input sequence x one digit at a time, and Player II's step-by-step answers are the output y . Since each output digit depends only on finitely many input digits, the map $x \mapsto y$ is continuous, and $x \in E$ if and only if $y \in F$, so $E \leq_W F$. If Player I has one, his strategy turns each sequence y of moves of Player II into a sequence x continuously, with $y \in F$ if and only if $x \notin E$, so $F \leq_W \omega^\omega \setminus E$. The game thus produces the reduction itself, and the comparability of Borel sets is literally the determinacy of a game.

Exercises.

- (1) For each of the following sets $E \subseteq \omega^\omega$, decide which player has a winning strategy in $G(E)$, and describe one: (a) the sequences with $x(1) > x(0)$; (b) the sequences with

- $x(1) < x(0)$; (c) the sequences for which $\{x(2n) : n \in \omega\}$ is finite; (d) the sequences with $x(2n+1) = x(0)$ for infinitely many n .
- (2) Show that $\leq_W$ is reflexive and transitive, and that $E \leq_W F$ implies $\omega^\omega \setminus E \leq_W \omega^\omega \setminus F$.
 - (3) Let $E \subseteq \omega^\omega$ be closed. Show that $G(E)$ is determined, by adapting the argument for open games with the roles of the players exchanged. (Hint: call a position good for Player II if Player I has no winning strategy from it.)
 - (4) Let E be the set of $x \in \omega^\omega$ with $x(n) = 0$ for some n , and let $F = N_{(0)}$ be the set of sequences beginning with 0. Show that Player I has a winning strategy in $G_W(E, F)$, and that E is not Wadge reducible to F . Find a continuous f with $F = f^{-1}(\omega^\omega \setminus E)$, as Theorem 15.1(2) predicts.
 - (5) (*) Let $E \subseteq \omega^\omega$ be open but not closed. Show that $F \leq_W E$ for every open set $F \subseteq \omega^\omega$. (Hint: fix $z \in \overline{E} \setminus E$; in $G_W(F, E)$, let Player II copy z until an initial segment p of Player I's sequence satisfies $N_p \subseteq F$.)

Part IV. Applications of non-topological games

The games in this part are not played with open sets at all: they are finite games on boards, or games played with elements of two structures. They are included because they show the same two features of games (local decisions, combinable strategies) at work far from topology.

16. APPLICATIONS TO THE PROBABILISTIC METHOD

The problem. The probabilistic method proves that combinatorial objects exist by showing that a random object has the required property with positive probability. For example, Erdős proved in 1963 that if a family $\mathcal{F}$ of n -element sets has fewer than 2^{n-1} members, then the elements can be coloured red and blue so that no member of $\mathcal{F}$ is monochromatic: a random colouring makes a given set monochromatic with probability 2^{1-n} , so the expected number of monochromatic sets is less than 1. Such proofs give no construction. The problem is to turn them into explicit, deterministic, efficient constructions.

- ✂ 16.1. **History of the problem and of the solution using combinatorial games.** The probabilistic method goes back to a 1943 paper of Szele on tournaments, and was made famous by Erdős in 1947 through his lower bound for diagonal Ramsey numbers [Erd47]. The colouring theorem above is from 1963 [Erd63]. In 1973 Erdős and Selfridge studied Maker–Breaker games, in which the players alternately claim elements of a board and Maker tries to claim all the elements of some winning set [ES73]. They gave a strategy for Breaker based on a potential function equal to the expected number of winning sets that Maker would complete if the rest of the game were played at random, and showed that Breaker can keep this expectation from growing. The same argument solves the problem for the colouring theorem and related results: the same potential argument, with one player instead of two, colours the elements one at a time and never lets the expected number of monochromatic sets reach 1.

The argument was recognized as a general derandomization technique, the method of conditional probabilities, systematized by Spencer in 1987 [Spe87] and extended by Raghavan's pessimistic estimators in 1988 [Rag88]. Beck built on the Erdős–Selfridge theorem a theory of positional games, presented in his book of 2008 [Bec08].

Recognition. Erdős received the Wolf Prize in 1983/84.

16.2. The game. Let $\mathcal{F}$ be a family of subsets of a finite set V . In the *Maker–Breaker game* on $(V, \mathcal{F})$, Maker is Player I and Breaker is Player II. Tic-tac-toe is the example to keep in mind: the set V is the board and $\mathcal{F}$ is the set of lines. The players alternately claim unclaimed elements of V , Player I first, until all elements are claimed. Player I wins if he has claimed all the elements of some $A \in \mathcal{F}$, and Player II wins otherwise.

16.3. The Main Result of this section. The main result is the Erdős–Selfridge theorem, together with two classical consequences: a derandomized colouring and a result on multidimensional tic-tac-toe.

Theorem 16.1 (Erdős–Selfridge, 1973). *Let $\mathcal{F}$ be a family of subsets of a finite set V .*

- (1) *If $\sum_{A \in \mathcal{F}} 2^{-|A|} < \frac{1}{2}$, then in the Maker–Breaker game on $(V, \mathcal{F})$ in which Maker moves first, Breaker has a winning strategy.*
- (2) *If $\sum_{A \in \mathcal{F}} 2^{1-|A|} < 1$, then a colouring of V in two colours with no monochromatic member of $\mathcal{F}$ can be computed in time polynomial in $|V|$ and $|\mathcal{F}|$.*
- (3) *In tic-tac-toe on the board $\{1, \dots, n\}^d$, the first player cannot force a win when $\frac{1}{2}((n+2)^d - n^d) < 2^{n-1}$.*

Part (3) follows from part (1) because the board has $\frac{1}{2}((n+2)^d - n^d)$ lines of n cells each: the second player uses Breaker’s strategy and occupies a cell of every line. By the Hales–Jewett theorem (1963), on the other hand, the first player wins when d is large compared with n [HJ63].

16.4. How the game solves the problem. The proof turns the expectation of the problem into a potential function that Player II never lets grow. At any stage we call $A \in \mathcal{F}$ *alive* if Player II has claimed none of the elements of A , let $u(A)$ be the number of unclaimed elements of A , and set

$$\Phi = \sum_{A \text{ alive}} 2^{-u(A)}, \quad w(v) = \sum_{A \text{ alive}, v \in A} 2^{-u(A)}.$$

If the remaining elements were assigned to the two players by fair coin tosses, the quantity Φ would be the expected number of sets completed by Player I. Initially $\Phi < \frac{1}{2}$, and after Player I’s first move $\Phi < 1$. A claim of v by Player I increases Φ by $w(v)$, and a claim by Player II decreases Φ by $w(v)$. Player II’s strategy is to claim, at each move, an element of maximal weight. Player I’s next claim increases Φ by at most that maximal weight, because weights only decrease when Player II moves, so Φ never increases over a pair of moves and stays below 1. If Player I completed a set A , the term of A in Φ would be $2^0 = 1$, which is impossible. For part (2) one player colours the elements one at a time, choosing at each step the colour that does not increase the expected number of monochromatic sets; the expectation starts below 1, never increases, and at the end is an integer, hence 0. This is not just an existence proof: the argument is an explicit polynomial-time algorithm, short enough to write in a few lines of Python.

The point of view of games adds robustness. A probabilistic argument shows that a random play is good on average; the game shows that a good outcome can be forced one move at a time, even against an adversary who chooses half of the moves. This is what makes the argument both constructive and efficient.

Exercises.

- (1) Check the bookkeeping in the proof of Theorem 16.1(1): a claim of v by Player I increases Φ by exactly $w(v)$, a claim of v by Player II decreases Φ by exactly $w(v)$, and, under the hypothesis of part (1), we have $\Phi < 1$ after the first move of Player I.
- (2) Show that the board $\{1, \dots, n\}^2$ has $2n + 2 = \frac{1}{2}((n+2)^2 - n^2)$ lines, and find the least n for which Theorem 16.1(3) applies with $d = 2$. Do the same for $d = 3$, where the number of lines is $3n^2 + 6n + 4$.
- (3) Consider the Maker–Breaker game on the 3×3 tic-tac-toe board, with $\mathcal{F}$ the set of its 8 lines. Show that $\sum_{A \in \mathcal{F}} 2^{-|A|} = 1$, so that Theorem 16.1(1) does not apply, and show that Maker has a winning strategy. (Hint: Maker takes the center and then creates two threats at once.) Why does this not contradict the fact that ordinary tic-tac-toe is a draw?
- (4) Let $\mathcal{F}$ be a family of n -element subsets of a finite set V with fewer than 2^{n-1} members. Explain why, in the proof of Theorem 16.1(2), one of the two colors for the next element never increases the conditional expected number of monochromatic members of $\mathcal{F}$.
- (5) (*) Let V be the set of vertices of a complete binary tree with n levels, and let $\mathcal{F}$ be the family of 2^{n-1} paths from the root to a leaf. Show that $\sum_{A \in \mathcal{F}} 2^{-|A|} = \frac{1}{2}$ and that Maker has a winning strategy. Thus the strict inequality in Theorem 16.1(1) cannot be weakened.

17. APPLICATIONS TO FIXED POINT THEORY

The problem. The *Brouwer fixed point theorem* says that every continuous map $f: [0, 1]^n \rightarrow [0, 1]^n$ has a fixed point. The original proofs used algebraic topology. The problem is to find an elementary, combinatorial proof that exposes the finite core of the theorem and, ideally, yields a method for locating approximate fixed points.

✂ 17.1. **History of the problem and of the solution using combinatorial games.** The theorem was proved in general by Hadamard in 1910 and Brouwer in 1912 [Bro12], by methods of the new algebraic topology. A combinatorial proof was given by Sperner in 1928 [Spe28], through his lemma on labelled triangulations, and Scarf turned this line of thought into an algorithm for computing approximate fixed points in 1967 [Sca67]. Meanwhile the game of Hex was invented by Piet Hein in 1942 and, independently, by John Nash at Princeton in 1948. Nash observed that the game cannot end in a draw and, by a strategy-stealing argument, that the first player has a winning strategy.

The game-theoretic solution was found by David Gale in 1979 [Gal79], who proved that the Brouwer fixed point theorem is an easy consequence of the fact that Hex cannot end in a draw, and extended both statements to n dimensions.

Recognition. Gale's paper, in the *American Mathematical Monthly*, received a Lester R. Ford Award of the Mathematical Association of America in 1980, the year in which Gale shared the John von Neumann Theory Prize. Nash received the Nobel Memorial Prize in Economic Sciences in 1994 and the Abel Prize in 2015.

17.2. **The game.** Hex is played on a rhombus-shaped board of hexagonal cells, and anyone who has played Hex knows the feeling that somebody *must* win: if you have blocked your opponent completely, you seem to have connected your own sides. To make this precise, fix $k \geq 1$ and describe the board by the centres of its cells, which are the points (i, j) with $1 \leq i, j \leq k$, and (i, j) is adjacent to $(i \pm 1, j)$, $(i, j \pm 1)$, $(i + 1, j + 1)$ and $(i - 1, j - 1)$.

Player I owns the bottom and top rows (where $j = 1$ and $j = k$), and Player II the left and right columns (where $i = 1$ and $i = k$). The players alternately place stones on empty cells, Player I first. Player I wins if his stones contain a chain of adjacent cells from bottom to top, and Player II wins if hers contain one from left to right.

17.3. The Main Result of this section. The main result is the Hex theorem and, as a consequence, the fixed point theorem in dimension two.

Theorem 17.1 (Gale, 1979).

- (1) The Hex theorem. *If every cell of a $k \times k$ Hex board is occupied by a stone of Player I or of Player II, then exactly one of the players has a chain of his or her stones joining his or her two sides of the board.*
- (2) *The Hex theorem implies the Brouwer fixed point theorem for $[0, 1]^2$.*

Gale proved the corresponding statements for an n -dimensional version of Hex and $[0, 1]^n$.

17.4. How the game solves the problem.

No draws. Gale follows the boundary between the two colours. Add an extra row of Player I's stones above and below the board, and an extra column of Player II's on either side. Start at a corner of the enlarged board, between the two colours, and walk along the edges of the hexagons that separate a stone of Player I from one of Player II. At each vertex three cells meet, two of one colour and one of the other, so the walk continues in exactly one way and never returns to a vertex. Since the board is finite, the walk must leave the board at another corner, and the stones of one colour along one side of the walk form a winning chain. The two players cannot both win, because a top-to-bottom chain and a left-to-right chain would have to cross, and on the hexagonal board crossing chains share a cell. Figure 14 shows the walk on a full board.

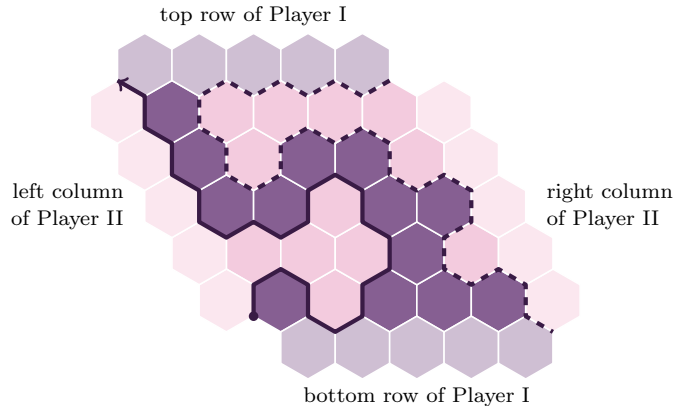


FIGURE 14. The no-draw argument on a full 5×5 board (see Theorem 17.1). The stones of Player I are plum and those of Player II are pink; the paler cells are the extra rows and columns. The walk starts at the lower left corner, between the two colours, and follows the edges that separate plum from pink. It cannot stop or turn back, so it leaves the board at another corner, and the plum stones along its right side form a chain from bottom to top. The dashed line is the walk that starts at the lower right corner.

Brouwer's theorem. Suppose that $f = (f_1, f_2): [0, 1]^2 \rightarrow [0, 1]^2$ has no fixed point. By compactness there is $\varepsilon > 0$ with $\max_i |f_i(z) - z_i| \geq \varepsilon$ for all z , and by uniform continuity there is $\delta < \varepsilon$ with $|f_i(z) - f_i(z')| < \varepsilon$ whenever $|z - z'|_\infty \leq \delta$. Choose k with $1/(k-1) \leq \delta$, and place at each cell (i, j) the point $z = ((i-1)/(k-1), (j-1)/(k-1))$. Give the cell to Player II if $|f_1(z) - z_1| \geq \varepsilon$, and to Player I otherwise, in which case $|f_2(z) - z_2| \geq \varepsilon$. By the Hex theorem one of the players has a winning chain. Say Player II has a chain from left to right. On the left side $z_1 = 0 \leq f_1(z)$, so there $f_1(z) - z_1 \geq \varepsilon$; on the right side $z_1 = 1 \geq f_1(z)$, so there $z_1 - f_1(z) \geq \varepsilon$. Along the chain the sign changes, so two adjacent cells z, z' satisfy $f_1(z) - z_1 \geq \varepsilon$ and $z'_1 - f_1(z') \geq \varepsilon$. Adding these inequalities, we obtain $(f_1(z) - f_1(z')) + (z'_1 - z_1) \geq 2\varepsilon$, which is impossible because both terms are less than ε . A chain of Player I leads to the same contradiction with f_2 .

The point of view of games is that the topological content of the theorem is the statement that a game cannot end in a draw. Read in the contrapositive, the argument shows that for every $\varepsilon > 0$ some point z of a sufficiently fine grid satisfies $|f(z) - z|_\infty < \varepsilon$, and the fixed point is obtained by compactness. This finite form is the one in which combinatorial proofs of fixed point theorems, from Sperner's lemma to Scarf's algorithm, are used in computation.

Exercises.

- (1) List the adjacent pairs of cells of the 2×2 Hex board, and check directly that in each of the 16 ways of filling the board exactly one player has a winning chain, as Theorem 17.1(1) asserts.
- (2) Show that on the 3×3 board Player I has a winning strategy. (Hint: open with the center cell $(2, 2)$; its neighbors $(1, 1)$ and $(2, 1)$ lie in the bottom row and $(2, 3)$ and $(3, 3)$ in the top row, and Player I answers a move of Player II in one of these pairs with the other cell of the pair.)
- (3) In the proof of Theorem 17.1(2), check that the points z and z' placed at adjacent cells satisfy $|z - z'|_\infty \leq \delta$, and carry out the argument for a winning chain of Player I, using f_2 .
- (4) (*) Show that Player I has a winning strategy in Hex on the $k \times k$ board for every $k \geq 1$. (Hint: the reflection $(i, j) \mapsto (j, i)$ preserves adjacency and exchanges the sides of the two players; combine Theorem 17.1(1), the determinacy of finite games, and the fact that an extra stone never hurts its owner, to show that Player II cannot have a winning strategy.)

18. APPLICATIONS TO MODEL THEORY

The problem. Two structures $\mathcal{A}$ and $\mathcal{B}$ in the same signature are *elementarily equivalent*, written $\mathcal{A} \equiv \mathcal{B}$, if they satisfy the same first-order sentences. The problem is to find a method, not requiring the inspection of all sentences, for deciding whether two given structures are elementarily equivalent, and, dually, for proving that a given property of structures cannot be expressed by a first-order sentence.

- ✂ **18.1. History of the problem and of the solution using combinatorial games.** Elementary equivalence was introduced by Tarski, whose programme of the 1930s to 1950s of classifying the complete first-order theories of familiar structures required tools for proving that two structures satisfy the same sentences. The back-and-forth method, which goes back to Cantor's theorem of 1895 that any two countable dense linear orders without endpoints are isomorphic, suggested the answer.

The problem was solved by Roland Fraïssé in 1954 [Fra54], who characterized elementary equivalence by systems of partial isomorphisms, and the solution was recast as a game by Andrzej Ehrenfeucht in 1961 [Ehr61], who used the game to prove completeness and decidability of theories. In finite model theory, which developed from the 1970s onward in connection with databases and computational complexity, these games became the principal method for proving that properties are not definable in a given logic; Fagin, for instance, used games of this kind in 1975 to show that connectivity of finite graphs is not expressible in monadic existential second-order logic [Fag75].

18.2. The game. Let $\mathcal{A}$ and $\mathcal{B}$ be structures in the same finite relational signature, and let $n \geq 0$ be given. The game $\text{EF}_n(\mathcal{A}, \mathcal{B})$ has n rounds. The names used in the literature describe the game well: Player I is the *Spoiler*, who wants to expose a difference between the two structures, and Player II is the *Duplicator*, who wants to show that they look alike. In round k , Player I chooses one of the two structures and an element of that structure, and Player II answers with an element of the other structure. After n rounds the players have produced $a_1, \dots, a_n \in \mathcal{A}$ and $b_1, \dots, b_n \in \mathcal{B}$, and Player II wins if the map $a_k \mapsto b_k$ is a partial isomorphism, that is, if the a_k and the b_k satisfy the same atomic formulas.

18.3. The Main Result of this section. The main result is the theorem of Fraïssé and Ehrenfeucht. Write $\mathcal{A} \equiv_n \mathcal{B}$ if $\mathcal{A}$ and $\mathcal{B}$ satisfy the same first-order sentences of quantifier rank at most n .

Theorem 18.1 (Fraïssé, 1954; Ehrenfeucht, 1961). *Let $\mathcal{A}$ and $\mathcal{B}$ be structures in a finite relational signature, and let $n \geq 0$ be given. Then $\mathcal{A} \equiv_n \mathcal{B}$ if and only if Player II has a winning strategy in the game $\text{EF}_n(\mathcal{A}, \mathcal{B})$ of the preceding subsection. Consequently, we have $\mathcal{A} \equiv \mathcal{B}$ if and only if Player II has a winning strategy in $\text{EF}_n(\mathcal{A}, \mathcal{B})$ for every n .*

Two standard applications show how the theorem is used: first, the orders $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are elementarily equivalent. Second, if L_m denotes the linear order with m elements, then $L_m \equiv_n L_{m'}$ whenever $m, m' \geq 2^n$; hence no first-order sentence is true exactly in the finite linear orders of even size. Hence connectivity of finite graphs is not first-order expressible either. Indeed, on a finite linear order $1 < 2 < \dots < m$ with $m \geq 3$, join each i to $i + 2$, and also join m to 2 and $m - 1$ to 1. This graph is first-order definable from the order, and it is connected exactly when m is odd: the odd and the even points form two paths, and for odd m the edge between m and 2 joins them, while for even m both extra edges stay inside one path. So a sentence expressing connectivity would yield one expressing that m is odd.

18.4. How the game solves the problem.

The theorem. The proof of Theorem 18.1 is by induction on n , and the game is a direct picture of the quantifiers. The induction needs a slightly stronger statement, for structures with finitely many named elements: for tuples $\bar{a}$ in $\mathcal{A}$ and $\bar{b}$ in $\mathcal{B}$ of the same length, the tuples satisfy the same formulas of quantifier rank at most n if and only if Player II wins the n -round game that starts from the position in which $\bar{a}$ and $\bar{b}$ have already been played. The case of empty tuples is the theorem. A formula of quantifier rank n about the named tuple is a Boolean combination of atomic formulas and formulas $\exists x \psi(\bar{a}, x)$ with ψ of quantifier rank $n - 1$. If $\mathcal{A} \models \exists x \psi(\bar{a}, x)$ but $\mathcal{B} \not\models \exists x \psi(\bar{b}, x)$, Player I plays a witness $a \in \mathcal{A}$, and whatever Player II answers in $\mathcal{B}$ fails ψ ; by the induction hypothesis, applied to the tuples extended by a and by Player II's answer, Player I wins the remaining $n - 1$ rounds. Conversely, in a

finite relational signature there are only finitely many formulas of each quantifier rank up to equivalence, so the rank- $(n - 1)$ type of an element over the named tuple is expressed by a single formula, and Player II answers each move with an element of the same type over the other tuple. A statement about all first-order sentences is thus replaced by a finite combinatorial task: finding a strategy.

The applications. In $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$, whatever element Player I chooses, Player II can choose an element in the same position relative to the elements already played, because both orders are dense and have no endpoints; this back-and-forth argument wins every EF_n . For finite linear orders, Player II keeps the following invariant after round k : any two consecutive chosen elements, including two imaginary elements placed just below the least and just above the greatest element, are at equal distances in the two orders, or at distances that are both at least 2^{n-k} . The intuition is that two elements at distance at least 2^{n-k} look infinitely far apart to a formula that has only $n - k$ quantifiers left to inspect them, since each quantifier can at best halve the distance. When Player I plays inside a long gap, Player II can split the corresponding gap so that the invariant persists, and the invariant guarantees that the final map preserves order. So a property that separates L_m from $L_{m'}$ with $m, m' \geq 2^n$ needs quantifier rank greater than n ; since the sizes of finite orders are unbounded, parity needs unbounded quantifier rank, and parity is not first-order expressible.

18.5. Further remarks (optional).

Ultrapowers. Ultrafilters give a second answer to the problem, algebraic where Theorem 18.1 is combinatorial: Keisler proved in 1961, assuming the generalized continuum hypothesis, and Shelah proved in ZFC in 1971, that two structures are elementarily equivalent if and only if they have isomorphic ultrapowers [Kei61, She71]. One direction is Łoś's theorem, which embeds every structure elementarily in each of its ultrapowers. For the other, Keisler lets λ be an infinite cardinal at least as large as the structures and the signature, and takes a countably incomplete λ^+ -good ultrafilter $\mathcal{D}$ on λ . Such ultrafilters exist under GCH by Keisler's theorem of 1964, and in ZFC by Kunen's theorem of 1972 [Kun72], whose proof manufactures them from an independent family of 2^λ subsets of λ . The ultrapowers $\mathcal{A}^\lambda/\mathcal{D}$ and $\mathcal{B}^\lambda/\mathcal{D}$ of elementarily equivalent infinite structures $\mathcal{A}$ and $\mathcal{B}$ are then λ^+ -saturated and have cardinality 2^λ , which equals λ^+ under GCH. Now the game comes back, played for λ^+ rounds instead of n . In the Ehrenfeucht–Fraïssé game of length λ^+ between the two ultrapowers, Player II answers each move by realizing, in the other structure, the type of the new element over the fewer than λ^+ elements already chosen; saturation guarantees that she can, and if both players enumerate their structures in turn, the play is an isomorphism. Shelah removed GCH by building the ultrafilter and the isomorphism together, in a single transfinite construction of length 2^λ .

Exercises.

- (1) Show that $L_m \equiv_1 L_{m'}$ for all $m, m' \geq 1$, by describing a winning strategy for Player II in $\text{EF}_1(L_m, L_{m'})$. Show that Player I wins $\text{EF}_2(L_1, L_2)$, and find a sentence of quantifier rank 2 true in L_2 and false in L_1 .
- (2) Show that Player I wins $\text{EF}_2(L_2, L_3)$, but that Player II wins $\text{EF}_2(L_3, L_4)$. Thus $L_3 \equiv_2 L_4$, although $3 < 2^2$.
- (3) Show that $(\mathbb{Z}, <) \equiv_2 (\mathbb{Q}, <)$, but that Player I wins $\text{EF}_3((\mathbb{Z}, <), (\mathbb{Q}, <))$. Find a sentence of quantifier rank 3 that distinguishes the two orders.
- (4) Draw the graph built from the linear order $1 < 2 < \dots < m$ in the application of Theorem 18.1 for $m = 5, 6$ and 7 , and check that it is connected exactly when m

is odd. Write down a first-order formula in the signature $\{<\}$ that defines its edge relation.

- (5) (*) Let $\mathbb{Z} + \mathbb{Z}$ denote the order obtained by placing one copy of $(\mathbb{Z}, <)$ after another. Show that $(\mathbb{Z}, <) \equiv (\mathbb{Z} + \mathbb{Z}, <)$. (Hint: adapt the invariant used for finite linear orders, regarding two elements in different copies as being infinitely far apart.)

Part V. The Scottish Book, von Neumann, Ulam, Texas, Toronto, and the triumph of games, logic, topology and combinatorics

This epilogue is about the people behind the mathematics of these notes: where they worked, what happened to them, and how their problems travelled. Many of them were very young. When the Scottish Book was started, in the summer of 1935, Stanisław Ulam was twenty-six, Stanisław Mazur was thirty, Mark Kac was twenty, and Stefan Banach, who led the discussions at the café table, was forty-three. Figure 15 shows the journey of the Scottish Book and of the people who carried it, and Figure 16 is a timeline to keep at hand while reading.

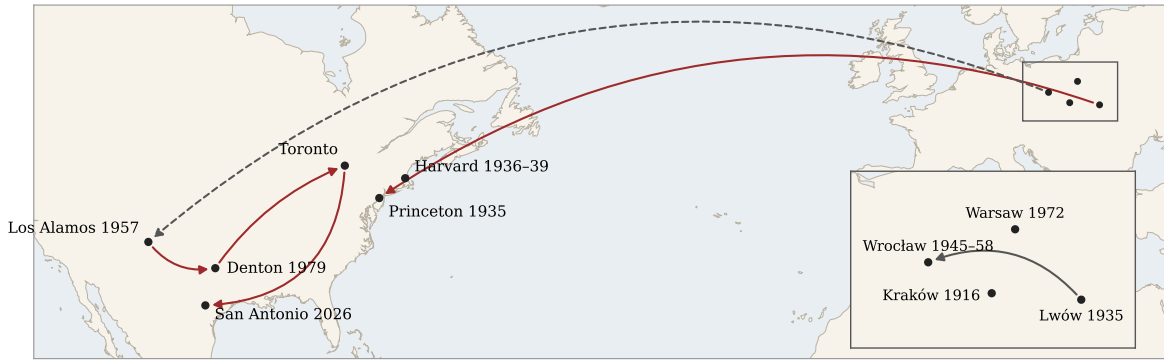


FIGURE 15. The journey of the Scottish Book. Red: Ulam's crossing in 1935, and the road from Los Alamos to Denton, Toronto and San Antonio. Dashed: the handwritten copy that Steinhaus sent from Wrocław to Ulam in 1956. The inset shows Poland: Kraków, where Steinhaus met Banach in 1916; Lwów, home of the Scottish Café; Wrocław, home of the New Scottish Book after the war; and Warsaw, where Enflo received his goose in 1972.

19. THE MARBLE TABLES OF THE SCOTTISH CAFÉ

In the interwar years, the city of Lwów (then in southeastern Poland, now Lviv, Ukraine) was home to one of the most vibrant mathematical communities in history. Led by Stefan Banach and Hugo Steinhaus, with Stanisław Mazur, Juliusz Schauder, Stanisław Ulam, Mark Kac and many others, the Lwów School did much to create modern functional analysis. Through Mazur's Problem 43 it also produced the first infinite game in topology.

The school began with a chance meeting. One evening in 1916, walking in the Planty, the park around the old town of Kraków, Steinhaus overheard the words "Lebesgue integral" and stopped to talk to the two young men who had said them. They were Stefan Banach and Otto Nikodym. Banach, then twenty-four, had studied engineering in Lwów without finishing his degree, and was teaching himself mathematics. Steinhaus told him about a problem he was working on; a few days later Banach brought him the solution, and their first joint paper

- 1916 Steinhaus meets Banach and Nikodym in the Planty park, Kraków.
- 1920 Banach's doctorate in Lwów, without a university degree.
- 1930 Ulam on measures defined on all subsets of a set (measurable cardinals).
- 1932 Banach's *Théorie des opérations linéaires*.
- 1935 17 July: Problem 1, by Banach. July: Mazur's Problem 43; 4 August: Banach's answer. End of the year: Ulam goes to Princeton.
- 1936 6 November: Mazur's Problem 153, the goose problem.
- 1939 20 August: Ulam leaves Poland for the last time. September: war.
- 1941 31 May: last problem, number 193, by Steinhaus. July: the Lwów professors are murdered.
- 1945 31 August: Banach dies in Lwów. Steinhaus moves to Wrocław.
- 1946 2 July: first entry in the New Scottish Book, by Steinhaus.
- 1946 Ulam's idea of the Monte Carlo method.
- 1957 Ulam's English translation, mimeographed at Los Alamos; Oxtoby publishes the first proof of Banach's solution of Problem 43.
- 1962 Mycielski and Steinhaus propose the Axiom of Determinacy.
- 1972 Mazur gives Enflo a live goose in Warsaw.
- 1975 Martin proves Borel determinacy.
- 1979 The Scottish Book Conference in Denton, organized by Mauldin.
- 1981 *The Scottish Book*, edited by Mauldin (second edition 2015).
- 1994–98 Distortion problem (Odell–Schlumprecht); Gowers's dichotomy; Fields Medal for Gowers.
- 2026 The UTSA–Toronto seminar.

FIGURE 16. A timeline of the Scottish Book.

followed. In 1920, with Steinhaus's support, Banach was allowed to submit a doctoral thesis in Lwów without a university degree; the thesis, published in 1922, contains what we now call Banach spaces. Steinhaus, who made many contributions to mathematics, later said that Banach was his greatest scientific discovery. Fifteen years later Banach opened the Scottish Book with Problem 1, on when a metric space can be remetrized so as to become complete and compact, and within its first weeks he answered Mazur's Problem 43.

Their real seminar room was the *Kawiarnia Szkocka*, the Scottish Café, on Plac Akademicki (see Figure 17). The mathematicians spent hours at its tables drinking coffee and writing formulas, as the stories go, directly on the marble tabletops, which were wiped clean when the café closed. To save at least the problems, Banach's wife, Lucja Banachowa, provided a large notebook, which was kept by the headwaiter and brought to the table on request. Ulam recalled one session with Mazur and Banach that lasted seventeen hours, interrupted only for meals [Ula76]. He also recorded Banach's view of what a mathematician does: "Good mathematicians see analogies between theorems or theories, the very best ones see analogies between analogies." Thus was born the *Scottish Book* (*Księga Szkocka*); its first problem, by Banach, is dated 17 July 1935. Problems came with prizes: a cup of coffee, a small beer, five beers, a bottle of wine, 100 grams of caviar, a dinner at the Hotel George, a fondue in Geneva, a bottle of whisky "of measure > 0 " (von Neumann, 1937), a kilo of bacon (Saks, 1940), and a live goose (see Section 25).

20. PROBLEM 43

Mazur's Problem 43, the subject of these notes, was written in July 1935, a few days after the book was started. The Introduction tells what it asks and how Banach answered it



FIGURE 17. Plac Akademicki in Lwów, the square of the Scottish Café, on a hand-coloured postcard of 1909 (Kraków: Wydawnictwo Salonu Malarzy Polskich). National Library of Poland (Polona), public domain.

within a month (see Figure 1), and Section 6 tells how Oxtoby’s proof turned the answer into a theorem. Ulam proposed a variant in which the players choose binary digits; this “Ulam game” (see Section 13) later led Jan Mycielski and Hugo Steinhaus, the founder of the Lwów School, to propose in 1962 a new axiom for set theory, the *Axiom of Determinacy* [MS62]. A question about the size of sets of real numbers, asked over coffee, had become a question in the foundations of mathematics.

21. VON NEUMANN’S INVITATION AND THE BRIDGE TO AMERICA

John von Neumann met Ulam in Warsaw in 1935 and invited him to spend a few months at the Institute for Advanced Study in Princeton. Ulam, then twenty-six, crossed the Atlantic at the end of 1935.

Ulam was the most prolific author of the *Scottish Book*: about forty problems are his alone, eleven more are joint with Banach and Mazur, and some fifteen more are joint with others. His problems have had a long life. Problem 19 asks whether a solid of uniform density that floats in water in equilibrium in every position must be a ball; in 2022 Dmitry Ryabogin showed that the answer is no: there is a convex body of revolution of density $1/2$, not a ball, with this property [Rya22]. His work of the same years on measures defined on all subsets of a set led to measurable cardinals (see Section 28 below).

At a tea in von Neumann’s house, G. D. Birkhoff asked him whether he would like to join the Society of Fellows at Harvard, and for the next few years Ulam divided his time between Harvard and Lwów, returning to Poland each summer and to the tables of the Scottish Café.

On 20 August 1939 he sailed for the United States for the last time, taking with him his seventeen-year-old brother Adam. Twelve days later Germany invaded Poland, and on 17 September the Soviet Union followed. Ulam's father Józef, his sister Stefania and much of his family were killed in the Holocaust.

22. WAR, AND THE SURVIVAL OF THE BOOK

The war destroyed the Lwów School. From September 1939 the city was under Soviet occupation, and the café meetings went on; Saks's kilo of bacon dates from 1940. German troops entered Lwów on 30 June 1941. In the night of 3–4 July, in a planned action against the Polish intelligentsia, they arrested professors of the city's universities together with members of their families, and shot them on the Wuleckie Hills: about twenty-five professors and as many relatives died. Among them were the mathematicians Antoni Łomnicki and Włodzimierz Stożek, together with two of Stożek's sons. Kazimierz Bartel, geometer and former prime minister of Poland, was arrested on 2 July and killed later that month.

The killing did not stop there. Stanisław Saks was murdered in Warsaw in 1942. Herman Auerbach took poison in 1942 rather than be deported to a death camp. Juliusz Schauder, the author of the Schauder fixed point theorem, was murdered in 1943. Mark Kac had left for the United States in 1938; his parents and brother were killed in 1942. Steinhaus survived by hiding in the countryside under a false name.

Banach survived because of Rudolf Weigl, a biologist who had developed a vaccine against typhus. The vaccine was grown in lice, and the lice had to be fed on human blood: "louse feeders" let cages of lice feed on their legs. The German army needed the vaccine, so the papers of Weigl's institute protected its workers, and Weigl used this to employ mathematicians, scientists, members of the underground and Jews. Banach and his son were louse feeders, as were several other members of the school. Weigl also smuggled vaccine into the Warsaw and Lwów ghettos. In 2003 Yad Vashem recognized him as Righteous Among the Nations. Banach lived to see the end of the war; he died of lung cancer in Lwów on 31 August 1945.

The book survived too. Ulam recalled that Mazur had once proposed, in case the city was bombed, to put the manuscripts and the Scottish Book in a case and bury it near a goal post of a football field outside the city; whether this was done is not known. What is known is that after Banach's death his son, Stefan Banach Jr., found the book and showed it to Steinhaus, who had moved to Wrocław. The last entry, Problem 193, had been written by Steinhaus on 31 May 1941.

23. ULAM AFTER LWÓW

The mathematics of the café went on in Ulam's life (see Figure 18) in unexpected ways. At the end of 1943, after asking von Neumann for war work, he received an invitation from Hans Bethe and joined the Manhattan Project at Los Alamos. In 1946, recovering from a serious illness and playing solitaire, Ulam wondered what the chance of winning a game was, and saw that it was easier to estimate it by playing many random games than by counting. With von Neumann he turned this idea into the *Monte Carlo method*, named by their colleague Nicholas Metropolis after the casino [MU49]; it is now among the most used methods in science. With Edward Teller he took part in the design of the hydrogen bomb. In 1955, with Enrico Fermi and John Pasta, he studied on the MANIAC computer a chain of oscillators that, unexpectedly, did not reach thermal equilibrium [FPU55]. The programs were written by Mary Tsingou, whose role was acknowledged only decades later, when the problem became known as the Fermi–Pasta–Ulam–Tsingou problem [Dau08].

24. THE 1957 LOS ALAMOS MIMEOGRAPH

Steinhaus copied the book by hand and in 1956 sent the copy to Ulam, who was by then at the Los Alamos Scientific Laboratory in New Mexico. Ulam translated it into English and, in his words, “had some three hundred mimeographed copies of the translation made. I had to pay for this myself—Los Alamos is a government laboratory, and one cannot use taxpayers’ money for such frivolous purposes.” The copies circulated among mathematicians around the world, and for a generation they were the way most people knew the problems.

25. PROBLEM 153: A LIVE GOOSE

The best known prize of the Scottish Book was paid thirty-six years late. Mazur’s Problem 153 of 6 November 1936 asked a question about approximating continuous functions, and offered a live goose. The problem turned out to be equivalent to the approximation problem formulated by Grothendieck in 1955: does every compact operator on a Banach space lie in the closure of the finite-rank operators? A negative answer would also answer Banach’s basis problem. In 1972 the young Swedish mathematician Per Enflo constructed a separable Banach space without the approximation property, and therefore without a Schauder basis [Enf73]. Mazur, then in Warsaw, kept his word: in 1972, in a ceremony at the Stefan Banach Center, he presented Enflo, then twenty-eight, with a live goose; according to several accounts, the ceremony was broadcast on Polish television.

Enflo is also a concert pianist. At eleven he won Sweden’s competition for young pianists of 1956, and he won it again in 1961; among his teachers was Géza Anda. He has kept performing ever since.

26. THE ROAD TO DENTON: DAN MAULDIN AND NORTH TEXAS

In the 1970s the story moved to Texas. R. Daniel Mauldin, a measure theorist and descriptive set theorist at North Texas State University (now the University of North Texas) in Denton, took on the task of finding out what had become of the 193 problems. In May 1979 he organized the *Scottish Book Conference* in Denton.

The list of participants reads like a family reunion. Ulam, Mark Kac and Antoni Zygmund, contributors to the original book, came, together with Paul Erdős and Andrzej Granas. R. D. Anderson and D. A. Martin spoke about work that went back to the Scottish Book; Martin had proved Borel determinacy in 1975, the theorem that the infinite games descended from Problem 43 are determined whenever the payoff set is Borel. John Oxtoby, who had published the first proof of Banach’s solution of Problem 43, gave the toast, and R. H. Bing, a pillar of Texas topology, also spoke.

From the conference came the book edited by Mauldin, *The Scottish Book: Mathematics from the Scottish Café* (Birkhäuser, 1981) [Mau81]. It contains Ulam’s “anecdotal history” of the book, the problems with commentaries on their later history, and papers from the conference. A second edition appeared in 2015 [Mau15].

27. THE NEW SCOTTISH BOOK

The tradition did not die in Lwów. In Wrocław, where much of the Lwów university was re-established after the war, Steinhaus started the *New Scottish Book* (*Nowa Księga Szkocka*); its first entry, by Steinhaus, is dated 2 July 1946, and problems were entered in it until 1979, and occasionally until 1987. The second edition of Mauldin’s book includes a history of the New Scottish Book and selected problems from it.

28. THE CONNECTION TO SET-THEORETIC TOPOLOGY AND RAMSEY SPACES AT TORONTO

The questions of the Scottish Book on measure, category and infinite games also run through set-theoretic topology, a field in which the University of Toronto has long been a leading centre.

In 1930 Ulam proved that a countably additive probability measure that is defined on all subsets of a set and vanishes on points can exist only if the set is very large: at least as large as a weakly inaccessible cardinal, and, if the measure takes only the values 0 and 1, at least as large as a strongly inaccessible one. In particular there is no such measure on all subsets of the real line unless the continuum is weakly inaccessible. Along the way he introduced what are now called Ulam matrices, and the cardinals that carry two-valued measures of this kind are the measurable cardinals [Ula30]. Large cardinals of this kind became a standard tool for consistency proofs in topology. At Toronto, Franklin D. Tall and his collaborators applied such set-theoretic methods to classical problems of point-set topology, most notably the normal Moore space problem. Infinite games of the Banach–Mazur type, and their relatives such as the Menger and Rothberger games of Part III, are now everyday tools in the field.

Stevo Todorcevic, at Toronto, developed the theory of Ramsey spaces, which unifies the Galvin–Prikry and Ellentuck theorems, Gowers’s theorem and much else in a single axiomatic framework [Tod10]. The circle closes in the present: Tall is a coauthor, with Eduardo Dueñez and José Iovino of UT San Antonio, of recent work on the complexity of deep computations through the topology of function spaces [DnIMW⁺26], the subject of the seminar for which these notes were written.

29. THE TRIUMPH OF GAMES: COMBINATORICS, LOGIC, AND THE REAL-VALUED SETTING

Texas also became a centre of the Banach space theory that Banach’s 1932 monograph and the Scottish Book had launched. Edward Odell at the University of Texas at Austin and Thomas Schlumprecht at Texas A&M University were at the heart of it.

In 1991 Schlumprecht constructed the first arbitrarily distortable Banach space [Sch91], and in 1994 Odell and Schlumprecht used his ideas to solve the distortion problem: the Hilbert space ℓ_2 is itself arbitrarily distortable [OS94]. Schlumprecht’s space was also the starting point of W. T. Gowers and B. Maurey, who in 1992 (published in 1993) answered the old unconditional basic sequence problem by constructing a “hereditarily indecomposable” space, a space in which no infinite-dimensional subspace has an unconditional basis [GM93].

The last step needed a different kind of mathematics. Banach’s homogeneous space problem, from his 1932 monograph, asks whether ℓ_2 is the only infinite-dimensional Banach space isomorphic to all of its infinite-dimensional closed subspaces. To answer it, Gowers turned to infinite combinatorics and logic.

The bridge was infinite-dimensional Ramsey theory, the subject of Section 11: the theorems of Galvin and Prikry, Silver and Ellentuck, proved by logicians, say that definable colourings of the infinite subsets of ω have infinite homogeneous sets. Gowers saw that this machinery could be imported into Banach space theory if it was translated into a continuous, real-valued setting, with block sequences of vectors in place of infinite sets of integers and a two-player game in place of the choice of integers. His Ramsey theorem for this game yields his dichotomy theorem, and, combined with a theorem of Komorowski and Tomczak-Jaegermann, shows that every homogeneous Banach space is isomorphic to ℓ_2 (see Section 12).

When Gowers received the Fields Medal in 1998, the cycle was complete. Infinite games, born at a café table in Lwów in 1935, had passed through the foundations of mathematics (determinacy), through the hands of logicians and combinatorialists (infinite Ramsey theory), and through Texas, Paris, Cambridge and Toronto, and had come back to analysis to settle a problem posed by Banach himself. The notebook that survived a war, and the people who survived it, or did not, and the colleagues who carried their problems across an ocean, are part of that triumph.

30. PROPOSAL: A SCOTTISH BOOK OF OUR OWN

The Scottish Book was not a journal. It was a notebook on a café table, open to anyone who sat down, where a problem could be written before it was understood and a prize could be a beer or a goose. That spirit is easy to revive. We propose that the participants of this seminar, in San Antonio and in Toronto, keep a Scottish Book of their own: a shared notebook of problems that come up in our discussions, each signed and dated, with a prize chosen by its author, and with room for the answers, partial results and counterexamples that follow. Students are especially welcome to contribute; Ulam was twenty-six when he wrote in the original book, and some of its best problems are short questions anyone can state.

FURTHER READING

Ulam's autobiography [Ula76] and Kac's [Kac85] are the best first-hand accounts of the Lwów School and of what came after. Kałuža's biography of Banach [Kal96] tells the story of Banach's life and of the city. Mauldin's book [Mau81, Mau15] contains the problems themselves, with commentaries on what became of them.

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FIGURE 18. Stanislaw Ulam in 1962. Photograph by Alfred Eisenstaedt for *LIFE*, from a series of portraits of mathematicians that Eisenstaedt made for the magazine in 1962–63, which also includes Kurt Gödel and Claude Shannon.

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